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Tau single domain antibodies

Abstract

The present disclosure is directed to single domain antibodies that bind to tau and the use of these antibodies for the treatment and diagnosis of tauopathies. The present disclosure is also directed to polynucleotides encoding the tau single domain antibodies, therapeutic vectors comprising these polynucleotides and methods of administering these therapeutic vectors for the treatment of tauopathies.

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Background/Summary

(1) This application is a national stage application under 35 U.S.C. § 371 of PCT Application No. PCT/US2019/018571, filed Feb. 19, 2019, which claims the benefit of U.S. Provisional Patent Application Ser. Nos. 62/632,261 filed Feb. 19, 2018, and 62/672,949 filed May 17, 2018, which are hereby incorporated by reference in their entirety.

FIELD OF THE INVENTION

(1) The present invention is directed to antibodies for the treatment and diagnosis of conditions involving pathological tau.

BACKGROUND OF THE INVENTION

(2) Immunotherapies targeting various protein aggregates such as amyloid- β (A β), tau and α -synuclein are in different stages of clinical development, and collectively are the most common approach by the pharmaceutical industry to tackle diseases characterized by such depositions (Golde T. E. "Open Questions for Alzheimer's Disease Immunotherapy," *Alzheimers Res Ther* 6:3 (2014); Valera et al., "Immunotherapy for Neurodegenerative Diseases: Focus on Alpha-Synucleinopathies," *Pharmacol Ther* 138:311-322 (2013); Pedersen et al. "Tau Immunotherapy for Alzheimer's Disease," *Trends Mol Med* 21:394-402 (2015)). The majority of these approaches involve whole antibodies and much less attention has been paid to antibody fragments which have certain advantages and their therapeutic and diagnostic potential should be explored further.

(3) The present invention is directed to overcoming these and other deficiencies in the art.

SUMMARY OF THE INVENTION

(4) A first aspect of the present disclosure is directed to an antibody or fragment thereof comprising a heavy chain variable region. The heavy chain variable region comprises a complementarity-determining region 1 (H-CDR1) comprising an amino acid sequence of any one of SEQ ID NOs: 1-38, 202, 206, 210, 214 or a modified amino acid sequence of any one of SEQ ID NOs: 1-38, 202, 206, 210 and 214 said modified sequence containing 1, 2, or 3 amino acid residue modifications as compared to any one of SEQ ID NOs: 1-38, 202, 206, 210 and 214; a complementarity-determining region 2 (H-CDR2) comprising an amino acid sequence of any one of SEQ ID NOs: 39-76, 203, 207, 211, 215 or a modified amino acid sequence of any one of SEQ ID NOs: 39-76, 203, 207, 211, and 215 said modified sequence containing 1, 2, or 3 amino acid residue modifications as compared to any one of SEQ ID NOs: 39-76, 203, 207, 211, and 215; and a complementarity-determining region 3 (H-CDR3) comprising an amino acid sequence of any one of SEQ ID NOs: 77-114, 204, 208, 212, 216 or a modified amino acid sequence of any one of SEQ ID NOs: 77-114, 204, 208, 212, and 216 said modified sequence containing 1, 2, or 3 amino acid residue modifications as compared to any one of SEQ ID NOs: 77-114, 204, 208, 212 and 216.

(5) Another aspect of the present disclosure is directed to an antibody or fragment thereof comprising a heavy chain variable region. The heavy chain variable region comprises a complementarity-determining region 1 (H-CDR1) comprising an amino acid sequence of any one of SEQ ID NOs: 153-164, 307, or a modified amino acid sequence of any one of SEQ ID NOs: 153-164 and 307, said modified sequence containing 1, 2, or 3 amino acid residue modifications as compared to any one of SEQ ID NOs: 153-164 and 307; a complementarity-determining region 2 (H-CDR2) comprising an amino acid sequence of any one of SEQ ID NOs: 165-176, 308, or a modified amino acid sequence of any one of SEQ ID NOs: 165-176 and 308, said modified sequence containing 1, 2, or 3 amino acid residue modifications as compared to any one of SEQ ID NOs: 165-176 and 308; and a complementarity-determining region 3 (H-CDR3) comprising an amino acid sequence of any one of SEQ ID NOs: 177-188, 309 or a modified amino acid sequence of any one of SEQ ID NOs: 177-188 and 309, said modified sequence containing 1, 2, or 3 amino acid residue modifications as compared to any one of SEQ ID NOs: 177-188 and 309.

(6) Another aspect of the present disclosure is directed to a method of inhibiting onset of one or more symptoms of a condition involving pathological tau protein in a subject. This method involves administering to the subject a pharmaceutical composition comprising one or more

antibodies or binding fragments thereof as described herein, or polynucleotides encoding the one or more antibodies or binding fragments thereof as described herein, in an amount effective to inhibit onset of one or more symptoms of the condition involving pathological tau protein in the subject.

(7) Another aspect of the present disclosure is directed to a method of treating a condition involving pathological tau protein in a subject. This method involves administering to the subject a pharmaceutical composition comprising one or more antibodies or binding fragments thereof as described herein, or polynucleotides encoding the one or more antibodies or binding fragments thereof as described herein, in an amount effective to treat the condition involving pathological tau protein in the subject.

(8) Another aspect of the present disclosure is directed to a method of diagnosing Alzheimer's disease or a tauopathy in a subject. This method involves detecting, in the subject, the presence of accumulated tau protein or peptide using the antibody or binding fragment as described herein, and diagnosing Alzheimer's disease or a tauopathy based on the detection of the accumulated tau protein or peptide in the subject.

(9) Another aspect of the present disclosure is directed to a method of monitoring the progression of Alzheimer's disease or a tauopathy in a subject. This method involves detecting, in the subject, the presence of accumulated tau protein or peptide using the antibody or binding fragment thereof as described herein, repeating the detecting step periodically; and monitoring the progression of Alzheimer's disease or the tauopathy in the subject as a result of the repeated detecting.

(10) Another aspect of the present disclosure is directed to a diagnostic kit. The diagnostic kit contains any one or more antibodies or binding fragments thereof as described herein, and a detectable label.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. 1A shows the tau single domain antibody (sdAb) titer generated in llama following 1-5 immunizations with recombinant tau441. FIG. 1B shows immunoreactivity of the tau sdAb with brain homogenates having different Progressive Supranuclear Palsy (PSP) human tauopathy.

(2) FIGS. 2A-2B shows enrichment of tau sdAbs by phage display library panning against biotinylated-tau solution phase (FIG. 2A) and solid phase (FIG. 2B).

(3) FIGS. 3A-3D show the binding profiles of tau sdAbs to recTau441 and paired helical filaments (PHF). FIG. 3A-B show the binding profile of tau sdAbs enriched by solution phase tau binding (round 2 enriched antibodies shown in FIG. 3A and round 3 enriched antibodies shown in FIG. 3B). FIGS. 3C-3D show the binding profile of tau sdAbs enriched by solid phase tau binding (round 2 enriched antibodies shown in FIG. 3C and round 3 enriched antibodies shown in FIG. 3D).

(4) FIGS. 4A-4B show anti-tau sdAb binding tau pathology in human tauopathy brains. FIG. 4A is human tauopathy brain section reacted with tau sdAb 2B8. FIG. 4B is an adjacent human tauopathy brain section reacted with PHF1 antibody for comparison.

(5) FIGS. 5A-5B show that tau sdAb clears tau from primary neurons derived from Tg JNPL3 tauopathy pups (Day 0) (FIG. 5B) without overt signs of toxicity (FIG. 5A).

(6) FIGS. 6A-6B are graphs showing anti-tau sdAb prevents toxicity of PHF both extracellularly and intracellularly (FIG. 6A), and clears human Alzheimer's brain derived tau protein (FIG. 6B) in tauopathy mouse primary neuronal culture. To examine extracellular effects, antibodies (1 μ g/ml) were administered with enriched human paired helical filaments (PHF: 1 μ g/ml) derived from AD brain to primary tauopathy JNPL3 mouse cultures (PHF+Ab) for 7 days. Under these conditions, monoclonal sdAb (sdAb1) and polyclonal sdAbs (p-sdAbs) work equally well extracellularly in preventing PHF neurotoxicity (FIG. 6A) and in clearing tau pathology (FIG. 6B). To examine

intracellular effects, antibodies were administered 24 h after PHF (PHF.fwdarw.Ab), which by then has been taken up into the neurons. Under these conditions (PHF.fwdarw.Ab), both the m-sdAb (sdAb1) and p-sdAbs were equally effective in preventing PHF neurotoxicity (FIG. 6A), and in clearing tau pathology (FIG. 6B), examined 7 days later, with comparable efficacy to the extracellular condition. ##, ###p<0.01, 0.001 compared to untreated control. ****p<0.0001 compared to PHF alone. sdAb1: 2B8; p-sdAbs: polyclonal sdAb

(7) FIGS. 7A-7C show that anti-tau sdAb prevents toxicity of and clears human Alzheimer's brain derived tau protein in tauopathy human neuron-like culture. Differentiated SH-SY5Y human neuroblastoma cells were treated with PHF-enriched tau for 24 h, washed to remove extracellular PHF, treated with sdAb for 72 h, and then processed for western blots. PHF showed a dose-dependent neurotoxicity that was partially alleviated by the sdAbs (FIG. 7A). Likewise, PHF treatment led to a dose-dependant increase in total- (FIG. 7B) and phospho-tau (FIG. 7C) that was strongly attenuated by the sdAbs. #, ##, ###, ####: p<0.05, 0.01, 0.001, 0.0001, compared to untreated control. **, ***, ****: p<0.01, 0.001, 0.0001, compared to PHF alone. sdAb1: 2B8; p-sdAbs: polyclonal sdAb

(8) FIGS. 8A-8C show that anti-tau sdAb provides strong specific brain signal in tauopathy mice but not in wildtype mice. Mice received an intravenous injection of 250 µg of sdAb labeled with a near infrared dye. IVIS signal was obtained from the anesthetized mice several times within the first hour. Brain signal was strong in JNPL3 mouse G47 (FIG. 8A; 17 min), and G48 (FIG. 8B; 10 min) with strong signal seen also presumably in the kidneys (FIG. 8A) reflecting probe clearance and in the spinal cord (FIG. 8A and FIG. 8B), which in the JNPL3 model has extensive tau pathology. Quantitative analysis of IVIS brain signal over time (FIG. 8C), shows strong (G48 and G47) and moderate (rest of animals) specific signal from the tauopathy mice, presumably reflecting their degree of tau pathology, and minimal signal in wt mice (WT1 and WT2). One mouse (F45) received both sdAb1 and p-sdAbs resulting in a similar brain signal. sdAb1: 2B8; p-sdAbs: polyclonal sdAb.

(9) FIG. 9 is a panel of immunocytochemical images showing anti-tau sdAb distribution to the endosomal-lysosomal system in the brain after intravenous injection in tauopathy mice. In a subset of IVIS imaged mice (near-infrared dye labeled sdAb (250 µg) injected intravenously (i.v)), brains were removed 1 h after injection, sectioned and stained with a nuclear stain (Hoechst; FIG. 9, far left), and an antibody against early endosomes (EEA1; FIG. 9 top panel, second image from left) or late endosomes/lysosomes (Rab; FIG. 9, bottom panel, second image from left). Merged images and their magnification (as per white boxes; FIG. 9, top/bottom panel right images) revealed that sdAb1 enters the brain following i.v. injection and is taken up into neurons into the endosomal-lysosomal system. sdAb1: 2B8

(10) FIG. 10. is a panel of immunocytochemical images showing anti-tau sdAb binding to pathological tau in the brain after intravenous injection in tauopathy mice. In a subset of IVIS imaged mice (near-infrared dye labeled sdAb (250 µg) injected intravenously (i.v)), brains were removed 1 h after injection, sectioned and stained with a nuclear stain (Hoechst; FIG. 10, far left images), and an antibody against hyperphosphorylated tau epitope (PHF1; FIG. 10, top panel, second image from left) or a conformational tau epitope (MC1; FIG. 10, bottom panel, second image from left). Merged images and their magnification (as per white boxes; FIG. 10, top/bottom panel right images) revealed that sdAb1 enters the brain following i.v. injection and is taken up into neurons, where it binds to pathological tau protein in the endosomal-lysosomal system as per FIG. 9 sdAb1: 2B8

(11) FIG. 11 is a sequence alignment of the anti-tau sdAbs selected via solid-phase panning. The three complementarity determining regions (CDRs) are shown. Sequence identifiers for each of the sdAbs is indicated in the figure. Omitted from each sequence in the Sequence Listing is the portion containing the histidine tag.

(12) FIG. 12 is a sequence alignment of the anti-tau sdAb selected via solution-phase panning. The

three complementarity determining regions (CDRs) are shown. Sequence identifiers for each of the sdAbs is indicated in the figure. Omitted from each sequence in the Sequence Listing is the portion containing the histidine tag.

DETAILED DESCRIPTION OF THE INVENTION

(13) A first aspect of the present invention is directed to an antibody or binding fragment thereof that binds tau protein. In particular, the antibody or binding fragment as disclosed herein binds to human tau protein. In one embodiment, the antibody or binding fragment thereof of the present disclosure binds to pathological aggregated forms of tau protein, such as tangles, paired helical filaments, and tau oligomers that are associated with the development and progression of a tauopathy or associated condition. In another embodiment, the antibody or binding fragment thereof of the present disclosure binds to the non-pathological forms of tau.

(14) In one embodiment, the antibodies described herein are single domain antibodies. The unique binding property or antigen binding specificity of a given antibody is determined by its complementarity determining regions (CDR) typically found in the light and heavy chain variable regions of an immunoglobulin. Single domain antibodies are antibodies whose CDRs are part of a single domain polypeptide. Examples include, but are not limited to, heavy chain variable domain antibodies (including three CDRs) and light chain variable domain antibodies (containing three CDRs). Single domain antibodies can be derived from heavy chains and light chains of conventional 4-chain antibodies, antibodies naturally devoid of light chains, engineered antibodies and single domain scaffolds other than those derived from antibodies. In one embodiment, the single domain antibody of the present invention is a heavy chain single domain antibody, i.e., a V.sub.HH or nanobody. In one embodiment, the heavy chain single domain antibodies described herein are derived from naturally occurring heavy chain antibodies devoid of light chains.

(15) Single domain antibodies are about 10-times smaller than conventional heavy chain and light chain containing IgG molecules. They are single polypeptides that are very stable, as they are resistant to extreme pH and temperature conditions. Moreover, unlike conventional antibodies, single domain antibodies are resistant to the action of proteases. In vitro expression of V.sub.HHs produces high yield, properly folded functional V.sub.HHs. Given their small size, V.sub.HHs are capable of recognizing unique epitopes of an antigen that cannot be bound by traditional full sized antibodies. As such, the anti-tau single domain antibodies described herein bind unique epitopes of tau and/or bind epitopes more efficiently than conventional tau antibodies.

(16) The single domain antibodies as described herein can be derived from antibodies raised in Camelidae species, for example in camel, dromedary, llama, vicuna, alpaca and guanaco. Single domain antibodies produced by other species are also within the scope of the invention. For example, single domain antibodies as disclosed herein may be derived from antibodies produced in any species including, but not limited to mouse, human, camel, llama, goat, rabbit, bovine, and cartilaginous fish.

(17) In another embodiment, the antibody as described herein is an antibody fragment. In one embodiment, an antibody fragment is a single-chain polypeptide comprising one CDR as described herein. In another embodiment, the antibody fragment is a single-chain polypeptide comprising two CDRs as described herein. In another embodiment, the antibody fragment is a single-chain polypeptide containing all three CDRs of the heavy chain variable regions as described herein. An antibody fragment as referred to herein may be devoid of one or more framework regions (FR1, FR2, FR3, or FR4) or any portion thereof of the heavy chain variable region.

(18) In one embodiment, the antibody as described herein comprises a V.sub.H domain coupled to one or more heavy chain constant regions (C.sub.H). Mammalian heavy chain immunoglobulins typically have three or four constant region domains. Accordingly, the heavy chain variable regions described herein may be coupled to one heavy chain constant region, two heavy chain constant regions, three heavy chain constant regions, or four heavy chain constant regions.

(19) In one embodiment, the antibody as described herein comprises a V.sub.H domain coupled to

an Fc region, i.e., the antibody is an Fc-fusion antibody. The Fc region can be composed of the second and third constant domain regions (as it is for IgG, IgA, and IgD antibody isotypes), or composed of the second, third, and fourth constant domain regions (as it is for IgM and IgE antibody isotypes). In one embodiment, the Fc domain is derived from a human immunoglobulin. In one embodiment, the Fc domain is derived from human IgG1 including the C.sub.H2 and C.sub.H3 regions.

(20) The Fc-region or domain of the fusion polypeptides described herein may impart non-antigen binding functions to the polypeptide, termed “effector functions”, such as complement binding, antibody-dependent cell cytotoxicity (ADCC), and other functions mediated through the binding of subregions of this dimeric structure with immune cell surface receptors, Fc-receptors. Certain natural and synthetic variants of the Fc-region polypeptide sequences with altered effector functions that are suitable for use in the fusion polypeptides described herein include the subclass variants; e.g. IgGi, IgG2i, IgG3i, IgG24; and mutant polypeptides as described in e.g. U.S. Pat. No. 5,624,821 to Winter, U.S. Pat. No. 6,528,624 to Idusogie, U.S. Pat. No. 7,183,387 to Presta, and U.S. Pat. No. 7,317,091 to Lazar et al., which are hereby incorporated by reference in their entirety.

(21) In another embodiment, the antibody or fragment thereof comprises two or more variable domain regions coupled together. For example, in one embodiment, the antibody as described herein comprises two, or three, or more heavy chain variable regions linked together in tandem. In another embodiment, the heavy chain variable region is fused together with a light chain variable region to form a single-chain variable domain antibody (scFv) or a single-chain variable domain with an Fc portion (i.e., a scFv-Fc, e.g., a minibody). In another embodiment, two or more single-chain antibodies are linked together either in tandem (i.e., tandem scFvs), or such that they dimerize to form diabodies or triabodies. In another embodiment, the antibody is a tetrabody single chain variable fragment. In another embodiment, the antibody is a “linear antibody”, which is an antibody comprising a pair of tandem Fd segments (V.sub.H—C.sub.H1-V.sub.H-C.sub.H1) that form a pair of antigen binding regions (see Zapata et al. *Protein Eng.* 8(10):1057-1062 (1995), which is hereby incorporated by reference in its entirety).

(22) In another embodiment, the antibody of the disclosure is a conventional immunoglobulin (Ig) molecule comprising four polypeptide chains, i.e., two heavy chains and two light chains linked by disulfide bonds. In accordance with this embodiment, the single-domain antibodies as described herein are coupled to constant domain regions and further coupled to Ig light chains to create a four chain conventional antibody.

(23) Antibody and antibody fragments disclosed herein can be mono-valent, bi-valent, or tri-valent with regard to binding domains, and the binding domains may be mono-specific, bi-specific, or tri-specific in binding specificity by design.

(24) In one embodiment, the antibody or fragment thereof is isolated. As used herein, the term “isolated” refers to an antibody which has been separated from a component of its natural environment. In some embodiments, an antibody is purified to greater than 95% or 99% purity as determined by, for example, electrophoretic separation (e.g., SDS-PAGE, isoelectric focusing (IEF), capillary electrophoresis) or chromatographic separation (e.g., gel filtration, ion exchange or reverse phase HPLC). Methods for assessing antibody purity are known in the art (see e.g., Flatman et al., *J. Chromatogr. B* 848:79-87 (2007), which is hereby incorporated by reference in its entirety), and are suitable for assessing the purity of the antibodies described herein.

(25) In one embodiment, the antibody or binding fragment thereof described herein is a chimeric antibody. A chimeric antibody is an antibody where one portion of the amino acid sequence of each of the heavy chains is homologous to corresponding sequences in an antibody derived from a particular species or belonging to a particular class, while the remaining segment of each chain is homologous to corresponding sequences in another species or class. Typically, the variable region mimics the variable region of an antibody derived from one species of mammals, while the constant portions are homologous to sequences of antibodies derived from another. For example,

the variable region can be any one of the heavy chain variable regions disclosed herein derived from a camelid antibody coupled to one or more constant regions derived from a human immunoglobulin. Methods of making chimeric antibodies are well known in the art, see e.g., U.S. Pat. No. 4,816,567; and Morrison et al., "Chimeric human antibody molecules: mouse antigen-binding domains with human constant region domains" *Proc. Natl. Acad. Sci. USA* 81:6851-6855 (1984), which are hereby incorporated by reference in their entirety).

(26) In another embodiment, the antibody or binding fragment thereof is a CDR-grafted antibody. A "CDR-grafted antibody" is an antibody which comprises variable region sequences of one species, where one or more of the CDR regions are replaced with CDR regions of another species. For example, in one embodiment the CDR grafted antibody comprises human or humanized heavy chain variable regions, where one or more of the CDRs within these regions is replaced with one or more CDRs disclosed herein that are derived from camelid heavy chain antibodies.

(27) In another embodiment, the antibody or binding fragment thereof is a humanized antibody. A humanized antibody is an antibody or a variant, derivative, analog or portion thereof which comprises a framework region having substantially the amino acid sequence of a human antibody and a complementary determining region having substantially the amino acid sequence of a non-human antibody. As used herein, the term "substantially" in the context of a CDR refers to a CDR having an amino acid sequence that is at least 65%, at least 70%, at least 75%, at least 80%, at least 85%, at least 90%, at least 95%, at least 98% or at least 99% identical to the amino acid sequence of a non-human antibody CDR. Likewise, the term "substantially" in the context of a FR refers to a FR having an amino acid sequence that is at least 65%, at least 70%, at least 75%, at least 80%, at least 85%, at least 90%, at least 95%, at least 98% or at least 99% identical to the amino acid sequence of a human FR. A humanized antibody in accordance with the present disclosure comprises, for example, substantially all of at least one variable domains (Fv) in which all or substantially all of the CDR regions correspond to those of a non-human immunoglobulin (i.e., the donor camelid antibody) and all or substantially all of the framework regions are those of a human or humanized immunoglobulin framework sequence (i.e., the acceptor antibody).

(28) Methods of humanizing antibodies are well known in the art, see e.g., Almagro and Fransson, "Humanization of Antibodies," *Frontiers in Bioscience* 13:1619-1633 (2008), U.S. Pat. No. 6,054,297 to Carter et al., U.S. Pat. No. 8,343,489, and U.S. Patent Application Publication No. US20100261620 to Almagro et al., which are hereby incorporated by reference in their entirety. The human or humanized framework sequences can be chosen based on known structure, i.e., a fixed framework sequence, sequence homology to the framework sequences of the donor antibody (e.g., the antibody from which the CDRs are derived), i.e., a best-fit framework sequence, or a combination of both approaches. Regardless of the method chosen to select the human framework sequence, the sequences can be selected from mature framework sequences, germline gene sequences, or consensus framework sequences. Compatible human framework sequences are those that are similar in both length and sequence to the framework sequence of the donor antibody sequence (i.e., the antibody from which the CDRs are derived) to ensure proper folding of the antibody and binding domain formation.

(29) In one embodiment, the humanized framework sequence of a humanized antibody of the disclosure comprises a consensus framework sequence. A consensus framework sequence is derived from a consensus immunoglobulin sequence, which is the sequence formed from the most frequently occurring amino acids (or nucleotides) in a family of related immunoglobulin sequences (see e.g., WINNAKER, "From Genes to Clones: Introduction to Gene Technology" (1987); Carter et al., *Proc. Natl. Acad. Sci. USA*, 89:4285 (1992); and Presta et al., *J. Immunol.*, 151:2623 (1993), which are hereby incorporated by reference in their entirety). In a family of immunoglobulins, each position in the consensus sequence is occupied by the amino acid residue occurring most frequently at that position in the family. If two amino acids occur equally frequently, either can be included in the consensus sequence.

(30) In another embodiment, a humanized antibody or binding fragment thereof as disclosed herein comprises a fixed framework region. Human heavy chain FR sequences known in the art can be used as heavy chain “acceptor” framework sequences (or simply, “acceptor” sequences) to humanize a non-human antibody using techniques known in the art (see e.g., Sims et al., *J. Immunol.*, 151:2296 (1993); Chothia et al., *J. Mol. Biol.*, 196:901 (1987), which are hereby incorporated by reference in their entirety). In one embodiment, human heavy chain acceptor sequences are selected from the framework sequences listed in publically available databases such as V-base or in the international ImMunoGeneTics® (IMGT®) information system.

(31) Humanized antibodies or binding fragments thereof as described herein may also comprise at least a portion of an immunoglobulin constant region (Fc), typically that of a human immunoglobulin. In one embodiment, the humanized antibody disclosed herein comprises the heavy chain variable domain. The humanized antibody may further comprise the CH1, hinge, CH2, CH3, and CH4 regions of a human heavy chain. In another embodiment, the humanized antibody comprises only a humanized heavy chain. Humanized antibodies and binding fragments thereof as described herein may be selected from any class of immunoglobulins, including IgM, IgG, IgD, IgA and IgE, and any isotype, including without limitation IgG1, IgG2, IgG3 and IgG4. The humanized antibody or binding fragment thereof may comprise sequences from more than one class or isotype, and particular constant domains may be selected to optimize desired effector functions using techniques well-known in the art.

(32) The antibodies and binding fragments thereof described herein can be humanized antibodies (fully or partially humanized) as described supra. Alternatively, the antibodies and binding fragments thereof can be animal antibodies such as, but not limited to, a bird (for example, a duck or a goose), a shark, a whale, or a mammal, including a non-primate (for example, a cow, a pig, a camel, a llama, a horse, a goat, a rabbit, a sheep, a hamster, a guinea pig, a cat, a dog, a rat, a mouse, etc.) or a non-human primate (for example, a monkey, a chimpanzee, etc.). In one embodiment, the antibodies and binding fragments thereof as described herein are derived from camelid antibodies.

(33) Methods of antibody production, in particular, monoclonal antibody production, may be carried out using the methods described herein and those well-known in the art (MONOCLONAL ANTIBODIES—PRODUCTION, ENGINEERING AND CLINICAL APPLICATIONS (Mary A. Ritter and Heather M. Ladyman eds., 1995), which is hereby incorporated by reference in its entirety). Generally, the process involves obtaining immune cells (lymphocytes) from the spleen of an animal which has been previously immunized with the antigen of interest (e.g., full length tau as set forth in the Examples herein) either in vivo or in vitro.

(34) The antibody-secreting lymphocytes are then fused with myeloma cells or transformed cells, which are capable of replicating indefinitely in cell culture, thereby producing an immortal, immunoglobulin-secreting cell line. Fusion with mammalian myeloma cells or other fusion partners capable of replicating indefinitely in cell culture is achieved by standard and well-known techniques, for example, by using polyethylene glycol (PEG) or other fusing agents (Milstein and Kohler, “Derivation of Specific Antibody-Producing Tissue Culture and Tumor Lines by Cell Fusion,” *Eur J Immunol* 6:511 (1976), which is hereby incorporated by reference in its entirety). The immortal cell line, which is preferably murine, but may also be derived from cells of other mammalian species, is selected to be deficient in enzymes necessary for the utilization of certain nutrients, to be capable of rapid growth, and have good fusion capability. The resulting fused cells, or hybridomas, are cultured, and the resulting colonies screened for the production of the desired monoclonal antibodies. Colonies producing such antibodies are cloned, and grown either in vivo or in vitro to produce large quantities of antibody.

(35) In another embodiment, monoclonal antibodies can be isolated from antibody phage libraries generated using the techniques described herein or known in the art, see e.g., McCafferty et al., “Phage Antibodies: Filamentous Phage Displaying Antibody Variable Domains,” *Nature* 348:552-

554 (1990), which is hereby incorporated by reference in its entirety. Clackson et al., "Making Antibody Fragments using Phage Display Libraries," *Nature* 352:624-628 (1991); and Marks et al., "By-Passing Immunization. Human Antibodies from V-Gene Libraries Displayed on Phage," *J. Mol. Biol.* 222:581-597 (1991), which are hereby incorporated by reference in their entirety, describe the isolation of murine and human antibodies, respectively, using phage libraries. Subsequent publications describe the production of high affinity (nM range) human antibodies by chain shuffling (Marks et al., *BioTechnology* 10:779-783 (1992), which is hereby incorporated by reference in its entirety), as well as combinatorial infection and in vivo recombination as a strategy for constructing very large phage libraries (Waterhouse et al., *Nuc. Acids. Res.* 21:2265-2266 (1993), which is hereby incorporated by reference in its entirety). Thus, these techniques are viable alternatives to traditional monoclonal antibody hybridoma techniques for isolation of monoclonal antibodies.

(36) Alternatively, monoclonal antibodies can be made using recombinant DNA methods as described in U.S. Pat. No. 4,816,567 to Cabilly et al, which is hereby incorporated by reference in its entirety. The polynucleotides encoding a monoclonal antibody are isolated from mature B-cells or hybridoma cells, for example, by RT-PCR using oligonucleotide primers that specifically amplify the genes encoding the heavy and light chains of the antibody. The isolated polynucleotides encoding the heavy and light chains are then cloned into suitable expression vectors, which when transfected into host cells such as *E. coli* cells, simian COS cells, Chinese hamster ovary (CHO) cells, or myeloma cells that do not otherwise produce immunoglobulin protein, generate monoclonal antibodies.

(37) The polynucleotide(s) encoding a monoclonal antibody can further be modified using recombinant DNA technology to generate alternative antibodies. For example, the heavy chain constant domains of a camelid monoclonal antibody can be substituted for those regions of a human antibody to generate a chimeric antibody. Alternatively, the heavy chain constant domains of a camelid monoclonal antibody can be substituted for a non-immunoglobulin polypeptide to generate a fusion antibody. In other embodiments, the constant regions are truncated or removed to generate the desired antibody fragment of a monoclonal antibody. Furthermore, site-directed or high-density mutagenesis of the variable region can be used to optimize specificity and affinity of a monoclonal antibody.

(38) In one embodiment of the present disclosure, the antibody or binding fragment thereof as disclosed herein comprises a heavy chain variable region (HCVR) having a H-CDR1 with an amino acid sequence selected from SEQ ID NOs: 1-38, 202, 206, 210, 214 or a modified amino acid sequence thereof containing 1, 2, 3, 4, 5, 6, 7, 8, 9, or 10 amino acid residue modifications as compared to SEQ ID NOs: 1-38, 202, 206, 210, and 214 that maintain or enhance binding specificity of the H-CDR1. In one embodiment, the amino acid sequence of the H-CDR1 contains no more than 1, 2, or 3 amino acid modifications as compared to SEQ ID NOs: 1-38, 202, 206, 210, and 214 respectively. The HCVR further comprises a H-CDR2 with an amino acid sequence selected from any one of SEQ ID NOs: 39-76, 203, 207, 211, 215, or a modified amino acid sequence thereof containing 1, 2, 3, 4, 5, 6 or 7, amino acid residue modifications as compared to SEQ ID NOs: 39-76, 203, 207, 211, and 215 that maintain or enhance binding specificity of the H-CDR2. In one embodiment, the amino acid sequence of the H-CDR2 contains no more than 1, 2, or 3, amino acid modifications as compared to SEQ ID NOs: 39-76, 203, 207, 211, and 215 respectively. The HCVR of the antibody or binding fragment thereof comprises a H-CDR3 with an amino acid sequence selected from any one of SEQ ID NOs: 77-114, 204, 208, 212, 216 or a modified amino acid sequence thereof containing 1, 2, 3, 4, 5, 6 or 7 amino acid residue modifications as compared to SEQ ID NOs: 77-114, 204, 208, 212, and 216 that maintain or enhance binding specificity of the H-CDR3. In one embodiment, the amino acid sequence of the H-CDR3 contains no more than 1, 2, or 3 amino acid modifications as compared to SEQ ID NOs: 77-114, 204, 208, 212, and 216 respectively. The amino acid sequences of SEQ ID NOs: 1-114, 202-

204, 206-208, 210-212, and 214-216 are provided in Table 1 below.

(39) TABLE-US-00001 TABLE 1 Tau Single Domain Antibody Complementarity
Determining Regions SEQ SEQ SEQ ID ID ID Antibody ID descr sequence NO: descr
sequence NO: descr sequence NO: 03-T-sR2-1D10 CDR1 TASGRTFRAYAM 1 CDR2
AISRTGGVTTY 39 CDR3 AYFRWGTRY 77 04-T-sR2-2A11 CDR1 AASGRIFSIWTM
2 CDR2 AITSVGNTDY 40 CDR3 VESRRGIGFLRKTYSY 78 11-T-sR2-1E4 CDR1
AASGSIFRINNM 3 CDR2 TITRGGNTNY 41 CDR3 ANYLIRSY 79 12-T-sR2-2B4
CDR1 AASGSIFRINNM 4 CDR2 TITRGGNTNY 42 CDR3 VKKHFGIRYDY 80 13-T-
sR2-2F8 CDR1 AASGRTFSNYAL 5 CDR2 SISWSGGVLYY 43 CDR3
ANAGLSLLRNWRTNEYAY 81 18-T-sR2-1A8 CDR1 AASGSIFRINAM 6 CDR2
RINTGGNTNY 44 CDR3 VQRFITTY 82 20-T-sR2-2B7 CDR1 AASGRTFSTYRM 7
CDR2 AIRWSTSY 45 CDR3 TGRAWSTLATTYVY 83 21-T-sR2-2G7 CDR1
AASGRTFSRYAT 8 CDR2 GISWSGTSY 46 CDR3 NFKYRYGLGPRDY 84 26-T-sR2-
1B2 CDR1 VASGSIFRFNAI 9 CDR2 RIRRLGSTSY 47 CDR3 ADTHFSTRNY 85 27-
T-sR2-1E9 CDR1 AASGSIFRINGM 10 CDR2 TITRGGSTNY 48 CDR3
ADPRWRLPFPGYGMDY 86 28-T-sR2-2B11; CDR1 AASGRILISSM 11 CDR2
TITRGGTTNY 49 CDR3 RVYGRVWSRPYDY 87 82-T-sR2-1D3 29-T-sR2-2G8 CDR1
TVSGRTFRINGI 12 CDR2 GISSTGSTNY 50 CDR3 ASRGLSGSWYLRSSYPY 88 34-T-
sR2-1B7 CDR1 AASRYIFGTM 13 CDR2 SISRGGSTNY 51 CDR3
AVPYRWGSSWYAGRY 89 35-T-sR2-1F5 CDR1 VASGSRFSINTM 14 CDR2
GITRGGSTNY 52 CDR3 ATLRAWALTFATSAY 90 36-T-sR2-2C1 CDR1
AASGRTFSRYAM 15 CDR2 RISWSGGWTYY 53 CDR3 AGSRAVLFGTYDY 91 37-T-
sR2-2G9 CDR1 AASGRTLSSYRM 16 CDR2 AINWRGSWTY 54 CDR3
RPTARWDLFREKYDFR 92 42-T-sR2-1B8 CDR1 AASGRTFSSYAM 17 CDR2
AISRSGGITSY 55 CDR3 VQRFITTY 93 43-T-sR2-1F8 CDR1 AASGRTFGLYTM 18
CDR2 AISWRGLSIMY 56 CDR3 ANAGLSLLRNWRTNEYAY 94 44-T-sR2-2C5 CDR1
EASARTFSSYAV 19 CDR2 AINWSGRRTNY 57 CDR3 ARSSLLEFWLGSRRGYDY 95
45-T-sR2-2H4 CDR1 AASGRIFSIWTM 20 CDR2 AITSGGSTNY 58 CDR3
ADIRPRIISFFKDY 96 50-T-sR2-1C5 CDR1 AASGRTFSRYAM 21 CDR2 AITWSGGHYY
59 CDR3 AYFRWGTRY 97 51-T-sR2-1G1 CDR1 AASGRSFSWLTM 22 CDR2
RITWRGTPYY 60 CDR3 AAKQILIRPDAYVY 98 52-T-sR2-2C9 CDR1 VASGSIFRFNAI
23 CDR2 RIRRLGSTSY 61 CDR3 AGRHPHFMSMDYPDL 99 53-T-sR2-2H5 CDR1
AASRYIFGTM 24 CDR2 SISRGGSTNY 62 CDR3 LALGFFTNYVYVRESSYRY 100 58-T-
sR2-1C6 CDR1 AASGRTFRLYSM 25 CDR2 SIRWNGGNIYY 63 CDR3
AGPRIAVWRYEYNY 101 59-T-sR2-1G5 CDR1 AASGSIGSFKTM 26 CDR2 TITRWGFTNY
64 CDR3 AAQGGFMKPRANWYNS 102 60-T-sR2-2D8 CDR1 VASGRTFSRYGM 27
CDR2 AISRSGAISYY 65 CDR3 ASSRLLGGPFAYDY 103 61-T-sR2-2H7 CDR1
AASRYIFGTM 28 CDR2 SISRGGSTNY 66 CDR3 ANRRGWNYY 104 66-T-sR2-1C7 CDR1
AASGTIFTMKNM 29 CDR2 AISTSGGV TWY 67 CDR3 VRSGPRIIPQLRREY 105 68-T-
sR2-2E6 CDR1 AASGRTFSSYAM 30 CDR2 AINWSGHSTYY 68 CDR3 LTRLLNTY 106
69-T-sR2-2H8 CDR1 AFSGRTFGLRTM 31 CDR2 SLTWRDNNAYY 69 CDR3
VHLVFTNRDY 107 74-T-sR2-1C10 CDR1 AASWRIFSPNAM 32 CDR2 RITWAGITNY 70
CDR3 ADDRSSYLGPFRDY 108 75-T-sR2-1G12 CDR1 AASRYIFGTM 33 CDR2
SISRGGSTNY 71 CDR3 ARSSLLEFWLGSRRGYDY 109 76-T-sR2-2E7 CDR1
AASGRFFRINAM 34 CDR2 TITRAGTTTY 72 CDR3 KYPTITWYGRHDYR 110 84-T-
sR2-2E8 CDR1 AASGSFFRINTM 35 CDR2 SITRGGSTNY 73 CDR3
ASRFGINYITARQYGY 111 90-T-sR2-1D9 CDR1 AASGRTFSRYPM 36 CDR2
RFGWSGLSTYY 74 CDR3 AGRFMGSYDY 112 91-T-sR2-1H12 CDR1 AASGRTFSWYAM
37 CDR2 AIRRSGGITIY 75 CDR3 ANLFQWRLNDNGNQYGS 113 92-T-sR2-2E9 CDR1
VVSRTFTSTSQM 38 CDR2 RISWRGKQHY 76 CDR3 AGPRAVLFGTYDY 114 T-sR2-

B28 CDR1 VSGRTFSTQ 202 CDR2 ARISWRGKQH 203 CDR3 AADRRRTYLGGQHD 204 T-sR2-2F12 CDR1 ASGSIFRING 206 CDR2 ATITRGGGIS 207 CDR3 AKYRRPLFYSGSNYREGD 208 FAS T-sR2-2B3 CDR1 ASGRTFGSYT 210 CDR2 AAISRSGGSTY 211 CDR3 NVRGRPFILSKPFDS 212 T-sR2-2C4 CDR1 ASGRTFSNYA 214 CDR2 AAITRNGGITY 215 CDR3 NIKARRGSFFNPVNN 216

(40) Suitable amino acid modifications to the heavy chain CDR sequences of the anti-tau antibodies disclosed herein include, for example, conservative substitutions or functionally equivalent amino acid residue substitutions that result in variant CDR sequences having similar or enhanced binding characteristics to those of the CDR sequences disclosed herein. Conservative substitutions are those that take place within a family of amino acids that are related in their side chains. Genetically encoded amino acids can be divided into four families: (1) acidic (aspartate, glutamate); (2) basic (lysine, arginine, histidine); (3) nonpolar (alanine, valine, leucine, isoleucine, proline, phenylalanine, methionine, tryptophan); and (4) uncharged polar (glycine, asparagine, glutamine, cysteine, serine, threonine, tyrosine). Phenylalanine, tryptophan, and tyrosine are sometimes classified jointly as aromatic amino acids. Alternatively, the amino acid repertoire can be grouped as (1) acidic (aspartate, glutamate); (2) basic (lysine, arginine, histidine), (3) aliphatic (glycine, alanine, valine, leucine, isoleucine, serine, threonine), with serine and threonine optionally grouped separately as aliphatic-hydroxyl; (4) aromatic (phenylalanine, tyrosine, tryptophan); (5) amide (asparagine, glutamine); and (6) sulfur-containing (cysteine and methionine) (Stryer (ed.), Biochemistry, 2nd ed, WH Freeman and Co., 1981, which is hereby incorporated by reference in its entirety). Non-conservative substitutions can also be made to the heavy chain CDR sequences and the light chain CDR sequences as disclosed herein. Non-conservative substitutions involve substituting one or more amino acid residues of the CDR with one or more amino acid residues from a different class of amino acids to improve or enhance the binding properties of CDR.

(41) The amino acid sequences of the heavy chain variable region CDRs of the anti-tau antibodies described herein may further comprise one or more internal neutral amino acid insertions or deletions that do not alter tau protein binding. In one embodiment, the H-CDR3 having an amino acid sequence of any one of SEQ ID NOs: 77-114, 204, 208, 212, and 216 further contains one or more internal neutral amino acid insertions or deletions that do not alter tau binding.

(42) In one embodiment, the antibody or binding fragment thereof has a heavy chain variable region with a H-CDR1 having the amino acid sequence of SEQ ID NO: 1, or a modified amino acid sequence thereof containing 1, 2, 3, or more amino acid residue modifications as compared to SEQ ID NO: 1; a H-CDR2 having the amino acid sequence of SEQ ID NO: 39, or a modified amino acid sequence thereof containing 1, 2, 3, or more amino acid residue modifications as compared to SEQ ID NO: 39; and a H-CDR3 comprising the amino acid sequence of SEQ ID NO: 77, or a modified amino acid sequence thereof, said modified amino acid sequence containing 1, 2, 3, or more amino acid modifications as compared to SEQ ID NO: 77.

(43) An exemplary single domain antibody having this heavy chain variable region is referred to herein as the 03-T-sR2-1D10 antibody. The 03-T-sR2-1D10 antibody comprises a V_{sub}.H chain amino acid sequence of SEQ ID NO: 115 as shown below.

(44) TABLE-US-00002 SEQ ID NO: 115
QVQLQESGGGLVQAGGSLRLSCTASGRTFRAYAMGWFRQAPGKERELVA
AISRTGGVTTYADSVKGRFTISRDNAKNTVYLQMNNLKTEDTAVYYCNA
YFRWGTRYWGQGTQVTVSS

(45) In another embodiment, the antibody or binding fragment thereof comprises a heavy chain variable region having an amino acid sequence that shares at least 60%, at least 65%, at least 70%, at least 75%, at least 80%, at least 85%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, at least 99% sequence identity to SEQ ID NO: 115

(46) In one embodiment, the antibody or binding fragment thereof of the present disclosure comprises a humanized variant of the heavy chain variable region of SEQ ID NO: 115, where the framework regions are humanized or replaced with human immunoglobulin framework sequences. As noted supra, suitable human or humanized framework sequences can be chosen based on their known structure, a consensus sequence, sequence homology to the framework sequences of donor antibody (e.g., the framework sequences of SEQ ID NO: 115), or a combination of these approaches. The humanized framework regions are designed to be similar in length and sequence to the parental framework sequences of SEQ ID NO: 115. In one embodiment, the humanized framework regions share 50%, 55%, 60%, 65%, 70%, 75%, 80%, 85%, 90%, 95% or more sequence identity to the framework regions of SEQ ID NO: 115. In another embodiment, the humanized framework regions are 60%, 65%, 70%, 75%, 80%, 85%, 90%, 95% or more similar in length to the framework regions of SEQ ID NO: 115. Humanized variants of the heavy chain variable region of SEQ ID NO: 115 share at least 50%, at least 55%, at least 60%, at least 65%, at least 70%, at least 75%, at least 80%, at least 85%, at least 90%, at least 95% sequence identity along the entire length of SEQ ID 115.

(47) Another aspect of the present disclosure relates to an antibody or binding portion thereof (e.g., a human antibody) that competes for binding to human tau protein with the anti-tau antibody comprising a heavy chain variable region with the amino acid sequence of SEQ ID NO: 115. In accordance with this aspect of the disclosure, a competitive binding assay, such as Bio-competes for binding to a tau protein with the enumerated monoclonal antibody. Other competitive binding assays known in the art can also be utilized to identify a competitive binding antibody in accordance with this aspect of the disclosure.

(48) In another embodiment, the antibody or binding fragment thereof has a heavy chain variable region with a H-CDR1 having the amino acid sequence of SEQ ID NO: 2, or a modified amino acid sequence thereof containing 1, 2, 3, or more amino acid residue modifications as compared to SEQ ID NO: 2; a H-CDR2 having the amino acid sequence of SEQ ID NO: 40, or a modified amino acid sequence thereof containing 1, 2, 3, or more amino acid residue modifications as compared to SEQ ID NO: 40; and a H-CDR3 comprising the amino acid sequence of SEQ ID NO: 78, or a modified amino acid sequence thereof, said modified amino acid sequence containing 1, 2, 3, or more amino acid modifications as compared to SEQ ID NO: 78.

(49) Exemplary single domain antibodies having this heavy chain variable region are referred to herein as the 04-T-sR2-2A11 antibody and the 04-T-sR2-2A11 antibody. These antibodies comprise a V.sub.H chain amino acid sequence of SEQ ID NO: 116 as shown below.

(50) TABLE-US-00003 SEQ ID NO: 116

QVQLQESGGLVQPGGSLRLSCAASGRIFSIWTMGWYRQAPGKQRELVA
AITSVGNTDYADSVKGRFTISRETAKKTVYLQMNSLPEDTAVYYCNVE
SRRGIGFLRKTYSYWGQGTQVTVSS

(51) In another embodiment, the antibody or binding fragment thereof comprises a heavy chain variable region having an amino acid sequence that shares at least 60%, at least 65%, at least 70%, at least 75%, at least 80%, at least 85%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94% at least 95%, at least 96%, at least 97%, at least 98%, at least 99% sequence identity to SEQ ID NO: 116.

(52) In one embodiment, the antibody or binding fragment thereof of the present disclosure comprises a humanized variant of the heavy chain variable region of SEQ ID NO: 116, where the framework regions are humanized or replaced with human immunoglobulin framework sequences. As noted supra, suitable human or humanized framework sequences can be chosen based on their known structure, a consensus sequence, sequence homology to the framework sequences of donor antibody (e.g., the framework sequences of SEQ ID NO: 116), or a combination of these approaches. The humanized framework regions are designed to be similar in length and sequence to the parental framework sequences of SEQ ID NO: 116. In one embodiment, the humanized

framework regions share 50%, 55%, 60%, 65%, 70%, 75%, 80%, 85%, 90%, 95% or more sequence identity to the framework regions of SEQ ID NO: 116. In another embodiment, the humanized framework regions are 60%, 65%, 70%, 75%, 80%, 85%, 90%, 95% or more similar in length to the framework regions of SEQ ID NO: 116. Humanized variants of the heavy chain variable region of SEQ ID NO: 116 share at least 50%, at least 55%, at least 60%, at least 65%, at least 70%, at least 75%, at least 80%, at least 85%, at least 90%, at least 95% sequence identity along the entire length of SEQ ID 116.

(53) Another aspect of the present disclosure relates to an antibody or binding portion thereof (e.g., a human antibody) that competes for binding to human tau protein with the anti-tau antibody comprising a heavy chain variable region with the amino acid sequence of SEQ ID NO: 116. In accordance with this aspect of the disclosure, a competitive binding assay, such as Bio-Layer Interferometry (BLI) can be utilized to identify an antibody or binding portion thereof that competes for binding to a tau protein with the enumerated monoclonal antibody. Other competitive binding assays known in the art can also be utilized to identify a competitive binding antibody in accordance with this aspect of the disclosure.

(54) In one embodiment, the antibody or binding fragment thereof has a heavy chain variable region with a H-CDR1 having the amino acid sequence of SEQ ID NO: 3, or a modified amino acid sequence thereof containing 1, 2, 3, or more amino acid residue modifications as compared to SEQ ID NO: 3; a H-CDR2 having the amino acid sequence of SEQ ID NO: 41, or a modified amino acid sequence thereof containing 1, 2, 3, or more amino acid residue modifications as compared to SEQ ID NO: 41; and a H-CDR3 comprising the amino acid sequence of SEQ ID NO: 79, or a modified amino acid sequence thereof, said modified amino acid sequence containing 1, 2, 3, or more amino acid modifications as compared to SEQ ID NO: 79.

(55) An exemplary single domain antibody having this heavy chain variable region is referred to herein as the 11-T-sR2-1E4 antibody. The 11-T-sR2-1E4 antibody comprises a V_{sub}.H chain amino acid sequence of SEQ ID NO: 117 as shown below.

(56) TABLE-US-00004 SEQ ID NO: 117

QVQLQESGGGLVQPGGSLTSCAASGSIFRINNMGWFRQAPGKQRELVA
TITRGGNTNYADSVKGRFTISRDNAAKKTLYLQMNSLKPEDTAVYYCNAN
YLIRSYWGQGTQVTVSS

(57) In another embodiment, the antibody or binding fragment thereof comprises a heavy chain variable region having an amino acid sequence that shares at least 60%, at least 65%, at least 70%, at least 75%, at least 80%, at least 85%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94% at least 95%, at least 96%, at least 97%, at least 98%, at least 99% sequence identity to SEQ ID NO: 117.

(58) In one embodiment, the antibody or binding fragment thereof of the present disclosure comprises a humanized variant of the heavy chain variable region of SEQ ID NO: 117, where the framework regions are humanized or replaced with human immunoglobulin framework sequences. As noted supra, suitable human or humanized framework sequences can be chosen based on their known structure, a consensus sequence, sequence homology to the framework sequences of donor antibody (e.g., the framework sequences of SEQ ID NO: 117), or a combination of these approaches. The humanized framework regions are designed to be similar in length and sequence to the parental framework sequences of SEQ ID NO: 117. In one embodiment, the humanized framework regions share 50%, 55%, 60%, 65%, 70%, 75%, 80%, 85%, 90%, 95% or more sequence identity to the framework regions of SEQ ID NO: 117. In another embodiment, the humanized framework regions are 60%, 65%, 70%, 75%, 80%, 85%, 90%, 95% or more similar in length to the framework regions of SEQ ID NO: 117. Humanized variants of the heavy chain variable region of SEQ ID NO: 117 share at least 50%, at least 55%, at least 60%, at least 65%, at least 70%, at least 75%, at least 80%, at least 85%, at least 90%, at least 95% sequence identity along the entire length of SEQ ID 117.

(59) Another aspect of the present disclosure relates to an antibody or binding portion thereof (e.g., a human antibody) that competes for binding to human tau protein with the anti-tau antibody comprising a heavy chain variable region with the amino acid sequence of SEQ ID NO: 117. In accordance with this aspect of the disclosure, a competitive binding assay, such as Bio-Layer Interferometry (BLI) can be utilized to identify an antibody or binding portion thereof that competes for binding to a tau protein with the enumerated monoclonal antibody. Other competitive binding assays known in the art can also be utilized to identify a competitive binding antibody in accordance with this aspect of the disclosure.

(60) In one embodiment, the antibody or binding fragment thereof has a heavy chain variable region with a H-CDR1 having the amino acid sequence of SEQ ID NO: 4, or a modified amino acid sequence thereof containing 1, 2, 3, or more amino acid residue modifications as compared to SEQ ID NO: 4; a H-CDR2 having the amino acid sequence of SEQ ID NO: 42, or a modified amino acid sequence thereof containing 1, 2, 3, or more amino acid residue modifications as compared to SEQ ID NO: 42; and a H-CDR3 comprising the amino acid sequence of SEQ ID NO: 80, or a modified amino acid sequence thereof, said modified amino acid sequence containing 1, 2, 3, or more amino acid modifications as compared to SEQ ID NO: 80.

(61) An exemplary single domain antibody having this heavy chain variable region is referred to herein as the 12-T-sR2-2B4 antibody. The 12-T-sR2-2B4 antibody comprises a V.sub.H chain amino acid sequence of SEQ ID NO: 118 as shown below.

(62) TABLE-US-00005 SEQ ID NO: 118

QVQLQESGGGLVQPGGSLTSCAASGSIFRINNMGWFRQAPGKQRELVA
TITRGGNTNYADSVKGRFTISRDNANKNTVYLQMDSLKPEDTAVYYCNVK
KHFGIRYDYWGQGTQVTVSS

(63) In another embodiment, the antibody or binding fragment thereof comprises a heavy chain variable region having an amino acid sequence that shares at least 60%, at least 65%, at least 70%, at least 75%, at least 80%, at least 85%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94% at least 95%, at least 96%, at least 97%, at least 98%, at least 99% sequence identity to SEQ ID NO: 118.

(64) In one embodiment, the antibody or binding fragment thereof of the present disclosure comprises a humanized variant of the heavy chain variable region of SEQ ID NO: 118, where the framework regions are humanized or replaced with human immunoglobulin framework sequences. As noted supra, suitable human or humanized framework sequences can be chosen based on their known structure, a consensus sequence, sequence homology to the framework sequences of donor antibody (e.g., the framework sequences of SEQ ID NO: 118), or a combination of these approaches. The humanized framework regions are designed to be similar in length and sequence to the parental framework sequences of SEQ ID NO: 118. In one embodiment, the humanized framework regions share 50%, 55%, 60%, 65%, 70%, 75%, 80%, 85%, 90%, 95% or more sequence identity to the framework regions of SEQ ID NO: 118. In another embodiment, the humanized framework regions are 60%, 65%, 70%, 75%, 80%, 85%, 90%, 95% or more similar in length to the framework regions of SEQ ID NO: 118. Humanized variants of the heavy chain variable region of SEQ ID NO: 118 share at least 50%, at least 55%, at least 60%, at least 65%, at least 70%, at least 75%, at least 80%, at least 85%, at least 90%, at least 95% sequence identity along the entire length of SEQ ID 118.

(65) Another aspect of the present disclosure relates to an antibody or binding portion thereof (e.g., a human antibody) that competes for binding to human tau protein with the anti-tau antibody comprising a heavy chain variable region with the amino acid sequence of SEQ ID NO: 118. In accordance with this aspect of the disclosure, a competitive binding assay, such as Bio-compete for binding to a tau protein with the enumerated monoclonal antibody. Other competitive binding assays known in the art can also be utilized to identify a competitive binding antibody in accordance with this aspect of the disclosure.

(66) In one embodiment, the antibody or binding fragment thereof has a heavy chain variable region with a H-CDR1 having the amino acid sequence of SEQ ID NO: 5, or a modified amino acid sequence thereof containing 1, 2, 3, or more amino acid residue modifications as compared to SEQ ID NO: 5; a H-CDR2 having the amino acid sequence of SEQ ID NO: 43, or a modified amino acid sequence thereof containing 1, 2, 3, or more amino acid residue modifications as compared to SEQ ID NO: 43; and a H-CDR3 comprising the amino acid sequence of SEQ ID NO: 81, or a modified amino acid sequence thereof, said modified amino acid sequence containing 1, 2, 3, or more amino acid modifications as compared to SEQ ID NO: 81.

(67) An exemplary single domain antibody having this heavy chain variable region is referred to herein as the 13-T-sR2-2F8 antibody. The 13-T-sR2-2F8 antibody comprises a V_{sub}.H chain amino acid sequence of SEQ ID NO: 119 as shown below.

(68) TABLE-US-00006 SEQ ID NO: 119

QVQLQESGGLVQAGGSLRLSCAASGRTFSNYALAWFRQAPGLEREFVS
SISWSGGVLYYADSVKGRFTMSRDNGKNTVYLMNSLKPEDTAVYYCAA
NAGLSLLRNWRTNEYAYWGQGTQVTVSS

(69) In another embodiment, the antibody or binding fragment thereof comprises a heavy chain variable region having an amino acid sequence that shares at least 60%, at least 65%, at least 70%, at least 75%, at least 80%, at least 85%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, at least 99% sequence identity to SEQ ID NO: 119.

(70) In one embodiment, the antibody or binding fragment thereof of the present disclosure comprises a humanized variant of the heavy chain variable region of SEQ ID NO: 119, where the framework regions are humanized or replaced with human immunoglobulin framework sequences. As noted supra, suitable human or humanized framework sequences can be chosen based on their known structure, a consensus sequence, sequence homology to the framework sequences of donor antibody (e.g., the framework sequences of SEQ ID NO: 119), or a combination of these approaches. The humanized framework regions are designed to be similar in length and sequence to the parental framework sequences of SEQ ID NO: 119. In one embodiment, the humanized framework regions share 50%, 55%, 60%, 65%, 70%, 75%, 80%, 85%, 90%, 95% or more sequence identity to the framework regions of SEQ ID NO: 119. In another embodiment, the humanized framework regions are 60%, 65%, 70%, 75%, 80%, 85%, 90%, 95% or more similar in length to the framework regions of SEQ ID NO: 119. Humanized variants of the heavy chain variable region of SEQ ID NO: 119 share at least 50%, at least 55%, at least 60%, at least 65%, at least 70%, at least 75%, at least 80%, at least 85%, at least 90%, at least 95% sequence identity along the entire length of SEQ ID 119.

(71) Another aspect of the present disclosure relates to an antibody or binding portion thereof (e.g., a human antibody) that competes for binding to human tau protein with the anti-tau antibody comprising a heavy chain variable region with the amino acid sequence of SEQ ID NO: 119. In accordance with this aspect of the disclosure, a competitive binding assay, such as Bio-Layer Interferometry (BLI) can be utilized to identify an antibody or binding portion thereof that competes for binding to a tau protein with the enumerated monoclonal antibody. Other competitive binding assays known in the art can also be utilized to identify a competitive binding antibody in accordance with this aspect of the disclosure.

(72) In one embodiment, the antibody or binding fragment thereof has a heavy chain variable region with a H-CDR1 having the amino acid sequence of SEQ ID NO: 6, or a modified amino acid sequence thereof containing 1, 2, 3, or more amino acid residue modifications as compared to SEQ ID NO: 6; a H-CDR2 having the amino acid sequence of SEQ ID NO: 44, or a modified amino acid sequence thereof containing 1, 2, 3, or more amino acid residue modifications as compared to SEQ ID NO: 44; and a H-CDR3 comprising the amino acid sequence of SEQ ID NO: 82, or a modified amino acid sequence thereof, said modified amino acid sequence containing 1, 2,

3, or more amino acid modifications as compared to SEQ ID NO: 82.

(73) An exemplary single domain antibody having this heavy chain variable region is referred to herein as the 18-T-sR2-1A8 antibody. The 18-T-sR2-1A8 antibody comprises a V.sub.H chain amino acid sequence of SEQ ID NO: 120 as shown below.

(74) TABLE-US-00007 SEQ ID NO: 120

QVQLQESGGGLVQAGGSLRLSCAASGSIFRINAMAWYRQAPGKQRELVA
RINTGGNTNYAGSVKGRFTISRDNKGKNTVYLMNSLKPEDTGVYYCNVQ
RFITTYWGQGTQVTVSS

(75) In another embodiment, the antibody or binding fragment thereof comprises a heavy chain variable region having an amino acid sequence that shares at least 60%, at least 65%, at least 70%, at least 75%, at least 80%, at least 85%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94% at least 95%, at least 96%, at least 97%, at least 98%, at least 99% sequence identity to SEQ ID NO: 120.

(76) In one embodiment, the antibody or binding fragment thereof of the present disclosure comprises a humanized variant of the heavy chain variable region of SEQ ID NO: 120 where the framework regions are humanized or replaced with human immunoglobulin framework sequences. As noted supra, suitable human or humanized framework sequences can be chosen based on their known structure, a consensus sequence, sequence homology to the framework sequences of donor antibody (e.g., the framework sequences of SEQ ID NO: 120), or a combination of these approaches. The humanized framework regions are designed to be similar in length and sequence to the parental framework sequences of SEQ ID NO: 120. In one embodiment, the humanized framework regions share 50%, 55%, 60%, 65%, 70%, 75%, 80%, 85%, 90%, 95% or more sequence identity to the framework regions of SEQ ID NO: 120. In another embodiment, the humanized framework regions are 60%, 65%, 70%, 75%, 80%, 85%, 90%, 95% or more similar in length to the framework regions of SEQ ID NO: 120. Humanized variants of the heavy chain variable region of SEQ ID NO: 120 share at least 50%, at least 55%, at least 60%, at least 65%, at least 70%, at least 75%, at least 80%, at least 85%, at least 90%, at least 95% sequence identity along the entire length of SEQ ID 120.

(77) Another aspect of the present disclosure relates to an antibody or binding portion thereof (e.g., a human antibody) that competes for binding to human tau protein with the anti-tau antibody comprising a heavy chain variable region with the amino acid sequence of SEQ ID NO: 120. In accordance with this aspect of the disclosure, a competitive binding assay, such as Bio-Layer Interferometry (BLI) can be utilized to identify an antibody or binding portion thereof that competes for binding to a tau protein with the enumerated monoclonal antibody. Other competitive binding assays known in the art can also be utilized to identify a competitive binding antibody in accordance with this aspect of the disclosure.

(78) In one embodiment, the antibody or binding fragment thereof has a heavy chain variable region with a H-CDR1 having the amino acid sequence of SEQ ID NO: 7, or a modified amino acid sequence thereof containing 1, 2, 3, or more amino acid residue modifications as compared to SEQ ID NO: 7; a H-CDR2 having the amino acid sequence of SEQ ID NO: 45, or a modified amino acid sequence thereof containing 1, 2, 3, or more amino acid residue modifications as compared to SEQ ID NO: 45; and a H-CDR3 comprising the amino acid sequence of SEQ ID NO: 83, or a modified amino acid sequence thereof, said modified amino acid sequence containing 1, 2, 3, or more amino acid modifications as compared to SEQ ID NO: 83.

(79) An exemplary single domain antibody having this heavy chain variable region is referred to herein as the 20-T-sR2-2B7 antibody. The 20-T-sR2-2B7 antibody comprises a V.sub.H chain amino acid sequence of SEQ ID NO: 121 as shown below.

(80) TABLE-US-00008 SEQ ID NO: 121

QVQLQESGGGLVQAGGSLRLSCAASGRFTSTYRMGWFRQAPGKEREFVA
AIRWSTSYADSVKGRFIISRDNKNTVYLMNSLKPEDTAVYYCATGRA

WSTLTATTVYWGQGTQVTVSS

(81) In another embodiment, the antibody or binding fragment thereof comprises a heavy chain variable region having an amino acid sequence that shares at least 60%, at least 65%, at least 70%, at least 75%, at least 80%, at least 85%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94% at least 95%, at least 96%, at least 97%, at least 98%, at least 99% sequence identity to SEQ ID NO: 121.

(82) In one embodiment, the antibody or binding fragment thereof of the present disclosure comprises a humanized variant of the heavy chain variable region of SEQ ID NO: 121, where the framework regions are humanized or replaced with human immunoglobulin framework sequences. As noted supra, suitable human or humanized framework sequences can be chosen based on their known structure, a consensus sequence, sequence homology to the framework sequences of donor antibody (e.g., the framework sequences of SEQ ID NO: 121), or a combination of these approaches. The humanized framework regions are designed to be similar in length and sequence to the parental framework sequences of SEQ ID NO: 121. In one embodiment, the humanized framework regions share 50%, 55%, 60%, 65%, 70%, 75%, 80%, 85%, 90%, 95% or more sequence identity to the framework regions of SEQ ID NO: 121. In another embodiment, the humanized framework regions are 60%, 65%, 70%, 75%, 80%, 85%, 90%, 95% or more similar in length to the framework regions of SEQ ID NO: 121. Humanized variants of the heavy chain variable region of SEQ ID NO: 121 share at least 50%, at least 55%, at least 60%, at least 65%, at least 70%, at least 75%, at least 80%, at least 85%, at least 90%, at least 95% sequence identity along the entire length of SEQ ID 121.

(83) Another aspect of the present disclosure relates to an antibody or binding portion thereof (e.g., a human antibody) that competes for binding to human tau protein with the anti-tau antibody comprising a heavy chain variable region with the amino acid sequence of SEQ ID NO: 121. In accordance with this aspect of the disclosure, a competitive binding assay, such as Bio-competes for binding to a tau protein with the enumerated monoclonal antibody. Other competitive binding assays known in the art can also be utilized to identify a competitive binding antibody in accordance with this aspect of the disclosure.

(84) In one embodiment, the antibody or binding fragment thereof has a heavy chain variable region with a H-CDR1 having the amino acid sequence of SEQ ID NO: 8, or a modified amino acid sequence thereof containing 1, 2, 3, or more amino acid residue modifications as compared to SEQ ID NO: 8; a H-CDR2 having the amino acid sequence of SEQ ID NO: 46, or a modified amino acid sequence thereof containing 1, 2, 3, or more amino acid residue modifications as compared to SEQ ID NO: 46; and a H-CDR3 comprising the amino acid sequence of SEQ ID NO: 84, or a modified amino acid sequence thereof, said modified amino acid sequence containing 1, 2, 3, or more amino acid modifications as compared to SEQ ID NO: 84.

(85) An exemplary single domain antibody having this heavy chain variable region is referred to herein as the 21-T-sR2-2G7 antibody. This antibody comprises a V.sub.H chain amino acid sequence of SEQ ID NO: 122 as shown below.

(86) TABLE-US-00009 SEQ ID NO: 122

QVQLQESGGLVQAGGSLRLSCAASGRTFSRYATAWFRQAPGKEREFVA
GISWSGTSYADSVKGRFTISRDNKNTLYLQMNSLKS EDTAVYYCANFK
YRYGLGPRDYWGQGTTLVTVSS

(87) In another embodiment, the antibody or binding fragment thereof comprises a heavy chain variable region having an amino acid sequence that shares at least 60%, at least 65%, at least 70%, at least 75%, at least 80%, at least 85%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94% at least 95%, at least 96%, at least 97%, at least 98%, at least 99% sequence identity to SEQ ID NO: 122.

(88) In one embodiment, the antibody or binding fragment thereof of the present disclosure comprises a humanized variant of the heavy chain variable region of SEQ ID NO: 122, where the

framework regions are humanized or replaced with human immunoglobulin framework sequences. As noted supra, suitable human or humanized framework sequences can be chosen based on their known structure, a consensus sequence, sequence homology to the framework sequences of donor antibody (e.g., the framework sequences of SEQ ID NO: 122), or a combination of these approaches. The humanized framework regions are designed to be similar in length and sequence to the parental framework sequences of SEQ ID NO: 122. In one embodiment, the humanized framework regions share 50%, 55%, 60%, 65%, 70%, 75%, 80%, 85%, 90%, 95% or more sequence identity to the framework regions of SEQ ID NO: 122. In another embodiment, the humanized framework regions are 60%, 65%, 70%, 75%, 80%, 85%, 90%, 95% or more similar in length to the framework regions of SEQ ID NO: 122. Humanized variants of the heavy chain variable region of SEQ ID NO: 122 share at least 50%, at least 55%, at least 60%, at least 65%, at least 70%, at least 75%, at least 80%, at least 85%, at least 90%, at least 95% sequence identity along the entire length of SEQ ID NO: 122, respectively.

(89) Another aspect of the present disclosure relates to an antibody or binding portion thereof (e.g., a human antibody) that competes for binding to human tau protein with the anti-tau antibody comprising a heavy chain variable region with the amino acid sequence of SEQ ID NO: 122. In accordance with this aspect of the disclosure, a competitive binding assay, such as Bio-Layer Interferometry (BLI) can be utilized to identify an antibody or binding portion thereof that competes for binding to tau protein with the enumerated monoclonal antibody. Other competitive binding assays known in the art can also be utilized to identify a competitive binding antibody in accordance with this aspect of the disclosure.

(90) In one embodiment, the antibody or binding fragment thereof has a heavy chain variable region with a H-CDR1 having the amino acid sequence of SEQ ID NO: 9, or a modified amino acid sequence thereof containing 1, 2, 3, or more amino acid residue modifications as compared to SEQ ID NO: 9; a H-CDR2 having the amino acid sequence of SEQ ID NO: 47, or a modified amino acid sequence thereof containing 1, 2, 3, or more amino acid residue modifications as compared to SEQ ID NO: 47; and a H-CDR3 comprising the amino acid sequence of SEQ ID NO: 85, or a modified amino acid sequence thereof, said modified amino acid sequence containing 1, 2, 3, or more amino acid modifications as compared to SEQ ID NO: 85.

(91) An exemplary single domain antibody having this heavy chain variable region is referred to herein as the 26-T-sR2-1B2 antibody. The 26-T-sR2-1B2 antibody comprises a V.sub.H chain amino acid sequence of SEQ ID NO: 123 as shown below.

(92) TABLE-US-00010 SEQ ID NO: 123

QVQLQESGGGLVQAGGSLRLSCVASGSIFRFNAIGWYRQAPGKERELVA
RIRRLGSTSYADSVKGRFSISRDSAKNTVYQLQMNSLKPEDTAVYYCNAD
THFSTRNYWGQGTQVTVSS

(93) In another embodiment, the antibody or binding fragment thereof comprises a heavy chain variable region having an amino acid sequence that shares at least 60%, at least 65%, at least 70%, at least 75%, at least 80%, at least 85%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94% at least 95%, at least 96%, at least 97%, at least 98%, at least 99% sequence identity to SEQ ID NO: 123.

(94) In one embodiment, the antibody or binding fragment thereof of the present disclosure comprises a humanized variant of the heavy chain variable region of SEQ ID NO: 123, where the framework regions are humanized or replaced with human immunoglobulin framework sequences. As noted supra, suitable human or humanized framework sequences can be chosen based on their known structure, a consensus sequence, sequence homology to the framework sequences of donor antibody (e.g., the framework sequences of SEQ ID NO: 123), or a combination of these approaches. The humanized framework regions are designed to be similar in length and sequence to the parental framework sequences of SEQ ID NO: 123. In one embodiment, the humanized framework regions share 50%, 55%, 60%, 65%, 70%, 75%, 80%, 85%, 90%, 95% or more

sequence identity to the framework regions of SEQ ID NO: 123. In another embodiment, the humanized framework regions are 60%, 65%, 70%, 75%, 80%, 85%, 90%, 95% or more similar in length to the framework regions of SEQ ID NO: 123. Humanized variants of the heavy chain variable region of SEQ ID NO: 123 share at least 50%, at least 55%, at least 60%, at least 65%, at least 70%, at least 75%, at least 80%, at least 85%, at least 90%, at least 95% sequence identity along the entire length of SEQ ID 123.

(95) Another aspect of the present disclosure relates to an antibody or binding portion thereof (e.g., a human antibody) that competes for binding to human tau protein with the anti-tau antibody comprising a heavy chain variable region with the amino acid sequence of SEQ ID NO: 123. In accordance with this aspect of the disclosure, a competitive binding assay, such as Bio-Layer Interferometry (BLI) can be utilized to identify an antibody or binding portion thereof that competes for binding to a tau protein with the enumerated monoclonal antibody. Other competitive binding assays known in the art can also be utilized to identify a competitive binding antibody in accordance with this aspect of the disclosure.

(96) In one embodiment, the antibody or binding fragment thereof has a heavy chain variable region with a H-CDR1 having the amino acid sequence of SEQ ID NO: 10, or a modified amino acid sequence thereof containing 1, 2, 3, or more amino acid residue modifications as compared to SEQ ID NO: 10; a H-CDR2 having the amino acid sequence of SEQ ID NO: 48, or a modified amino acid sequence thereof containing 1, 2, 3, or more amino acid residue modifications as compared to SEQ ID NO: 48; and a H-CDR3 comprising the amino acid sequence of SEQ ID NO: 86, or a modified amino acid sequence thereof, said modified amino acid sequence containing 1, 2, 3, or more amino acid modifications as compared to SEQ ID NO: 86.

(97) An exemplary single domain antibody having this heavy chain variable region is referred to herein as the 27-T-sR2-1E9 antibody. The 27-T-sR2-1E9 antibody comprises a V.sub.H chain amino acid sequence of SEQ ID NO: 124 as shown below.

(98) TABLE-US-00011 SEQ ID NO: 124

QVQLQESGGGLVQAGGSLRLSCAASGSIFRINGMGWHRQAPGKERELVA
TITRGGSTNYADSVKGRFTISRDNKNTVYLQMNSLPEDTAVYYCAAD
PRWRLPFPGYGMDYWGKGQTQVTSS

(99) In another embodiment, the antibody or binding fragment thereof comprises a heavy chain variable region having an amino acid sequence that shares at least 60%, at least 65%, at least 70%, at least 75%, at least 80%, at least 85%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94% at least 95%, at least 96%, at least 97%, at least 98%, at least 99% sequence identity to SEQ ID NO: 124.

(100) In one embodiment, the antibody or binding fragment thereof of the present disclosure comprises a humanized variant of the heavy chain variable region of SEQ ID NO: 124, where the framework regions are humanized or replaced with human immunoglobulin framework sequences. As noted supra, suitable human or humanized framework sequences can be chosen based on their known structure, a consensus sequence, sequence homology to the framework sequences of donor antibody (e.g., the framework sequences of SEQ ID NO: 124), or a combination of these approaches. The humanized framework regions are designed to be similar in length and sequence to the parental framework sequences of SEQ ID NO: 124. In one embodiment, the humanized framework regions share 50%, 55%, 60%, 65%, 70%, 75%, 80%, 85%, 90%, 95% or more sequence identity to the framework regions of SEQ ID NO: 124. In another embodiment, the humanized framework regions are 60%, 65%, 70%, 75%, 80%, 85%, 90%, 95% or more similar in length to the framework regions of SEQ ID NO: 124. Humanized variants of the heavy chain variable region of SEQ ID NO: 124 share at least 50%, at least 55%, at least 60%, at least 65%, at least 70%, at least 75%, at least 80%, at least 85%, at least 90%, at least 95% sequence identity along the entire length of SEQ ID 124.

(101) Another aspect of the present disclosure relates to an antibody or binding portion thereof

(e.g., a human antibody) that competes for binding to human tau protein with the anti-tau antibody comprising a heavy chain variable region with the amino acid sequence of SEQ ID NO: 124. In accordance with this aspect of the disclosure, a competitive binding assay, such as Bio-competes for binding to a tau protein with the enumerated monoclonal antibody. Other competitive binding assays known in the art can also be utilized to identify a competitive binding antibody in accordance with this aspect of the disclosure.

(102) In one embodiment, the antibody or binding fragment thereof has a heavy chain variable region with a H-CDR1 having the amino acid sequence of SEQ ID NO: 11, or a modified amino acid sequence thereof containing 1, 2, 3, or more amino acid residue modifications as compared to SEQ ID NO: 11; a H-CDR2 having the amino acid sequence of SEQ ID NO: 49, or a modified amino acid sequence thereof containing 1, 2, 3, or more amino acid residue modifications as compared to SEQ ID NO: 49; and a H-CDR3 comprising the amino acid sequence of SEQ ID NO: 87, or a modified amino acid sequence thereof, said modified amino acid sequence containing 1, 2, 3, or more amino acid modifications as compared to SEQ ID NO: 87.

(103) Exemplary single domain antibodies having this heavy chain variable region are the 28-T-sR2-2B11 and 82-T-sR2-1D3 antibodies. The 28-T-sR2-2B11 and 82-T-sR2-1D3 antibodies comprise a V.sub.H chain amino acid sequence of SEQ ID NO: 125 as shown below.

(104) TABLE-US-00012 SEQ ID NO: 125

QVQLQESGGLVQPGGSLRLSCAASGRILISSMGWYRQAQGEQRELVAT
ITRGTTNYADSVKGRFTISRDNAMVYLQMKNLKS EDTAVYYCARVY
GRVWSRPYDYWGQGTQVTVSS

(105) In another embodiment, the antibody or binding fragment thereof comprises a heavy chain variable region having an amino acid sequence that shares at least 60%, at least 65%, at least 70%, at least 75%, at least 80%, at least 85%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94% at least 95%, at least 96%, at least 97%, at least 98%, at least 99% sequence identity to SEQ ID NO: 125.

(106) In one embodiment, the antibody or binding fragment thereof of the present disclosure comprises a humanized variant of the heavy chain variable region of SEQ ID NO: 125, where the framework regions are humanized or replaced with human immunoglobulin framework sequences. As noted supra, suitable human or humanized framework sequences can be chosen based on their known structure, a consensus sequence, sequence homology to the framework sequences of donor antibody (e.g., the framework sequences of SEQ ID NO: 125), or a combination of these approaches. The humanized framework regions are designed to be similar in length and sequence to the parental framework sequences of SEQ ID NO: 125. In one embodiment, the humanized framework regions share 50%, 55%, 60%, 65%, 70%, 75%, 80%, 85%, 90%, 95% or more sequence identity to the framework regions of SEQ ID NO: 125. In another embodiment, the humanized framework regions are 60%, 65%, 70%, 75%, 80%, 85%, 90%, 95% or more similar in length to the framework regions of SEQ ID NO: 125. Humanized variants of the heavy chain variable region of SEQ ID NO: 125 share at least 50%, at least 55%, at least 60%, at least 65%, at least 70%, at least 75%, at least 80%, at least 85%, at least 90%, at least 95% sequence identity along the entire length of SEQ ID 125.

(107) Another aspect of the present disclosure relates to an antibody or binding portion thereof (e.g., a human antibody) that competes for binding to human tau protein with the anti-tau antibody comprising a heavy chain variable region with the amino acid sequence of SEQ ID NO: 125. In accordance with this aspect of the disclosure, a competitive binding assay, such as Bio-Layer Interferometry (BLI) can be utilized to identify an antibody or binding portion thereof that competes for binding to a tau protein with the enumerated monoclonal antibody. Other competitive binding assays known in the art can also be utilized to identify a competitive binding antibody in accordance with this aspect of the disclosure.

(108) In one embodiment, the antibody or binding fragment thereof has a heavy chain variable

region with a H-CDR1 having the amino acid sequence of SEQ ID NO: 12, or a modified amino acid sequence thereof containing 1, 2, 3, or more amino acid residue modifications as compared to SEQ ID NO: 12; a H-CDR2 having the amino acid sequence of SEQ ID NO: 50, or a modified amino acid sequence thereof containing 1, 2, 3, or more amino acid residue modifications as compared to SEQ ID NO: 50; and a H-CDR3 comprising the amino acid sequence of SEQ ID NO: 88, or a modified amino acid sequence thereof, said modified amino acid sequence containing 1, 2, 3, or more amino acid modifications as compared to SEQ ID NO: 88.

(109) An exemplary single domain antibody having this heavy chain variable region is referred to herein as the 29-T-sR2-2G8 antibody. The 29-T-sR2-2G8 antibody comprises a V.sub.H chain amino acid sequence of SEQ ID NO: 126 as shown below.

(110) TABLE-US-00013 SEQ ID NO: 126

QVQLQESGGGFVQAGGSLRLSCTVSGRTRFRINGIDWYRQAPGKQRELVA
GISSTGSTNYADSVKGRFTISRDNAGNAVYLMNNLKPEDTGRYYCAAS
RGLSGSWYLRSSYPYWGQGTQVTVSS

(111) In another embodiment, the antibody or binding fragment thereof comprises a heavy chain variable region having an amino acid sequence that shares at least 60%, at least 65%, at least 70%, at least 75%, at least 80%, at least 85%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, at least 99% sequence identity to SEQ ID NO: 126.

(112) In one embodiment, the antibody or binding fragment thereof of the present disclosure comprises a humanized variant of the heavy chain variable region of SEQ ID NO: 126, where the framework regions are humanized or replaced with human immunoglobulin framework sequences. As noted supra, suitable human or humanized framework sequences can be chosen based on their known structure, a consensus sequence, sequence homology to the framework sequences of donor antibody (e.g., the framework sequences of SEQ ID NO: 126), or a combination of these approaches. The humanized framework regions are designed to be similar in length and sequence to the parental framework sequences of SEQ ID NO: 126. In one embodiment, the humanized framework regions share 50%, 55%, 60%, 65%, 70%, 75%, 80%, 85%, 90%, 95% or more sequence identity to the framework regions of SEQ ID NO: 126. In another embodiment, the humanized framework regions are 60%, 65%, 70%, 75%, 80%, 85%, 90%, 95% or more similar in length to the framework regions of SEQ ID NO: 126. Humanized variants of the heavy chain variable region of SEQ ID NO: 126 share at least 50%, at least 55%, at least 60%, at least 65%, at least 70%, at least 75%, at least 80%, at least 85%, at least 90%, at least 95% sequence identity along the entire length of SEQ ID 126.

(113) Another aspect of the present disclosure relates to an antibody or binding portion thereof (e.g., a human antibody) that competes for binding to human tau protein with the anti-tau antibody comprising a heavy chain variable region with the amino acid sequence of SEQ ID NO: 126. In accordance with this aspect of the disclosure, a competitive binding assay, such as Bio-Layer Interferometry (BLI) can be utilized to identify an antibody or binding portion thereof that competes for binding to a tau protein with the enumerated monoclonal antibody. Other competitive binding assays known in the art can also be utilized to identify a competitive binding antibody in accordance with this aspect of the disclosure.

(114) In one embodiment, the antibody or binding fragment thereof has a heavy chain variable region with a H-CDR1 having the amino acid sequence of SEQ ID NO: 13, or a modified amino acid sequence thereof containing 1, 2, 3, or more amino acid residue modifications as compared to SEQ ID NO: 13; a H-CDR2 having the amino acid sequence of SEQ ID NO: 51, or a modified amino acid sequence thereof containing 1, 2, 3, or more amino acid residue modifications as compared to SEQ ID NO: 51; and a H-CDR3 comprising the amino acid sequence of SEQ ID NO: 89, or a modified amino acid sequence thereof, said modified amino acid sequence containing 1, 2, 3, or more amino acid modifications as compared to SEQ ID NO: 89.

(115) An exemplary single domain antibody having this heavy chain variable region is referred to herein as the 34-T-sR2-1B7 antibody. The 34-T-sR2-1B7 antibody comprises a V.sub.H chain amino acid sequence of SEQ ID NO: 127 as shown below.

(116) TABLE-US-00014 SEQ ID NO: 127

QVQLQESGGGLVQPGGSLRLSCAASRYIFGTMGWYRQAPGLQRELVASI
SRGGSTNYADSVKGRFAISRDNAAKKTIVYLQMNSLKPEDTAVYYCNAV
PYRWGSSWYAGRYWGQGTLVTVSS

(117) In another embodiment, the antibody or binding fragment thereof comprises a heavy chain variable region having an amino acid sequence that shares at least 60%, at least 65%, at least 70%, at least 75%, at least 80%, at least 85%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94% at least 95%, at least 96%, at least 97%, at least 98%, at least 99% sequence identity to SEQ ID NO: 127.

(118) In one embodiment, the antibody or binding fragment thereof of the present disclosure comprises a humanized variant of the heavy chain variable region of SEQ ID NO: 127, where the framework regions are humanized or replaced with human immunoglobulin framework sequences. As noted supra, suitable human or humanized framework sequences can be chosen based on their known structure, a consensus sequence, sequence homology to the framework sequences of donor antibody (e.g., the framework sequences of SEQ ID NO: 127), or a combination of these approaches. The humanized framework regions are designed to be similar in length and sequence to the parental framework sequences of SEQ ID NO: 127. In one embodiment, the humanized framework regions share 50%, 55%, 60%, 65%, 70%, 75%, 80%, 85%, 90%, 95% or more sequence identity to the framework regions of SEQ ID NO: 127. In another embodiment, the humanized framework regions are 60%, 65%, 70%, 75%, 80%, 85%, 90%, 95% or more similar in length to the framework regions of SEQ ID NO: 127. Humanized variants of the heavy chain variable region of SEQ ID NO: 127 share at least 50%, at least 55%, at least 60%, at least 65%, at least 70%, at least 75%, at least 80%, at least 85%, at least 90%, at least 95% sequence identity along the entire length of SEQ ID 127.

(119) Another aspect of the present disclosure relates to an antibody or binding portion thereof (e.g., a human antibody) that competes for binding to human tau protein with the anti-tau antibody comprising a heavy chain variable region with the amino acid sequence of SEQ ID NO: 127. In accordance with this aspect of the disclosure, a competitive binding assay, such as Bio-competes for binding to a tau protein with the enumerated monoclonal antibody. Other competitive binding assays known in the art can also be utilized to identify a competitive binding antibody in accordance with this aspect of the disclosure.

(120) In one embodiment, the antibody or binding fragment thereof has a heavy chain variable region with a H-CDR1 having the amino acid sequence of SEQ ID NO: 14, or a modified amino acid sequence thereof containing 1, 2, 3, or more amino acid residue modifications as compared to SEQ ID NO: 14; a H-CDR2 having the amino acid sequence of SEQ ID NO: 52, or a modified amino acid sequence thereof containing 1, 2, 3, or more amino acid residue modifications as compared to SEQ ID NO: 52; and a H-CDR3 comprising the amino acid sequence of SEQ ID NO: 90 or a modified amino acid sequence thereof, said modified amino acid sequence containing 1, 2, 3, or more amino acid modifications as compared to SEQ ID NO: 90.

(121) An exemplary single domain antibody having this heavy chain variable region is referred to herein as the 35-T-sR2-1F5 antibody. The 35-T-sR2-1F5 antibody comprises a V.sub.H chain amino acid sequence of SEQ ID NO: 128 as shown below.

(122) TABLE-US-00015 SEQ ID NO: 128

QVQLQQSGGGLVQAGGSLTLSCVASGSRFSINTMGWYRQAPGKQRELVA
GITRGGSTNYADSVKGRFTISRENAKNTIVYLQMNSLKPEDTAVYYCAAT
LRAWALTFATSYAYWGQGTTQVTVSS

(123) In another embodiment, the antibody or binding fragment thereof comprises a heavy chain

variable region having an amino acid sequence that shares at least 60%, at least 65%, at least 70%, at least 75%, at least 80%, at least 85%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94% at least 95%, at least 96%, at least 97%, at least 98%, at least 99% sequence identity to SEQ ID NO: 128.

(124) In one embodiment, the antibody or binding fragment thereof of the present disclosure comprises a humanized variant of the heavy chain variable region of SEQ ID NO: 128, where the framework regions are humanized or replaced with human immunoglobulin framework sequences. As noted supra, suitable human or humanized framework sequences can be chosen based on their known structure, a consensus sequence, sequence homology to the framework sequences of donor antibody (e.g., the framework sequences of SEQ ID NO: 128), or a combination of these approaches. The humanized framework regions are designed to be similar in length and sequence to the parental framework sequences of SEQ ID NO: 128. In one embodiment, the humanized framework regions share 50%, 55%, 60%, 65%, 70%, 75%, 80%, 85%, 90%, 95% or more sequence identity to the framework regions of SEQ ID NO: 128. In another embodiment, the humanized framework regions are 60%, 65%, 70%, 75%, 80%, 85%, 90%, 95% or more similar in length to the framework regions of SEQ ID NO: 128. Humanized variants of the heavy chain variable region of SEQ ID NO: 128 share at least 50%, at least 55%, at least 60%, at least 65%, at least 70%, at least 75%, at least 80%, at least 85%, at least 90%, at least 95% sequence identity along the entire length of SEQ ID 128.

(125) Another aspect of the present disclosure relates to an antibody or binding portion thereof (e.g., a human antibody) that competes for binding to human tau protein with the anti-tau antibody comprising a heavy chain variable region with the amino acid sequence of SEQ ID NO: 128. In accordance with this aspect of the disclosure, a competitive binding assay, such as Bio-Layer Interferometry (BLI) can be utilized to identify an antibody or binding portion thereof that competes for binding to a tau protein with the enumerated monoclonal antibody. Other competitive binding assays known in the art can also be utilized to identify a competitive binding antibody in accordance with this aspect of the disclosure.

(126) In one embodiment, the antibody or binding fragment thereof has a heavy chain variable region with a H-CDR1 having the amino acid sequence of SEQ ID NO: 15, or a modified amino acid sequence thereof containing 1, 2, 3, or more amino acid residue modifications as compared to SEQ ID NO: 15; a H-CDR2 having the amino acid sequence of SEQ ID NO: 53, or a modified amino acid sequence thereof containing 1, 2, 3, or more amino acid residue modifications as compared to SEQ ID NO: 53; and a H-CDR3 comprising the amino acid sequence of SEQ ID NO: 91, or a modified amino acid sequence thereof, said modified amino acid sequence containing 1, 2, 3, or more amino acid modifications as compared to SEQ ID NO: 91.

(127) An exemplary single domain antibody having this heavy chain variable region is referred to herein as the 36-T-sR2-2C1 antibody. The 36-T-sR2-2C1 antibody comprises a V.sub.H chain amino acid sequence of SEQ ID NO: 129 as shown below.

(128) TABLE-US-00016 SEQ ID NO: 129

QVQLQASGGGLVQPGGSLTLSCAASGRTFSTRYAMGWFRQAPGKEREFVA
RISWSGGWTTYADSVKGRFAISRDNANKNTVYLMNSLKPEDTAVYYCAA
GSRAVLFGTYDYWGQGQTQVTVSS

(129) In another embodiment, the antibody or binding fragment thereof comprises a heavy chain variable region having an amino acid sequence that shares at least 60%, at least 65%, at least 70%, at least 75%, at least 80%, at least 85%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94% at least 95%, at least 96%, at least 97%, at least 98%, at least 99% sequence identity to SEQ ID NO: 129.

(130) In one embodiment, the antibody or binding fragment thereof of the present disclosure comprises a humanized variant of the heavy chain variable region of SEQ ID NO: 129, where the framework regions are humanized or replaced with human immunoglobulin framework sequences.

As noted supra, suitable human or humanized framework sequences can be chosen based on their known structure, a consensus sequence, sequence homology to the framework sequences of donor antibody (e.g., the framework sequences of SEQ ID NO: 129), or a combination of these approaches. The humanized framework regions are designed to be similar in length and sequence to the parental framework sequences of SEQ ID NO: 129. In one embodiment, the humanized framework regions share 50%, 55%, 60%, 65%, 70%, 75%, 80%, 85%, 90%, 95% or more sequence identity to the framework regions of SEQ ID NO: 129. In another embodiment, the humanized framework regions are 60%, 65%, 70%, 75%, 80%, 85%, 90%, 95% or more similar in length to the framework regions of SEQ ID NO: 129. Humanized variants of the heavy chain variable region of SEQ ID NO: 129 share at least 50%, at least 55%, at least 60%, at least 65%, at least 70%, at least 75%, at least 80%, at least 85%, at least 90%, at least 95% sequence identity along the entire length of SEQ ID 129.

(131) Another aspect of the present disclosure relates to an antibody or binding portion thereof (e.g., a human antibody) that competes for binding to human tau protein with the anti-tau antibody comprising a heavy chain variable region with the amino acid sequence of SEQ ID NO: 129. In accordance with this aspect of the disclosure, a competitive binding assay, such as Bio-Layer Interferometry (BLI) can be utilized to identify an antibody or binding portion thereof that competes for binding to a tau protein with the enumerated monoclonal antibody. Other competitive binding assays known in the art can also be utilized to identify a competitive binding antibody in accordance with this aspect of the disclosure.

(132) In one embodiment, the antibody or binding fragment thereof has a heavy chain variable region with a H-CDR1 having the amino acid sequence of SEQ ID NO: 16, or a modified amino acid sequence thereof containing 1, 2, 3, or more amino acid residue modifications as compared to SEQ ID NO: 16; a H-CDR2 having the amino acid sequence of SEQ ID NO: 54, or a modified amino acid sequence thereof containing 1, 2, 3, or more amino acid residue modifications as compared to SEQ ID NO: 54; and a H-CDR3 comprising the amino acid sequence of SEQ ID NO: 92, or a modified amino acid sequence thereof, said modified amino acid sequence containing 1, 2, 3, or more amino acid modifications as compared to SEQ ID NO: 92.

(133) An exemplary single domain antibody having this heavy chain variable region is referred to herein as the 37-T-sR2-2G9 antibody. The 37-T-sR2-2G9 antibody comprises a V.sub.H chain amino acid sequence of SEQ ID NO: 130 as shown below.

(134) TABLE-US-00017 SEQ ID NO: 130

QVQLQESGGGLVQAGGSLRLSCAASGRTLSSYRMGWFRQVPGKERELVA
AINWRGSWTYYADSVKGRVTISRDNAKNTVYLMNSLQPEDTALYYCAR
PTARWDLFREKYDFRGQGTQVTVSS

(135) In another embodiment, the antibody or binding fragment thereof comprises a heavy chain variable region having an amino acid sequence that shares at least 60%, at least 65%, at least 70%, at least 75%, at least 80%, at least 85%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94% at least 95%, at least 96%, at least 97%, at least 98%, at least 99% sequence identity to SEQ ID NO: 130.

(136) In one embodiment, the antibody or binding fragment thereof of the present disclosure comprises a humanized variant of the heavy chain variable region of SEQ ID NO: 130, where the framework regions are humanized or replaced with human immunoglobulin framework sequences. As noted supra, suitable human or humanized framework sequences can be chosen based on their known structure, a consensus sequence, sequence homology to the framework sequences of donor antibody (e.g., the framework sequences of SEQ ID NO: 130), or a combination of these approaches. The humanized framework regions are designed to be similar in length and sequence to the parental framework sequences of SEQ ID NO: 130. In one embodiment, the humanized framework regions share 50%, 55%, 60%, 65%, 70%, 75%, 80%, 85%, 90%, 95% or more sequence identity to the framework regions of SEQ ID NO: 130. In another embodiment, the

humanized framework regions are 60%, 65%, 70%, 75%, 80%, 85%, 90%, 95% or more similar in length to the framework regions of SEQ ID NO: 130. Humanized variants of the heavy chain variable region of SEQ ID NO: 130 share at least 50%, at least 55%, at least 60%, at least 65%, at least 70%, at least 75%, at least 80%, at least 85%, at least 90%, at least 95% sequence identity along the entire length of SEQ ID 130.

(137) Another aspect of the present disclosure relates to an antibody or binding portion thereof (e.g., a human antibody) that competes for binding to human tau protein with the anti-tau antibody comprising a heavy chain variable region with the amino acid sequence of SEQ ID NO: 130. In accordance with this aspect of the disclosure, a competitive binding assay, such as Bio-competes for binding to a tau protein with the enumerated monoclonal antibody. Other competitive binding assays known in the art can also be utilized to identify a competitive binding antibody in accordance with this aspect of the disclosure.

(138) In one embodiment, the antibody or binding fragment thereof has a heavy chain variable region with a H-CDR1 having the amino acid sequence of SEQ ID NO: 17, or a modified amino acid sequence thereof containing 1, 2, 3, or more amino acid residue modifications as compared to SEQ ID NO: 17; a H-CDR2 having the amino acid sequence of SEQ ID NO: 55, or a modified amino acid sequence thereof containing 1, 2, 3, or more amino acid residue modifications as compared to SEQ ID NO: 55; and a H-CDR3 comprising the amino acid sequence of SEQ ID NO: 93, or a modified amino acid sequence thereof, said modified amino acid sequence containing 1, 2, 3, or more amino acid modifications as compared to SEQ ID NO: 93.

(139) An exemplary single domain antibody having this heavy chain variable region is referred to herein as the 42-T-sR2-1B8 antibody. The 42-T-sR2-1B8 antibody comprises a V.sub.H chain amino acid sequence of SEQ ID NO: 131 as shown below.

(140) TABLE-US-00018 SEQ ID NO: 131

QVQLQESGGGLVQAGGSLRLSCAASGRTFSSYAMGWFRQAPGKEREFVA
AISRSGGITSYADSVKGRFAISRDNKNTVYLMNSLKPEDTGVYYCNV
QRFITTYWGQGTLLTVSS

(141) In another embodiment, the antibody or binding fragment thereof comprises a heavy chain variable region having an amino acid sequence that shares at least 60%, at least 65%, at least 70%, at least 75%, at least 80%, at least 85%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94% at least 95%, at least 96%, at least 97%, at least 98%, at least 99% sequence identity to SEQ ID NO: 131.

(142) In one embodiment, the antibody or binding fragment thereof of the present disclosure comprises a humanized variant of the heavy chain variable region of SEQ ID NO: 131, where the framework regions are humanized or replaced with human immunoglobulin framework sequences. As noted supra, suitable human or humanized framework sequences can be chosen based on their known structure, a consensus sequence, sequence homology to the framework sequences of donor antibody (e.g., the framework sequences of SEQ ID NO: 131), or a combination of these approaches. The humanized framework regions are designed to be similar in length and sequence to the parental framework sequences of SEQ ID NO: 131. In one embodiment, the humanized framework regions share 50%, 55%, 60%, 65%, 70%, 75%, 80%, 85%, 90%, 95% or more sequence identity to the framework regions of SEQ ID NO: 131. In another embodiment, the humanized framework regions are 60%, 65%, 70%, 75%, 80%, 85%, 90%, 95% or more similar in length to the framework regions of SEQ ID NO: 131. Humanized variants of the heavy chain variable region of SEQ ID NO: 131 share at least 50%, at least 55%, at least 60%, at least 65%, at least 70%, at least 75%, at least 80%, at least 85%, at least 90%, at least 95% sequence identity along the entire length of SEQ ID 131.

(143) Another aspect of the present disclosure relates to an antibody or binding portion thereof (e.g., a human antibody) that competes for binding to human tau protein with the anti-tau antibody comprising a heavy chain variable region with the amino acid sequence of SEQ ID NO: 131. In

accordance with this aspect of the disclosure, a competitive binding assay, such as Bio-Layer Interferometry (BLI) can be utilized to identify an antibody or binding portion thereof that competes for binding to a tau protein with the enumerated monoclonal antibody. Other competitive binding assays known in the art can also be utilized to identify a competitive binding antibody in accordance with this aspect of the disclosure.

(144) In one embodiment, the antibody or binding fragment thereof has a heavy chain variable region with a H-CDR1 having the amino acid sequence of SEQ ID NO: 18, or a modified amino acid sequence thereof containing 1, 2, 3, or more amino acid residue modifications as compared to SEQ ID NO: 18; a H-CDR2 having the amino acid sequence of SEQ ID NO: 56, or a modified amino acid sequence thereof containing 1, 2, 3, or more amino acid residue modifications as compared to SEQ ID NO: 56; and a H-CDR3 comprising the amino acid sequence of SEQ ID NO: 94, or a modified amino acid sequence thereof, said modified amino acid sequence containing 1, 2, 3, or more amino acid modifications as compared to SEQ ID NO: 94.

(145) An exemplary single domain antibody having this heavy chain variable region is referred to herein as the 43-T-sR2-1F8 antibody. The 43-T-sR2-1F8 antibody comprises a V.sub.H chain amino acid sequence of SEQ ID NO: 132 as shown below.

(146) TABLE-US-00019 SEQ ID NO: 132

QVQLQASGGGLVQAGGSLSLSCAASGRFTGLYTMGWFRQAPEKEREFFVA
AISWRGLSIMYADSVKGRFTISRDNVKNNTVYLQMNSLPEDTAVYYCAA
NAGLSLLRNWRTNEYAYWGQGTQVTVSS

(147) In another embodiment, the antibody or binding fragment thereof comprises a heavy chain variable region having an amino acid sequence that shares at least 60%, at least 65%, at least 70%, at least 75%, at least 80%, at least 85%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94% at least 95%, at least 96%, at least 97%, at least 98%, at least 99% sequence identity to SEQ ID NO: 132.

(148) In one embodiment, the antibody or binding fragment thereof of the present disclosure comprises a humanized variant of the heavy chain variable region of SEQ ID NO: 132, where the framework regions are humanized or replaced with human immunoglobulin framework sequences. As noted supra, suitable human or humanized framework sequences can be chosen based on their known structure, a consensus sequence, sequence homology to the framework sequences of donor antibody (e.g., the framework sequences of SEQ ID NO: 132), or a combination of these approaches. The humanized framework regions are designed to be similar in length and sequence to the parental framework sequences of SEQ ID NO: 132. In one embodiment, the humanized framework regions share 50%, 55%, 60%, 65%, 70%, 75%, 80%, 85%, 90%, 95% or more sequence identity to the framework regions of SEQ ID NO: 132. In another embodiment, the humanized framework regions are 60%, 65%, 70%, 75%, 80%, 85%, 90%, 95% or more similar in length to the framework regions of SEQ ID NO: 132. Humanized variants of the heavy chain variable region of SEQ ID NO: 132 share at least 50%, at least 55%, at least 60%, at least 65%, at least 70%, at least 75%, at least 80%, at least 85%, at least 90%, at least 95% sequence identity along the entire length of SEQ ID 132.

(149) Another aspect of the present disclosure relates to an antibody or binding portion thereof (e.g., a human antibody) that competes for binding to human tau protein with the anti-tau antibody comprising a heavy chain variable region with the amino acid sequence of SEQ ID NO: 132. In accordance with this aspect of the disclosure, a competitive binding assay, such as Bio-Layer Interferometry (BLI) can be utilized to identify an antibody or binding portion thereof that competes for binding to a tau protein with the enumerated monoclonal antibody. Other competitive binding assays known in the art can also be utilized to identify a competitive binding antibody in accordance with this aspect of the disclosure.

(150) In one embodiment, the antibody or binding fragment thereof has a heavy chain variable region with a H-CDR1 having the amino acid sequence of SEQ ID NO: 19, or a modified amino

acid sequence thereof containing 1, 2, 3, or more amino acid residue modifications as compared to SEQ ID NO: 19; a H-CDR2 having the amino acid sequence of SEQ ID NO: 57, or a modified amino acid sequence thereof containing 1, 2, 3, or more amino acid residue modifications as compared to SEQ ID NO: 57; and a H-CDR3 comprising the amino acid sequence of SEQ ID NO: 95, or a modified amino acid sequence thereof, said modified amino acid sequence containing 1, 2, 3, or more amino acid modifications as compared to SEQ ID NO: 95.

(151) An exemplary single domain antibody having this heavy chain variable region is referred to herein as the 44-T-sR2-2C5 antibody. The 44-T-sR2-2C5 antibody comprises a V.sub.H chain amino acid sequence of SEQ ID NO: 133 as shown below.

(152) TABLE-US-00020 SEQ ID NO: 133

QVQLQESGGGLVQAGGSLRLSCEASARTFSSYAVGWFRQAPGKEREFVA
AINWSGRRTNYADSVKGRFSISRDNAKNTMYLQMNSLKPEDTAVYYCAA
RSSLLEFWLGSRRGYDYWGQGTQVTVSS

(153) In another embodiment, the antibody or binding fragment thereof comprises a heavy chain variable region having an amino acid sequence that shares at least 60%, at least 65%, at least 70%, at least 75%, at least 80%, at least 85%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94% at least 95%, at least 96%, at least 97%, at least 98%, at least 99% sequence identity to SEQ ID NO: 133.

(154) In one embodiment, the antibody or binding fragment thereof of the present disclosure comprises a humanized variant of the heavy chain variable region of SEQ ID NO: 133, where the framework regions are humanized or replaced with human immunoglobulin framework sequences. As noted supra, suitable human or humanized framework sequences can be chosen based on their known structure, a consensus sequence, sequence homology to the framework sequences of donor antibody (e.g., the framework sequences of SEQ ID NO: 133), or a combination of these approaches. The humanized framework regions are designed to be similar in length and sequence to the parental framework sequences of SEQ ID NO: 133. In one embodiment, the humanized framework regions share 50%, 55%, 60%, 65%, 70%, 75%, 80%, 85%, 90%, 95% or more sequence identity to the framework regions of SEQ ID NO: 133. In another embodiment, the humanized framework regions are 60%, 65%, 70%, 75%, 80%, 85%, 90%, 95% or more similar in length to the framework regions of SEQ ID NO: 133. Humanized variants of the heavy chain variable region of SEQ ID NO: 133 share at least 50%, at least 55%, at least 60%, at least 65%, at least 70%, at least 75%, at least 80%, at least 85%, at least 90%, at least 95% sequence identity along the entire length of SEQ ID 133.

(155) Another aspect of the present disclosure relates to an antibody or binding portion thereof (e.g., a human antibody) that competes for binding to human tau protein with the anti-tau antibody comprising a heavy chain variable region with the amino acid sequence of SEQ ID NO: 133. In accordance with this aspect of the disclosure, a competitive binding assay, such as Bio-competes for binding to a tau protein with the enumerated monoclonal antibody. Other competitive binding assays known in the art can also be utilized to identify a competitive binding antibody in accordance with this aspect of the disclosure.

(156) In one embodiment, the antibody or binding fragment thereof has a heavy chain variable region with a H-CDR1 having the amino acid sequence of SEQ ID NO: 20, or a modified amino acid sequence thereof containing 1, 2, 3, or more amino acid residue modifications as compared to SEQ ID NO: 20; a H-CDR2 having the amino acid sequence of SEQ ID NO: 58, or a modified amino acid sequence thereof containing 1, 2, 3, or more amino acid residue modifications as compared to SEQ ID NO: 58; and a H-CDR3 comprising the amino acid sequence of SEQ ID NO: 96, or a modified amino acid sequence thereof, said modified amino acid sequence containing 1, 2, 3, or more amino acid modifications as compared to SEQ ID NO: 96.

(157) An exemplary single domain antibody having this heavy chain variable region is referred to herein as the 45-T-sR2-2H4 antibody. The 45-T-sR2-2H4 antibody comprises a V.sub.H chain

amino acid sequence of SEQ ID NO: 134 as shown below.

(158) TABLE-US-00021 SEQ ID NO: 134

QVQLQQFGGGLVQPGGSLRLSCAASGRIFSIWTMGWYRQAPGKQRELVA
AITSGGSTNYADSVKGRFTISRDN AENTVYLQMNSLKPEDTAVYYCNAD
IRPRIISFFKDYWGQGTQVTVSS

(159) In another embodiment, the antibody or binding fragment thereof comprises a heavy chain variable region having an amino acid sequence that shares at least 60%, at least 65%, at least 70%, at least 75%, at least 80%, at least 85%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94% at least 95%, at least 96%, at least 97%, at least 98%, at least 99% sequence identity to SEQ ID NO: 134.

(160) In one embodiment, the antibody or binding fragment thereof of the present disclosure comprises a humanized variant of the heavy chain variable region of SEQ ID NO: 134, where the framework regions are humanized or replaced with human immunoglobulin framework sequences. As noted supra, suitable human or humanized framework sequences can be chosen based on their known structure, a consensus sequence, sequence homology to the framework sequences of donor antibody (e.g., the framework sequences of SEQ ID NO: 134), or a combination of these approaches. The humanized framework regions are designed to be similar in length and sequence to the parental framework sequences of SEQ ID NO: 134. In one embodiment, the humanized framework regions share 50%, 55%, 60%, 65%, 70%, 75%, 80%, 85%, 90%, 95% or more sequence identity to the framework regions of SEQ ID NO: 134. In another embodiment, the humanized framework regions are 60%, 65%, 70%, 75%, 80%, 85%, 90%, 95% or more similar in length to the framework regions of SEQ ID NO: 134. Humanized variants of the heavy chain variable region of SEQ ID NO: 134 share at least 50%, at least 55%, at least 60%, at least 65%, at least 70%, at least 75%, at least 80%, at least 85%, at least 90%, at least 95% sequence identity along the entire length of SEQ ID 134.

(161) Another aspect of the present disclosure relates to an antibody or binding portion thereof (e.g., a human antibody) that competes for binding to human tau protein with the anti-tau antibody comprising a heavy chain variable region with the amino acid sequence of SEQ ID NO: 134. In accordance with this aspect of the disclosure, a competitive binding assay, such as Bio-Layer Interferometry (BLI) can be utilized to identify an antibody or binding portion thereof that competes for binding to a tau protein with the enumerated monoclonal antibody. Other competitive binding assays known in the art can also be utilized to identify a competitive binding antibody in accordance with this aspect of the disclosure.

(162) In one embodiment, the antibody or binding fragment thereof has a heavy chain variable region with a H-CDR1 having the amino acid sequence of SEQ ID NO: 21, or a modified amino acid sequence thereof containing 1, 2, 3, or more amino acid residue modifications as compared to SEQ ID NO: 21; a H-CDR2 having the amino acid sequence of SEQ ID NO: 59, or a modified amino acid sequence thereof containing 1, 2, 3, or more amino acid residue modifications as compared to SEQ ID NO: 59; and a H-CDR3 comprising the amino acid sequence of SEQ ID NO: 97, or a modified amino acid sequence thereof, said modified amino acid sequence containing 1, 2, 3, or more amino acid modifications as compared to SEQ ID NO: 97.

(163) An exemplary single domain antibody having this heavy chain variable region is referred to herein as the 50-T-sR2-1C5 antibody. The 50-T-sR2-1C5 antibody comprises a V.sub.H chain amino acid sequence of SEQ ID NO: 135 as shown below.

(164) TABLE-US-00022 SEQ ID NO: 135

QVQLQESGGGLVQPGGSLRLSCAASGRTF SRYAMGWFRQAPGKEREFVA
AITWSGGIYYADSVKGRFTISRDN AKNTVYLQMNNLKTEDTAVYYCNA
YFRWGTRYWGQGTTLVTVSS

(165) In another embodiment, the antibody or binding fragment thereof comprises a heavy chain variable region having an amino acid sequence that shares at least 60%, at least 65%, at least 70%,

at least 75%, at least 80%, at least 85%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94% at least 95%, at least 96%, at least 97%, at least 98%, at least 99% sequence identity to SEQ ID NO: 135.

(166) In one embodiment, the antibody or binding fragment thereof of the present disclosure comprises a humanized variant of the heavy chain variable region of SEQ ID NO: 135, where the framework regions are humanized or replaced with human immunoglobulin framework sequences. As noted supra, suitable human or humanized framework sequences can be chosen based on their known structure, a consensus sequence, sequence homology to the framework sequences of donor antibody (e.g., the framework sequences of SEQ ID NO: 135), or a combination of these approaches. The humanized framework regions are designed to be similar in length and sequence to the parental framework sequences of SEQ ID NO: 135. In one embodiment, the humanized framework regions share 50%, 55%, 60%, 65%, 70%, 75%, 80%, 85%, 90%, 95% or more sequence identity to the framework regions of SEQ ID NO: 135. In another embodiment, the humanized framework regions are 60%, 65%, 70%, 75%, 80%, 85%, 90%, 95% or more similar in length to the framework regions of SEQ ID NO: 135. Humanized variants of the heavy chain variable region of SEQ ID NO: 135 share at least 50%, at least 55%, at least 60%, at least 65%, at least 70%, at least 75%, at least 80%, at least 85%, at least 90%, at least 95% sequence identity along the entire length of SEQ ID 135.

(167) Another aspect of the present disclosure relates to an antibody or binding portion thereof (e.g., a human antibody) that competes for binding to human tau protein with the anti-tau antibody comprising a heavy chain variable region with the amino acid sequence of SEQ ID NO: 135. In accordance with this aspect of the disclosure, a competitive binding assay, such as Bio-Layer Interferometry (BLI) can be utilized to identify an antibody or binding portion thereof that competes for binding to a tau protein with the enumerated monoclonal antibody. Other competitive binding assays known in the art can also be utilized to identify a competitive binding antibody in accordance with this aspect of the disclosure.

(168) In one embodiment, the antibody or binding fragment thereof has a heavy chain variable region with a H-CDR1 having the amino acid sequence of SEQ ID NO: 22, or a modified amino acid sequence thereof containing 1, 2, 3, or more amino acid residue modifications as compared to SEQ ID NO: 22; a H-CDR2 having the amino acid sequence of SEQ ID NO: 60, or a modified amino acid sequence thereof containing 1, 2, 3, or more amino acid residue modifications as compared to SEQ ID NO: 60; and a H-CDR3 comprising the amino acid sequence of SEQ ID NO: 98, or a modified amino acid sequence thereof, said modified amino acid sequence containing 1, 2, 3, or more amino acid modifications as compared to SEQ ID NO: 98.

(169) An exemplary single domain antibody having this heavy chain variable region is referred to herein as the 51-T-sR2-1G1 antibody. The 51-T-sR2-1G1 antibody comprises a V.sub.H chain amino acid sequence of SEQ ID NO: 136 as shown below.

(170) TABLE-US-00023 SEQ ID NO: 136

QVQLQESGGLVQAGGSLRLSCAASGRSFSWLTMAWFRQAPGKEREIVA
RITWRGTPYYADSVKGRFAISRDNKNTVYLQMNSLKPEDTAIYYCAA
KQILIRPDAYVYWGQGTQVTVSS

(171) In another embodiment, the antibody or binding fragment thereof comprises a heavy chain variable region having an amino acid sequence that shares at least 60%, at least 65%, at least 70%, at least 75%, at least 80%, at least 85%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94% at least 95%, at least 96%, at least 97%, at least 98%, at least 99% sequence identity to SEQ ID NO: 136.

(172) In one embodiment, the antibody or binding fragment thereof of the present disclosure comprises a humanized variant of the heavy chain variable region of SEQ ID NO: 136, where the framework regions are humanized or replaced with human immunoglobulin framework sequences. As noted supra, suitable human or humanized framework sequences can be chosen based on their

known structure, a consensus sequence, sequence homology to the framework sequences of donor antibody (e.g., the framework sequences of SEQ ID NO: 136), or a combination of these approaches. The humanized framework regions are designed to be similar in length and sequence to the parental framework sequences of SEQ ID NO: 136. In one embodiment, the humanized framework regions share 50%, 55%, 60%, 65%, 70%, 75%, 80%, 85%, 90%, 95% or more sequence identity to the framework regions of SEQ ID NO: 136. In another embodiment, the humanized framework regions are 60%, 65%, 70%, 75%, 80%, 85%, 90%, 95% or more similar in length to the framework regions of SEQ ID NO: 136. Humanized variants of the heavy chain variable region of SEQ ID NO: 136 share at least 50%, at least 55%, at least 60%, at least 65%, at least 70%, at least 75%, at least 80%, at least 85%, at least 90%, at least 95% sequence identity along the entire length of SEQ ID 136.

(173) Another aspect of the present disclosure relates to an antibody or binding portion thereof (e.g., a human antibody) that competes for binding to human tau protein with the anti-tau antibody comprising a heavy chain variable region with the amino acid sequence of SEQ ID NO: 136. In accordance with this aspect of the disclosure, a competitive binding assay, such as Bio-competes for binding to a tau protein with the enumerated monoclonal antibody. Other competitive binding assays known in the art can also be utilized to identify a competitive binding antibody in accordance with this aspect of the disclosure.

(174) In one embodiment, the antibody or binding fragment thereof has a heavy chain variable region with a H-CDR1 having the amino acid sequence of SEQ ID NO: 23, or a modified amino acid sequence thereof containing 1, 2, 3, or more amino acid residue modifications as compared to SEQ ID NO: 23; a H-CDR2 having the amino acid sequence of SEQ ID NO: 61, or a modified amino acid sequence thereof containing 1, 2, 3, or more amino acid residue modifications as compared to SEQ ID NO: 61; and a H-CDR3 comprising the amino acid sequence of SEQ ID NO: 99, or a modified amino acid sequence thereof, said modified amino acid sequence containing 1, 2, 3, or more amino acid modifications as compared to SEQ ID NO: 99.

(175) An exemplary single domain antibody having this heavy chain variable region is referred to herein as the 52-T-sR2-2C9 antibody. The 52-T-sR2-2C9 antibody comprises a V.sub.H chain amino acid sequence of SEQ ID NO: 137 as shown below.

(176) TABLE-US-00024 SEQ ID NO: 137

QVQLQASGGGLVQAGGSLRLSCVASGSIFRFNAIGWYRQAPGKERELVA
RIRRLGSTSYADSVKGRFSISRDSAKNTVYVLQMDNLKPEDTAVYYCKAG
RHPHFSMDYPDLGQGTQVTVSS

(177) In another embodiment, the antibody or binding fragment thereof comprises a heavy chain variable region having an amino acid sequence that shares at least 60%, at least 65%, at least 70%, at least 75%, at least 80%, at least 85%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94% at least 95%, at least 96%, at least 97%, at least 98%, at least 99% sequence identity to SEQ ID NO: 137.

(178) In one embodiment, the antibody or binding fragment thereof of the present disclosure comprises a humanized variant of the heavy chain variable region of SEQ ID NO: 137, where the framework regions are humanized or replaced with human immunoglobulin framework sequences. As noted supra, suitable human or humanized framework sequences can be chosen based on their known structure, a consensus sequence, sequence homology to the framework sequences of donor antibody (e.g., the framework sequences of SEQ ID NO: 137), or a combination of these approaches. The humanized framework regions are designed to be similar in length and sequence to the parental framework sequences of SEQ ID NO: 137. In one embodiment, the humanized framework regions share 50%, 55%, 60%, 65%, 70%, 75%, 80%, 85%, 90%, 95% or more sequence identity to the framework regions of SEQ ID NO: 137. In another embodiment, the humanized framework regions are 60%, 65%, 70%, 75%, 80%, 85%, 90%, 95% or more similar in length to the framework regions of SEQ ID NO: 137. Humanized variants of the heavy chain

variable region of SEQ ID NO: 137 share at least 50%, at least 55%, at least 60%, at least 65%, at least 70%, at least 75%, at least 80%, at least 85%, at least 90%, at least 95% sequence identity along the entire length of SEQ ID 137.

(179) Another aspect of the present disclosure relates to an antibody or binding portion thereof (e.g., a human antibody) that competes for binding to human tau protein with the anti-tau antibody comprising a heavy chain variable region with the amino acid sequence of SEQ ID NO: 137. In accordance with this aspect of the disclosure, a competitive binding assay, such as Bio-Layer Interferometry (BLI) can be utilized to identify an antibody or binding portion thereof that competes for binding to a tau protein with the enumerated monoclonal antibody. Other competitive binding assays known in the art can also be utilized to identify a competitive binding antibody in accordance with this aspect of the disclosure.

(180) In one embodiment, the antibody or binding fragment thereof has a heavy chain variable region with a H-CDR1 having the amino acid sequence of SEQ ID NO: 24, or a modified amino acid sequence thereof containing 1, 2, 3, or more amino acid residue modifications as compared to SEQ ID NO: 24; a H-CDR2 having the amino acid sequence of SEQ ID NO: 62, or a modified amino acid sequence thereof containing 1, 2, 3, or more amino acid residue modifications as compared to SEQ ID NO: 62; and a H-CDR3 comprising the amino acid sequence of SEQ ID NO: 100, or a modified amino acid sequence thereof, said modified amino acid sequence containing 1, 2, 3, or more amino acid modifications as compared to SEQ ID NO: 100.

(181) An exemplary single domain antibody having this heavy chain variable region is referred to herein as the 53-T-sR2-2H5 antibody. The 53-T-sR2-2H5 antibody comprises a V.sub.H chain amino acid sequence of SEQ ID NO: 138 as shown below.

(182) TABLE-US-00025 SEQ ID NO: 138

QVQLQESGGGLVQPGGSLRLSCAASRYIFGTMGWYRQAPGLQRELVASI
SRGGSTNYADSVKGRFAISRDNKNTVYLQMNNLKPEDTAVYYCALALG
FFTNYVRESSYRYWGQGTQVTVSS

(183) In another embodiment, the antibody or binding fragment thereof comprises a heavy chain variable region having an amino acid sequence that shares at least 60%, at least 65%, at least 70%, at least 75%, at least 80%, at least 85%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94% at least 95%, at least 96%, at least 97%, at least 98%, at least 99% sequence identity to SEQ ID NO: 138.

(184) In one embodiment, the antibody or binding fragment thereof of the present disclosure comprises a humanized variant of the heavy chain variable region of SEQ ID NO: 138, where the framework regions are humanized or replaced with human immunoglobulin framework sequences. As noted supra, suitable human or humanized framework sequences can be chosen based on their known structure, a consensus sequence, sequence homology to the framework sequences of donor antibody (e.g., the framework sequences of SEQ ID NO: 138), or a combination of these approaches. The humanized framework regions are designed to be similar in length and sequence to the parental framework sequences of SEQ ID NO: 138. In one embodiment, the humanized framework regions share 50%, 55%, 60%, 65%, 70%, 75%, 80%, 85%, 90%, 95% or more sequence identity to the framework regions of SEQ ID NO: 138. In another embodiment, the humanized framework regions are 60%, 65%, 70%, 75%, 80%, 85%, 90%, 95% or more similar in length to the framework regions of SEQ ID NO: 138. Humanized variants of the heavy chain variable region of SEQ ID NO: 138 share at least 50%, at least 55%, at least 60%, at least 65%, at least 70%, at least 75%, at least 80%, at least 85%, at least 90%, at least 95% sequence identity along the entire length of SEQ ID 138.

(185) Another aspect of the present disclosure relates to an antibody or binding portion thereof (e.g., a human antibody) that competes for binding to human tau protein with the anti-tau antibody comprising a heavy chain variable region with the amino acid sequence of SEQ ID NO: 138. In accordance with this aspect of the disclosure, a competitive binding assay, such as Bio-Layer

Interferometry (BLI) can be utilized to identify an antibody or binding portion thereof that competes for binding to a tau protein with the enumerated monoclonal antibody. Other competitive binding assays known in the art can also be utilized to identify a competitive binding antibody in accordance with this aspect of the disclosure.

(186) In one embodiment, the antibody or binding fragment thereof has a heavy chain variable region with a H-CDR1 having the amino acid sequence of SEQ ID NO: 25, or a modified amino acid sequence thereof containing 1, 2, 3, or more amino acid residue modifications as compared to SEQ ID NO: 25; a H-CDR2 having the amino acid sequence of SEQ ID NO: 63, or a modified amino acid sequence thereof containing 1, 2, 3, or more amino acid residue modifications as compared to SEQ ID NO: 63; and a H-CDR3 comprising the amino acid sequence of SEQ ID NO: 101, or a modified amino acid sequence thereof, said modified amino acid sequence containing 1, 2, 3, or more amino acid modifications as compared to SEQ ID NO: 101.

(187) An exemplary single domain antibody having this heavy chain variable region is referred to herein as the 58-T-sR2-1C6 antibody. The 58-T-sR2-1C6 antibody comprises a V.sub.H chain amino acid sequence of SEQ ID NO: 139 as shown below.

(188) TABLE-US-00026 SEQ ID NO: 139

QVQLQESGGGLVQTGGSLRLSCAASGRTRFLYSMAWFRQAPGKEREFLLG
SIRWNGGNIYYTDSVKGRFTISRDNKNTAYLQMNSLTPEDTAVYYCAA
GPRIAVWRYEYNYWGQGTQVTVSS

(189) In another embodiment, the antibody or binding fragment thereof comprises a heavy chain variable region having an amino acid sequence that shares at least 60%, at least 65%, at least 70%, at least 75%, at least 80%, at least 85%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94% at least 95%, at least 96%, at least 97%, at least 98%, at least 99% sequence identity to SEQ ID NO: 139.

(190) In one embodiment, the antibody or binding fragment thereof of the present disclosure comprises a humanized variant of the heavy chain variable region of SEQ ID NO: 139, where the framework regions are humanized or replaced with human immunoglobulin framework sequences. As noted supra, suitable human or humanized framework sequences can be chosen based on their known structure, a consensus sequence, sequence homology to the framework sequences of donor antibody (e.g., the framework sequences of SEQ ID NO: 139), or a combination of these approaches. The humanized framework regions are designed to be similar in length and sequence to the parental framework sequences of SEQ ID NO: 139. In one embodiment, the humanized framework regions share 50%, 55%, 60%, 65%, 70%, 75%, 80%, 85%, 90%, 95% or more sequence identity to the framework regions of SEQ ID NO: 139. In another embodiment, the humanized framework regions are 60%, 65%, 70%, 75%, 80%, 85%, 90%, 95% or more similar in length to the framework regions of SEQ ID NO: 139. Humanized variants of the heavy chain variable region of SEQ ID NO: 139 share at least 50%, at least 55%, at least 60%, at least 65%, at least 70%, at least 75%, at least 80%, at least 85%, at least 90%, at least 95% sequence identity along the entire length of SEQ ID 139.

(191) Another aspect of the present disclosure relates to an antibody or binding portion thereof (e.g., a human antibody) that competes for binding to human tau protein with the anti-tau antibody comprising a heavy chain variable region with the amino acid sequence of SEQ ID NO: 139. In accordance with this aspect of the disclosure, a competitive binding assay, such as Bio-competes for binding to a tau protein with the enumerated monoclonal antibody. Other competitive binding assays known in the art can also be utilized to identify a competitive binding antibody in accordance with this aspect of the disclosure.

(192) In one embodiment, the antibody or binding fragment thereof has a heavy chain variable region with a H-CDR1 having the amino acid sequence of SEQ ID NO: 26, or a modified amino acid sequence thereof containing 1, 2, 3, or more amino acid residue modifications as compared to SEQ ID NO: 26; a H-CDR2 having the amino acid sequence of SEQ ID NO: 64, or a modified

amino acid sequence thereof containing 1, 2, 3, or more amino acid residue modifications as compared to SEQ ID NO: 64; and a H-CDR3 comprising the amino acid sequence of SEQ ID NO: 102, or a modified amino acid sequence thereof, said modified amino acid sequence containing 1, 2, 3, or more amino acid modifications as compared to SEQ ID NO: 102.

(193) An exemplary single domain antibody having this heavy chain variable region is referred to herein as the 59-T-sR2-1G5 antibody. The 59-T-sR2-1G5 antibody comprises a V.sub.H chain amino acid sequence of SEQ ID NO: 140 as shown below.

(194) TABLE-US-00027 SEQ ID NO: 140

QVQLQESGGGLVQPGGSLRLSCAASGSGISFKTMGWYRQAPGKQRELVA
TITRWGFTNYADSVKGRFTIARDNAKSTLYLQMNSLKPEDTAIYYCAA
QGGFMKPRANWYNSWGQGTLLTVSS

(195) In another embodiment, the antibody or binding fragment thereof comprises a heavy chain variable region having an amino acid sequence that shares at least 60%, at least 65%, at least 70%, at least 75%, at least 80%, at least 85%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, at least 99% sequence identity to SEQ ID NO: 140.

(196) In one embodiment, the antibody or binding fragment thereof of the present disclosure comprises a humanized variant of the heavy chain variable region of SEQ ID NO: 140, where the framework regions are humanized or replaced with human immunoglobulin framework sequences. As noted supra, suitable human or humanized framework sequences can be chosen based on their known structure, a consensus sequence, sequence homology to the framework sequences of donor antibody (e.g., the framework sequences of SEQ ID NO: 140), or a combination of these approaches. The humanized framework regions are designed to be similar in length and sequence to the parental framework sequences of SEQ ID NO: 140. In one embodiment, the humanized framework regions share 50%, 55%, 60%, 65%, 70%, 75%, 80%, 85%, 90%, 95% or more sequence identity to the framework regions of SEQ ID NO: 140. In another embodiment, the humanized framework regions are 60%, 65%, 70%, 75%, 80%, 85%, 90%, 95% or more similar in length to the framework regions of SEQ ID NO: 140. Humanized variants of the heavy chain variable region of SEQ ID NO: 140 share at least 50%, at least 55%, at least 60%, at least 65%, at least 70%, at least 75%, at least 80%, at least 85%, at least 90%, at least 95% sequence identity along the entire length of SEQ ID 140.

(197) Another aspect of the present disclosure relates to an antibody or binding portion thereof (e.g., a human antibody) that competes for binding to human tau protein with the anti-tau antibody comprising a heavy chain variable region with the amino acid sequence of SEQ ID NO: 140. In accordance with this aspect of the disclosure, a competitive binding assay, such as Bio-Layer Interferometry (BLI) can be utilized to identify an antibody or binding portion thereof that competes for binding to a tau protein with the enumerated monoclonal antibody. Other competitive binding assays known in the art can also be utilized to identify a competitive binding antibody in accordance with this aspect of the disclosure.

(198) In one embodiment, the antibody or binding fragment thereof has a heavy chain variable region with a H-CDR1 having the amino acid sequence of SEQ ID NO: 27, or a modified amino acid sequence thereof containing 1, 2, 3, or more amino acid residue modifications as compared to SEQ ID NO: 27; a H-CDR2 having the amino acid sequence of SEQ ID NO: 65, or a modified amino acid sequence thereof containing 1, 2, 3, or more amino acid residue modifications as compared to SEQ ID NO: 65; and a H-CDR3 comprising the amino acid sequence of SEQ ID NO: 103, or a modified amino acid sequence thereof, said modified amino acid sequence containing 1, 2, 3, or more amino acid modifications as compared to SEQ ID NO: 103.

(199) An exemplary single domain antibody having this heavy chain variable region is referred to herein as the 60-T-sR2-2D8 antibody. The 60-T-sR2-2D8 antibody comprises a V.sub.H chain amino acid sequence of SEQ ID NO: 141 as shown below.

(200) TABLE-US-00028 SEQ ID NO: 141

QVQLQESGGGLVQAGASLRRLSCVASGRTFSRYGMGWFRQAPGKEREFVA
AISRSGAISYYADSVKGRFTISRGDATNTVVLQMSSLKPGDTAVYYCAA
SSRLLGGPFAYDYWGQGTTLTVSS

(201) In another embodiment, the antibody or binding fragment thereof comprises a heavy chain variable region having an amino acid sequence that shares at least 60%, at least 65%, at least 70%, at least 75%, at least 80%, at least 85%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94% at least 95%, at least 96%, at least 97%, at least 98%, at least 99% sequence identity to SEQ ID NO: 141.

(202) In one embodiment, the antibody or binding fragment thereof of the present disclosure comprises a humanized variant of the heavy chain variable region of SEQ ID NO: 141, where the framework regions are humanized or replaced with human immunoglobulin framework sequences. As noted supra, suitable human or humanized framework sequences can be chosen based on their known structure, a consensus sequence, sequence homology to the framework sequences of donor antibody (e.g., the framework sequences of SEQ ID NO: 141), or a combination of these approaches. The humanized framework regions are designed to be similar in length and sequence to the parental framework sequences of SEQ ID NO: 141. In one embodiment, the humanized framework regions share 50%, 55%, 60%, 65%, 70%, 75%, 80%, 85%, 90%, 95% or more sequence identity to the framework regions of SEQ ID NO: 141. In another embodiment, the humanized framework regions are 60%, 65%, 70%, 75%, 80%, 85%, 90%, 95% or more similar in length to the framework regions of SEQ ID NO: 141. Humanized variants of the heavy chain variable region of SEQ ID NO: 141 share at least 50%, at least 55%, at least 60%, at least 65%, at least 70%, at least 75%, at least 80%, at least 85%, at least 90%, at least 95% sequence identity along the entire length of SEQ ID 141.

(203) Another aspect of the present disclosure relates to an antibody or binding portion thereof (e.g., a human antibody) that competes for binding to human tau protein with the anti-tau antibody comprising a heavy chain variable region with the amino acid sequence of SEQ ID NO: 141. In accordance with this aspect of the disclosure, a competitive binding assay, such as Bio-Layer Interferometry (BLI) can be utilized to identify an antibody or binding portion thereof that competes for binding to a tau protein with the enumerated monoclonal antibody. Other competitive binding assays known in the art can also be utilized to identify a competitive binding antibody in accordance with this aspect of the disclosure.

(204) In one embodiment, the antibody or binding fragment thereof has a heavy chain variable region with a H-CDR1 having the amino acid sequence of SEQ ID NO: 28, or a modified amino acid sequence thereof containing 1, 2, 3, or more amino acid residue modifications as compared to SEQ ID NO: 28; a H-CDR2 having the amino acid sequence of SEQ ID NO: 66, or a modified amino acid sequence thereof containing 1, 2, 3, or more amino acid residue modifications as compared to SEQ ID NO: 66; and a H-CDR3 comprising the amino acid sequence of SEQ ID NO: 104, or a modified amino acid sequence thereof, said modified amino acid sequence containing 1, 2, 3, or more amino acid modifications as compared to SEQ ID NO: 104.

(205) An exemplary single domain antibody having this heavy chain variable region is referred to herein as the 61-T-sR2-2H7 antibody. The 61-T-sR2-2H7 antibody comprises a V.sub.H chain amino acid sequence of SEQ ID NO: 142 as shown below.

(206) TABLE-US-00029 SEQ ID NO: 142

QVQLQESGGGLVQPGGSLRLSCAASRYIFGTMGWYRQAPGLQRELVASI
SRGGSTNYADSVKGRFAISRDNKNTVYLQMNSLKPEDTAVYYCNANRR
GWNWYGQGTQVTVSS

(207) In another embodiment, the antibody or binding fragment thereof comprises a heavy chain variable region having an amino acid sequence that shares at least 60%, at least 65%, at least 70%, at least 75%, at least 80%, at least 85%, at least 90%, at least 91%, at least 92%, at least 93%, at

least 94% at least 95%, at least 96%, at least 97%, at least 98%, at least 99% sequence identity to SEQ ID NO: 142.

(208) In one embodiment, the antibody or binding fragment thereof of the present disclosure comprises a humanized variant of the heavy chain variable region of SEQ ID NO: 142, where the framework regions are humanized or replaced with human immunoglobulin framework sequences. As noted supra, suitable human or humanized framework sequences can be chosen based on their known structure, a consensus sequence, sequence homology to the framework sequences of donor antibody (e.g., the framework sequences of SEQ ID NO: 142), or a combination of these approaches. The humanized framework regions are designed to be similar in length and sequence to the parental framework sequences of SEQ ID NO: 142. In one embodiment, the humanized framework regions share 50%, 55%, 60%, 65%, 70%, 75%, 80%, 85%, 90%, 95% or more sequence identity to the framework regions of SEQ ID NO: 142. In another embodiment, the humanized framework regions are 60%, 65%, 70%, 75%, 80%, 85%, 90%, 95% or more similar in length to the framework regions of SEQ ID NO: 142. Humanized variants of the heavy chain variable region of SEQ ID NO: 142 share at least 50%, at least 55%, at least 60%, at least 65%, at least 70%, at least 75%, at least 80%, at least 85%, at least 90%, at least 95% sequence identity along the entire length of SEQ ID 142.

(209) Another aspect of the present disclosure relates to an antibody or binding portion thereof (e.g., a human antibody) that competes for binding to human tau protein with the anti-tau antibody comprising a heavy chain variable region with the amino acid sequence of SEQ ID NO: 142. In accordance with this aspect of the disclosure, a competitive binding assay, such as Bio-Layer Interferometry (BLI) can be utilized to identify an antibody or binding portion thereof that competes for binding to a tau protein with the enumerated monoclonal antibody. Other competitive binding assays known in the art can also be utilized to identify a competitive binding antibody in accordance with this aspect of the disclosure.

(210) In one embodiment, the antibody or binding fragment thereof has a heavy chain variable region with a H-CDR1 having the amino acid sequence of SEQ ID NO: 29, or a modified amino acid sequence thereof containing 1, 2, 3, or more amino acid residue modifications as compared to SEQ ID NO: 29; a H-CDR2 having the amino acid sequence of SEQ ID NO: 67, or a modified amino acid sequence thereof containing 1, 2, 3, or more amino acid residue modifications as compared to SEQ ID NO: 67; and a H-CDR3 comprising the amino acid sequence of SEQ ID NO: 105, or a modified amino acid sequence thereof, said modified amino acid sequence containing 1, 2, 3, or more amino acid modifications as compared to SEQ ID NO: 105.

(211) An exemplary single domain antibody having this heavy chain variable region is referred to herein as the 66-T-sR2-1C7 antibody. The 66-T-sR2-1C7 antibody comprises a V.sub.H chain amino acid sequence of SEQ ID NO: 143 as shown below.

(212) TABLE-US-00030 SEQ ID NO: 143

QVQLQESGGLVQAGGSLRLSCAASGTIFTMKNMAWYRQAPGKEREFVA
AISTSGGVVTWYADSSVKGRFTISRDNKNTLYLQMDSLKPEDTAVYYCK
VRSGPRIIPQLRREYWGQGTQVTVSS

(213) In another embodiment, the antibody or binding fragment thereof comprises a heavy chain variable region having an amino acid sequence that shares at least 60%, at least 65%, at least 70%, at least 75%, at least 80%, at least 85%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94% at least 95%, at least 96%, at least 97%, at least 98%, at least 99% sequence identity to SEQ ID NO: 143.

(214) In one embodiment, the antibody or binding fragment thereof of the present disclosure comprises a humanized variant of the heavy chain variable region of SEQ ID NO: 143, where the framework regions are humanized or replaced with human immunoglobulin framework sequences. As noted supra, suitable human or humanized framework sequences can be chosen based on their known structure, a consensus sequence, sequence homology to the framework sequences of donor

antibody (e.g., the framework sequences of SEQ ID NO: 143), or a combination of these approaches. The humanized framework regions are designed to be similar in length and sequence to the parental framework sequences of SEQ ID NO: 143. In one embodiment, the humanized framework regions share 50%, 55%, 60%, 65%, 70%, 75%, 80%, 85%, 90%, 95% or more sequence identity to the framework regions of SEQ ID NO: 143. In another embodiment, the humanized framework regions are 60%, 65%, 70%, 75%, 80%, 85%, 90%, 95% or more similar in length to the framework regions of SEQ ID NO: 143. Humanized variants of the heavy chain variable region of SEQ ID NO: 143 share at least 50%, at least 55%, at least 60%, at least 65%, at least 70%, at least 75%, at least 80%, at least 85%, at least 90%, at least 95% sequence identity along the entire length of SEQ ID 143.

(215) Another aspect of the present disclosure relates to an antibody or binding portion thereof (e.g., a human antibody) that competes for binding to human tau protein with the anti-tau antibody comprising a heavy chain variable region with the amino acid sequence of SEQ ID NO: 143. In accordance with this aspect of the disclosure, a competitive binding assay, such as Bio-Layer Interferometry (BLI) can be utilized to identify an antibody or binding portion thereof that competes for binding to a tau protein with the enumerated monoclonal antibody. Other competitive binding assays known in the art can also be utilized to identify a competitive binding antibody in accordance with this aspect of the disclosure.

(216) In one embodiment, the antibody or binding fragment thereof has a heavy chain variable region with a H-CDR1 having the amino acid sequence of SEQ ID NO: 30, or a modified amino acid sequence thereof containing 1, 2, 3, or more amino acid residue modifications as compared to SEQ ID NO: 30; a H-CDR2 having the amino acid sequence of SEQ ID NO: 68, or a modified amino acid sequence thereof containing 1, 2, 3, or more amino acid residue modifications as compared to SEQ ID NO: 68; and a H-CDR3 comprising the amino acid sequence of SEQ ID NO: 106, or a modified amino acid sequence thereof, said modified amino acid sequence containing 1, 2, 3, or more amino acid modifications as compared to SEQ ID NO: 106.

(217) An exemplary single domain antibody having this heavy chain variable region is referred to herein as the 68-T-sR2-2E6 antibody. The 68-T-sR2-2E6 antibody comprises a V.sub.H chain amino acid sequence of SEQ ID NO: 144 as shown below.

(218) TABLE-US-00031 SEQ ID NO: 144

QVQLQESGGGLVQPGDSLRLSCAASGRTFSSYAMGWFRQAPGKEREFVA
AINWSGHSTYYADSVKGRFTISRDNANKNTVYLQMNMLKPEDTAVYYCKL
TRLNTYWGQGTLVTVSS

(219) In another embodiment, the antibody or binding fragment thereof comprises a heavy chain variable region having an amino acid sequence that shares at least 60%, at least 65%, at least 70%, at least 75%, at least 80%, at least 85%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94% at least 95%, at least 96%, at least 97%, at least 98%, at least 99% sequence identity to SEQ ID NO: 144.

(220) In one embodiment, the antibody or binding fragment thereof of the present disclosure comprises a humanized variant of the heavy chain variable region of SEQ ID NO: 144, where the framework regions are humanized or replaced with human immunoglobulin framework sequences. As noted supra, suitable human or humanized framework sequences can be chosen based on their known structure, a consensus sequence, sequence homology to the framework sequences of donor antibody (e.g., the framework sequences of SEQ ID NO: 144), or a combination of these approaches. The humanized framework regions are designed to be similar in length and sequence to the parental framework sequences of SEQ ID NO: 144. In one embodiment, the humanized framework regions share 50%, 55%, 60%, 65%, 70%, 75%, 80%, 85%, 90%, 95% or more sequence identity to the framework regions of SEQ ID NO: 144. In another embodiment, the humanized framework regions are 60%, 65%, 70%, 75%, 80%, 85%, 90%, 95% or more similar in length to the framework regions of SEQ ID NO: 144. Humanized variants of the heavy chain

variable region of SEQ ID NO: 144 share at least 50%, at least 55%, at least 60%, at least 65%, at least 70%, at least 75%, at least 80%, at least 85%, at least 90%, at least 95% sequence identity along the entire length of SEQ ID 144.

(221) Another aspect of the present disclosure relates to an antibody or binding portion thereof (e.g., a human antibody) that competes for binding to human tau protein with the anti-tau antibody comprising a heavy chain variable region with the amino acid sequence of SEQ ID NO: 144. In accordance with this aspect of the disclosure, a competitive binding assay, such as Bio-Layer Interferometry (BLI) can be utilized to identify an antibody or binding portion thereof that competes for binding to a tau protein with the enumerated monoclonal antibody. Other competitive binding assays known in the art can also be utilized to identify a competitive binding antibody in accordance with this aspect of the disclosure.

(222) In one embodiment, the antibody or binding fragment thereof has a heavy chain variable region with a H-CDR1 having the amino acid sequence of SEQ ID NO: 31, or a modified amino acid sequence thereof containing 1, 2, 3, or more amino acid residue modifications as compared to SEQ ID NO: 31; a H-CDR2 having the amino acid sequence of SEQ ID NO: 69, or a modified amino acid sequence thereof containing 1, 2, 3, or more amino acid residue modifications as compared to SEQ ID NO: 69; and a H-CDR3 comprising the amino acid sequence of SEQ ID NO: 107, or a modified amino acid sequence thereof, said modified amino acid sequence containing 1, 2, 3, or more amino acid modifications as compared to SEQ ID NO: 107.

(223) An exemplary single domain antibody having this heavy chain variable region is referred to herein as the 69-T-sR2-2H8 antibody. The 69-T-sR2-2H8 antibody comprises a V.sub.H chain amino acid sequence of SEQ ID NO: 145 as shown below.

(224) TABLE-US-00032 SEQ ID NO: 145

QVQLQESGGGLVQAGDSLRLSCAFSGRTFGLRTMGWFRQAPGKEREFVS
SLTWRDNNAYYADSVKGRFTISRDNAAKKTLYLQMNSLKPEDTAVYFCNV
HLVFTNRDYYWGQGTQVTVSS

(225) In another embodiment, the antibody or binding fragment thereof comprises a heavy chain variable region having an amino acid sequence that shares at least 60%, at least 65%, at least 70%, at least 75%, at least 80%, at least 85%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94% at least 95%, at least 96%, at least 97%, at least 98%, at least 99% sequence identity to SEQ ID NO: 145.

(226) In one embodiment, the antibody or binding fragment thereof of the present disclosure comprises a humanized variant of the heavy chain variable region of SEQ ID NO: 145, where the framework regions are humanized or replaced with human immunoglobulin framework sequences. As noted supra, suitable human or humanized framework sequences can be chosen based on their known structure, a consensus sequence, sequence homology to the framework sequences of donor antibody (e.g., the framework sequences of SEQ ID NO: 145), or a combination of these approaches. The humanized framework regions are designed to be similar in length and sequence to the parental framework sequences of SEQ ID NO: 145. In one embodiment, the humanized framework regions share 50%, 55%, 60%, 65%, 70%, 75%, 80%, 85%, 90%, 95% or more sequence identity to the framework regions of SEQ ID NO: 145. In another embodiment, the humanized framework regions are 60%, 65%, 70%, 75%, 80%, 85%, 90%, 95% or more similar in length to the framework regions of SEQ ID NO: 145. Humanized variants of the heavy chain variable region of SEQ ID NO: 145 share at least 50%, at least 55%, at least 60%, at least 65%, at least 70%, at least 75%, at least 80%, at least 85%, at least 90%, at least 95% sequence identity along the entire length of SEQ ID 145.

(227) Another aspect of the present disclosure relates to an antibody or binding portion thereof (e.g., a human antibody) that competes for binding to human tau protein with the anti-tau antibody comprising a heavy chain variable region with the amino acid sequence of SEQ ID NO: 145. In accordance with this aspect of the disclosure, a competitive binding assay, such as Bio-competes

for binding to a tau protein with the enumerated monoclonal antibody. Other competitive binding assays known in the art can also be utilized to identify a competitive binding antibody in accordance with this aspect of the disclosure.

(228) In one embodiment, the antibody or binding fragment thereof has a heavy chain variable region with a H-CDR1 having the amino acid sequence of SEQ ID NO: 32, or a modified amino acid sequence thereof containing 1, 2, 3, or more amino acid residue modifications as compared to SEQ ID NO: 32; a H-CDR2 having the amino acid sequence of SEQ ID NO: 70, or a modified amino acid sequence thereof containing 1, 2, 3, or more amino acid residue modifications as compared to SEQ ID NO: 70; and a H-CDR3 comprising the amino acid sequence of SEQ ID NO: 108, or a modified amino acid sequence thereof, said modified amino acid sequence containing 1, 2, 3, or more amino acid modifications as compared to SEQ ID NO: 108.

(229) An exemplary single domain antibody having this heavy chain variable region is referred to herein as the 74-T-sR2-1C10 antibody. The 74-T-sR2-1C10 antibody comprises a V.sub.H chain amino acid sequence of SEQ ID NO: 146 as shown below.

(230) TABLE-US-00033 SEQ ID NO: 146

QVQLQQSGGGLVQPGGSLRLSCAASWRIFSPNAMA WYRQAPGKQRELVA
RITWAGITNYADSVKGRFTISRDN AKNTVYLQMHS LKPEDTAIYYCVAD
RRSSYLGPRFDYWGGQTQVTVSS

(231) In another embodiment, the antibody or binding fragment thereof comprises a heavy chain variable region having an amino acid sequence that shares at least 60%, at least 65%, at least 70%, at least 75%, at least 80%, at least 85%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94% at least 95%, at least 96%, at least 97%, at least 98%, at least 99% sequence identity to SEQ ID NO: 146.

(232) In one embodiment, the antibody or binding fragment thereof of the present disclosure comprises a humanized variant of the heavy chain variable region of SEQ ID NO: 146, where the framework regions are humanized or replaced with human immunoglobulin framework sequences. As noted supra, suitable human or humanized framework sequences can be chosen based on their known structure, a consensus sequence, sequence homology to the framework sequences of donor antibody (e.g., the framework sequences of SEQ ID NO: 146), or a combination of these approaches. The humanized framework regions are designed to be similar in length and sequence to the parental framework sequences of SEQ ID NO: 146. In one embodiment, the humanized framework regions share 50%, 55%, 60%, 65%, 70%, 75%, 80%, 85%, 90%, 95% or more sequence identity to the framework regions of SEQ ID NO: 146. In another embodiment, the humanized framework regions are 60%, 65%, 70%, 75%, 80%, 85%, 90%, 95% or more similar in length to the framework regions of SEQ ID NO: 146. Humanized variants of the heavy chain variable region of SEQ ID NO: 146 share at least 50%, at least 55%, at least 60%, at least 65%, at least 70%, at least 75%, at least 80%, at least 85%, at least 90%, at least 95% sequence identity along the entire length of SEQ ID 146.

(233) Another aspect of the present disclosure relates to an antibody or binding portion thereof (e.g., a human antibody) that competes for binding to human tau protein with the anti-tau antibody comprising a heavy chain variable region with the amino acid sequence of SEQ ID NO: 146. In accordance with this aspect of the disclosure, a competitive binding assay, such as Bio-Layer Interferometry (BLI) can be utilized to identify an antibody or binding portion thereof that competes for binding to a tau protein with the enumerated monoclonal antibody. Other competitive binding assays known in the art can also be utilized to identify a competitive binding antibody in accordance with this aspect of the disclosure.

(234) In one embodiment, the antibody or binding fragment thereof has a heavy chain variable region with a H-CDR1 having the amino acid sequence of SEQ ID NO: 33, or a modified amino acid sequence thereof containing 1, 2, 3, or more amino acid residue modifications as compared to SEQ ID NO: 33; a H-CDR2 having the amino acid sequence of SEQ ID NO: 71, or a modified

amino acid sequence thereof containing 1, 2, 3, or more amino acid residue modifications as compared to SEQ ID NO: 71; and a H-CDR3 comprising the amino acid sequence of SEQ ID NO: 109, or a modified amino acid sequence thereof, said modified amino acid sequence containing 1, 2, 3, or more amino acid modifications as compared to SEQ ID NO: 109.

(235) An exemplary single domain antibody having this heavy chain variable region is referred to herein as the 75-T-sR2-1G12 antibody. The 75-T-sR2-1G12 antibody comprises a V.sub.H chain amino acid sequence of SEQ ID NO: 147 as shown below.

(236) TABLE-US-00034 SEQ ID NO: 147

QVQLQESGGGLVQPGGSLRLSCAASRYIFGTMGWYRQAPGLQRELVASI
SRGGSTNYADSVKGRFAISRDNKNTVYLQMNSLPEDTAVYYCAARSS
LLEFWLGSRRGYDYWGQGTQVTVSS

(237) In another embodiment, the antibody or binding fragment thereof comprises a heavy chain variable region having an amino acid sequence that shares at least 60%, at least 65%, at least 70%, at least 75%, at least 80%, at least 85%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94% at least 95%, at least 96%, at least 97%, at least 98%, at least 99% sequence identity to SEQ ID NO: 147.

(238) In one embodiment, the antibody or binding fragment thereof of the present disclosure comprises a humanized variant of the heavy chain variable region of SEQ ID NO: 147, where the framework regions are humanized or replaced with human immunoglobulin framework sequences. As noted supra, suitable human or humanized framework sequences can be chosen based on their known structure, a consensus sequence, sequence homology to the framework sequences of donor antibody (e.g., the framework sequences of SEQ ID NO: 147), or a combination of these approaches. The humanized framework regions are designed to be similar in length and sequence to the parental framework sequences of SEQ ID NO: 147. In one embodiment, the humanized framework regions share 50%, 55%, 60%, 65%, 70%, 75%, 80%, 85%, 90%, 95% or more sequence identity to the framework regions of SEQ ID NO: 147. In another embodiment, the humanized framework regions are 60%, 65%, 70%, 75%, 80%, 85%, 90%, 95% or more similar in length to the framework regions of SEQ ID NO: 147. Humanized variants of the heavy chain variable region of SEQ ID NO: 147 share at least 50%, at least 55%, at least 60%, at least 65%, at least 70%, at least 75%, at least 80%, at least 85%, at least 90%, at least 95% sequence identity along the entire length of SEQ ID 147.

(239) Another aspect of the present disclosure relates to an antibody or binding portion thereof (e.g., a human antibody) that competes for binding to human tau protein with the anti-tau antibody comprising a heavy chain variable region with the amino acid sequence of SEQ ID NO: 147. In accordance with this aspect of the disclosure, a competitive binding assay, such as Bio-Layer Interferometry (BLI) can be utilized to identify an antibody or binding portion thereof that competes for binding to a tau protein with the enumerated monoclonal antibody. Other competitive binding assays known in the art can also be utilized to identify a competitive binding antibody in accordance with this aspect of the disclosure.

(240) In one embodiment, the antibody or binding fragment thereof has a heavy chain variable region with a H-CDR1 having the amino acid sequence of SEQ ID NO: 34, or a modified amino acid sequence thereof containing 1, 2, 3, or more amino acid residue modifications as compared to SEQ ID NO: 34; a H-CDR2 having the amino acid sequence of SEQ ID NO: 72, or a modified amino acid sequence thereof containing 1, 2, 3, or more amino acid residue modifications as compared to SEQ ID NO: 72; and a H-CDR3 comprising the amino acid sequence of SEQ ID NO: 110, or a modified amino acid sequence thereof, said modified amino acid sequence containing 1, 2, 3, or more amino acid modifications as compared to SEQ ID NO: 110.

(241) An exemplary single domain antibody having this heavy chain variable region is referred to herein as the 76-T-sR2-2E7 antibody. The 76-T-sR2-2E7 antibody comprises a V.sub.H chain amino acid sequence of SEQ ID NO: 148 as shown below.

(242) TABLE-US-00035 SEQ ID NO: 148

QVQLQQSGGGLVQPGGSLRLACAASGRFFRINAMAWYRQAPGKQRELVA
TITRAGTTTYADSVKGRFTISRDNKNTVYLQMISLKPEDTAVYYCAKY
PTITWYGRHDYRGQGTQVTVSS

(243) In another embodiment, the antibody or binding fragment thereof comprises a heavy chain variable region having an amino acid sequence that shares at least 60%, at least 65%, at least 70%, at least 75%, at least 80%, at least 85%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94% at least 95%, at least 96%, at least 97%, at least 98%, at least 99% sequence identity to SEQ ID NO: 148.

(244) In one embodiment, the antibody or binding fragment thereof of the present disclosure comprises a humanized variant of the heavy chain variable region of SEQ ID NO: 148, where the framework regions are humanized or replaced with human immunoglobulin framework sequences. As noted supra, suitable human or humanized framework sequences can be chosen based on their known structure, a consensus sequence, sequence homology to the framework sequences of donor antibody (e.g., the framework sequences of SEQ ID NO: 148), or a combination of these approaches. The humanized framework regions are designed to be similar in length and sequence to the parental framework sequences of SEQ ID NO: 148. In one embodiment, the humanized framework regions share 50%, 55%, 60%, 65%, 70%, 75%, 80%, 85%, 90%, 95% or more sequence identity to the framework regions of SEQ ID NO: 148. In another embodiment, the humanized framework regions are 60%, 65%, 70%, 75%, 80%, 85%, 90%, 95% or more similar in length to the framework regions of SEQ ID NO: 148. Humanized variants of the heavy chain variable region of SEQ ID NO: 148 share at least 50%, at least 55%, at least 60%, at least 65%, at least 70%, at least 75%, at least 80%, at least 85%, at least 90%, at least 95% sequence identity along the entire length of SEQ ID 148.

(245) Another aspect of the present disclosure relates to an antibody or binding portion thereof (e.g., a human antibody) that competes for binding to human tau protein with the anti-tau antibody comprising a heavy chain variable region with the amino acid sequence of SEQ ID NO: 148. In accordance with this aspect of the disclosure, a competitive binding assay, such as Bio-competes for binding to a tau protein with the enumerated monoclonal antibody. Other competitive binding assays known in the art can also be utilized to identify a competitive binding antibody in accordance with this aspect of the disclosure.

(246) In one embodiment, the antibody or binding fragment thereof has a heavy chain variable region with a H-CDR1 having the amino acid sequence of SEQ ID NO: 35, or a modified amino acid sequence thereof containing 1, 2, 3, or more amino acid residue modifications as compared to SEQ ID NO: 35; a H-CDR2 having the amino acid sequence of SEQ ID NO: 73, or a modified amino acid sequence thereof containing 1, 2, 3, or more amino acid residue modifications as compared to SEQ ID NO: 73; and a H-CDR3 comprising the amino acid sequence of SEQ ID NO: 111, or a modified amino acid sequence thereof, said modified amino acid sequence containing 1, 2, 3, or more amino acid modifications as compared to SEQ ID NO: 111.

(247) An exemplary single domain antibody having this heavy chain variable region is referred to herein as the 84-T-sR2-2E8 antibody. The 84-T-sR2-2E8 antibody comprises a V.sub.H chain amino acid sequence of SEQ ID NO: 149 as shown below.

(248) TABLE-US-00036 SEQ ID NO: 149

QVQLQESGGGLVQHGGSLRLSCAASGSFFRINTMAWHRQAPGKQRELVA
SITRGGSTNYADSVKGRFTISRDNKNTVYLQMNSLKSEDTAVYYCAAS
RFGINYITARQYGYWGQGTQVTVSS

(249) In another embodiment, the antibody or binding fragment thereof comprises a heavy chain variable region having an amino acid sequence that shares at least 60%, at least 65%, at least 70%, at least 75%, at least 80%, at least 85%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94% at least 95%, at least 96%, at least 97%, at least 98%, at least 99% sequence identity to

SEQ ID NO: 149.

(250) In one embodiment, the antibody or binding fragment thereof of the present disclosure comprises a humanized variant of the heavy chain variable region of SEQ ID NO: 149, where the framework regions are humanized or replaced with human immunoglobulin framework sequences. As noted supra, suitable human or humanized framework sequences can be chosen based on their known structure, a consensus sequence, sequence homology to the framework sequences of donor antibody (e.g., the framework sequences of SEQ ID NO: 149), or a combination of these approaches. The humanized framework regions are designed to be similar in length and sequence to the parental framework sequences of SEQ ID NO: 149. In one embodiment, the humanized framework regions share 50%, 55%, 60%, 65%, 70%, 75%, 80%, 85%, 90%, 95% or more sequence identity to the framework regions of SEQ ID NO: 149. In another embodiment, the humanized framework regions are 60%, 65%, 70%, 75%, 80%, 85%, 90%, 95% or more similar in length to the framework regions of SEQ ID NO: 149. Humanized variants of the heavy chain variable region of SEQ ID NO: 149 share at least 50%, at least 55%, at least 60%, at least 65%, at least 70%, at least 75%, at least 80%, at least 85%, at least 90%, at least 95% sequence identity along the entire length of SEQ ID 149.

(251) Another aspect of the present disclosure relates to an antibody or binding portion thereof (e.g., a human antibody) that competes for binding to human tau protein with the anti-tau antibody comprising a heavy chain variable region with the amino acid sequence of SEQ ID NO: 149. In accordance with this aspect of the disclosure, a competitive binding assay, such as Bio-Layer Interferometry (BLI) can be utilized to identify an antibody or binding portion thereof that competes for binding to a tau protein with the enumerated monoclonal antibody. Other competitive binding assays known in the art can also be utilized to identify a competitive binding antibody in accordance with this aspect of the disclosure.

(252) In one embodiment, the antibody or binding fragment thereof has a heavy chain variable region with a H-CDR1 having the amino acid sequence of SEQ ID NO: 36, or a modified amino acid sequence thereof containing 1, 2, 3, or more amino acid residue modifications as compared to SEQ ID NO: 36; a H-CDR2 having the amino acid sequence of SEQ ID NO: 74, or a modified amino acid sequence thereof containing 1, 2, 3, or more amino acid residue modifications as compared to SEQ ID NO: 74; and a H-CDR3 comprising the amino acid sequence of SEQ ID NO: 112, or a modified amino acid sequence thereof, said modified amino acid sequence containing 1, 2, 3, or more amino acid modifications as compared to SEQ ID NO: 112.

(253) An exemplary single domain antibody having this heavy chain variable region is referred to herein as the 90-T-sR2-1D9 antibody. The 90-T-sR2-1D9 antibody comprises a V.sub.H chain amino acid sequence of SEQ ID NO: 150 as shown below.

(254) TABLE-US-00037 SEQ ID NO: 150

QVQLQASGGGLVQAGGSLRLSCAASGRTFSRYPMGWFRQAPGKEREFVA
RFGWSGLSTYYADSVKGRFTISRDNANKNTVYLMNSLKPEDTAVYYCAA
GRFMGSYDYWGQGTQVTVSS

(255) In another embodiment, the antibody or binding fragment thereof comprises a heavy chain variable region having an amino acid sequence that shares at least 60%, at least 65%, at least 70%, at least 75%, at least 80%, at least 85%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94% at least 95%, at least 96%, at least 97%, at least 98%, at least 99% sequence identity to SEQ ID NO: 150.

(256) In one embodiment, the antibody or binding fragment thereof of the present disclosure comprises a humanized variant of the heavy chain variable region of SEQ ID NO: 150, where the framework regions are humanized or replaced with human immunoglobulin framework sequences. As noted supra, suitable human or humanized framework sequences can be chosen based on their known structure, a consensus sequence, sequence homology to the framework sequences of donor antibody (e.g., the framework sequences of SEQ ID NO: 150), or a combination of these

approaches. The humanized framework regions are designed to be similar in length and sequence to the parental framework sequences of SEQ ID NO: 150. In one embodiment, the humanized framework regions share 50%, 55%, 60%, 65%, 70%, 75%, 80%, 85%, 90%, 95% or more sequence identity to the framework regions of SEQ ID NO: 150. In another embodiment, the humanized framework regions are 60%, 65%, 70%, 75%, 80%, 85%, 90%, 95% or more similar in length to the framework regions of SEQ ID NO: 150. Humanized variants of the heavy chain variable region of SEQ ID NO: 150 share at least 50%, at least 55%, at least 60%, at least 65%, at least 70%, at least 75%, at least 80%, at least 85%, at least 90%, at least 95% sequence identity along the entire length of SEQ ID 150.

(257) Another aspect of the present disclosure relates to an antibody or binding portion thereof (e.g., a human antibody) that competes for binding to human tau protein with the anti-tau antibody comprising a heavy chain variable region with the amino acid sequence of SEQ ID NO: 150. In accordance with this aspect of the disclosure, a competitive binding assay, such as Bio-Layer Interferometry (BLI) can be utilized to identify an antibody or binding portion thereof that competes for binding to a tau protein with the enumerated monoclonal antibody. Other competitive binding assays known in the art can also be utilized to identify a competitive binding antibody in accordance with this aspect of the disclosure.

(258) In one embodiment, the antibody or binding fragment thereof has a heavy chain variable region with a H-CDR1 having the amino acid sequence of SEQ ID NO: 37, or a modified amino acid sequence thereof containing 1, 2, 3, or more amino acid residue modifications as compared to SEQ ID NO: 37; a H-CDR2 having the amino acid sequence of SEQ ID NO: 75, or a modified amino acid sequence thereof containing 1, 2, 3, or more amino acid residue modifications as compared to SEQ ID NO: 75; and a H-CDR3 comprising the amino acid sequence of SEQ ID NO: 113, or a modified amino acid sequence thereof, said modified amino acid sequence containing 1, 2, 3, or more amino acid modifications as compared to SEQ ID NO: 113.

(259) An exemplary single domain antibody having this heavy chain variable region is referred to herein as the 91-T-sR2-1H12 antibody. The 91-T-sR2-1H12 antibody comprises a V.sub.H chain amino acid sequence of SEQ ID NO: 151 as shown below.

(260) TABLE-US-00038 SEQ ID NO: 151

QVQLQESGGGLVQAGGSLRLSCAASGRTFSWYAMGWFRQAPGREREFVA
AIRRSGGITYADSVKGRFAVSRDNAKNTVYLQMNSLKPEDTAVYYCAA
NLFQWRLNDNGNQYGSWGQGTLLTVSS

(261) In another embodiment, the antibody or binding fragment thereof comprises a heavy chain variable region having an amino acid sequence that shares at least 60%, at least 65%, at least 70%, at least 75%, at least 80%, at least 85%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94% at least 95%, at least 96%, at least 97%, at least 98%, at least 99% sequence identity to SEQ ID NO: 151.

(262) In one embodiment, the antibody or binding fragment thereof of the present disclosure comprises a humanized variant of the heavy chain variable region of SEQ ID NO: 151, where the framework regions are humanized or replaced with human immunoglobulin framework sequences. As noted supra, suitable human or humanized framework sequences can be chosen based on their known structure, a consensus sequence, sequence homology to the framework sequences of donor antibody (e.g., the framework sequences of SEQ ID NO: 151), or a combination of these approaches. The humanized framework regions are designed to be similar in length and sequence to the parental framework sequences of SEQ ID NO: 151. In one embodiment, the humanized framework regions share 50%, 55%, 60%, 65%, 70%, 75%, 80%, 85%, 90%, 95% or more sequence identity to the framework regions of SEQ ID NO: 151. In another embodiment, the humanized framework regions are 60%, 65%, 70%, 75%, 80%, 85%, 90%, 95% or more similar in length to the framework regions of SEQ ID NO: 151. Humanized variants of the heavy chain variable region of SEQ ID NO: 151 share at least 50%, at least 55%, at least 60%, at least 65%, at

least 70%, at least 75%, at least 80%, at least 85%, at least 90%, at least 95% sequence identity along the entire length of SEQ ID 151.

(263) Another aspect of the present disclosure relates to an antibody or binding portion thereof (e.g., a human antibody) that competes for binding to human tau protein with the anti-tau antibody comprising a heavy chain variable region with the amino acid sequence of SEQ ID NO: 151. In accordance with this aspect of the disclosure, a competitive binding assay, such as Bio-competes for binding to a tau protein with the enumerated monoclonal antibody. Other competitive binding assays known in the art can also be utilized to identify a competitive binding antibody in accordance with this aspect of the disclosure.

(264) In one embodiment, the antibody or binding fragment thereof has a heavy chain variable region with a H-CDR1 having the amino acid sequence of SEQ ID NO: 38, or a modified amino acid sequence thereof containing 1, 2, 3, or more amino acid residue modifications as compared to SEQ ID NO: 38; a H-CDR2 having the amino acid sequence of SEQ ID NO: 76, or a modified amino acid sequence thereof containing 1, 2, 3, or more amino acid residue modifications as compared to SEQ ID NO: 76; and a H-CDR3 comprising the amino acid sequence of SEQ ID NO: 114, or a modified amino acid sequence thereof, said modified amino acid sequence containing 1, 2, 3, or more amino acid modifications as compared to SEQ ID NO: 114.

(265) An exemplary single domain antibody having this heavy chain variable region is referred to herein as the 92-T-sR2-2E9 antibody. The 92-T-sR2-2E9 antibody comprises a V.sub.H chain amino acid sequence of SEQ ID NO: 152 as shown below.

(266) TABLE-US-00039 SEQ ID NO: 152

QVQLQESGGLVQAGGSLRLSCVVSGRFTSTSQMGWFRQPPGKERELVA
RISWRGKQHYADSVKGRFTISRDYAKNTVYLQMNGLKSEDTAVYYCAAG
PRAVLFGTYDYWGQGTQVTVSS

(267) In another embodiment, the antibody or binding fragment thereof comprises a heavy chain variable region having an amino acid sequence that shares at least 60%, at least 65%, at least 70%, at least 75%, at least 80%, at least 85%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94% at least 95%, at least 96%, at least 97%, at least 98%, at least 99% sequence identity to SEQ ID NO: 152.

(268) In one embodiment, the antibody or binding fragment thereof of the present disclosure comprises a humanized variant of the heavy chain variable region of SEQ ID NO: 152, where the framework regions are humanized or replaced with human immunoglobulin framework sequences. As noted supra, suitable human or humanized framework sequences can be chosen based on their known structure, a consensus sequence, sequence homology to the framework sequences of donor antibody (e.g., the framework sequences of SEQ ID NO: 152), or a combination of these approaches. The humanized framework regions are designed to be similar in length and sequence to the parental framework sequences of SEQ ID NO: 152. In one embodiment, the humanized framework regions share 50%, 55%, 60%, 65%, 70%, 75%, 80%, 85%, 90%, 95% or more sequence identity to the framework regions of SEQ ID NO: 152. In another embodiment, the humanized framework regions are 60%, 65%, 70%, 75%, 80%, 85%, 90%, 95% or more similar in length to the framework regions of SEQ ID NO: 152. Humanized variants of the heavy chain variable region of SEQ ID NO: 152 share at least 50%, at least 55%, at least 60%, at least 65%, at least 70%, at least 75%, at least 80%, at least 85%, at least 90%, at least 95% sequence identity along the entire length of SEQ ID 152.

(269) Another aspect of the present disclosure relates to an antibody or binding portion thereof (e.g., a human antibody) that competes for binding to human tau protein with the anti-tau antibody comprising a heavy chain variable region with the amino acid sequence of SEQ ID NO: 152. In accordance with this aspect of the disclosure, a competitive binding assay, such as Bio-Layer Interferometry (BLI) can be utilized to identify an antibody or binding portion thereof that competes for binding to a tau protein with the enumerated monoclonal antibody. Other competitive

binding assays known in the art can also be utilized to identify a competitive binding antibody in accordance with this aspect of the disclosure.

(270) In another embodiment, the antibody or binding fragment thereof has a heavy chain variable region with a H-CDR1 having the amino acid sequence of SEQ ID NO: 202, or a modified amino acid sequence thereof containing 1, 2, 3, or more amino acid residue modifications as compared to SEQ ID NO: 202; a H-CDR2 having the amino acid sequence of SEQ ID NO: 203, or a modified amino acid sequence thereof containing 1, 2, 3, or more amino acid residue modifications as compared to SEQ ID NO: 203; and a H-CDR3 comprising the amino acid sequence of SEQ ID NO: 204, or a modified amino acid sequence thereof, said modified amino acid sequence containing 1, 2, 3, or more amino acid modifications as compared to SEQ ID NO: 204.

(271) An exemplary single domain antibody having this heavy chain variable region is referred to herein as the T-sR2-2B8 antibody. This antibody comprises a V.sub.H chain amino acid sequence of SEQ ID NO: 205 as shown below.

(272) TABLE-US-00040 SEQ ID NO: 205

QVQLAESGGGLVQAGGSLRLSCVVSGRFTSTSQMGWFRQPPGKERELVA
RISWRGKQHYADSVKGRFTISRDIYAKNTVYLMNGLKSEDTAVYYCAAD
RRRTYLGGQHDYWGQGTLVTVSS

(273) In another embodiment, the antibody or binding fragment thereof comprises a heavy chain variable region having an amino acid sequence that shares at least 60%, at least 65%, at least 70%, at least 75%, at least 80%, at least 85%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94% at least 95%, at least 96%, at least 97%, at least 98%, at least 99% sequence identity to SEQ ID NO: 205.

(274) In one embodiment, the antibody or binding fragment thereof of the present disclosure comprises a humanized variant of the heavy chain variable region of SEQ ID NO: 205, where the framework regions are humanized or replaced with human immunoglobulin framework sequences. As noted supra, suitable human or humanized framework sequences can be chosen based on their known structure, a consensus sequence, sequence homology to the framework sequences of donor antibody (e.g., the framework sequences of SEQ ID NO: 205), or a combination of these approaches. The humanized framework regions are designed to be similar in length and sequence to the parental framework sequences of SEQ ID NO: 205. In one embodiment, the humanized framework regions share 50%, 55%, 60%, 65%, 70%, 75%, 80%, 85%, 90%, 95% or more sequence identity to the framework regions of SEQ ID NO: 205. In another embodiment, the humanized framework regions are 60%, 65%, 70%, 75%, 80%, 85%, 90%, 95% or more similar in length to the framework regions of SEQ ID NO: 205. Humanized variants of the heavy chain variable region of SEQ ID NO: 205 share at least 50%, at least 55%, at least 60%, at least 65%, at least 70%, at least 75%, at least 80%, at least 85%, at least 90%, at least 95% sequence identity along the entire length of SEQ ID 205.

(275) Another aspect of the present disclosure relates to an antibody or binding portion thereof (e.g., a human antibody) that competes for binding to human tau protein with the anti-tau antibody comprising a heavy chain variable region with the amino acid sequence of SEQ ID NO: 205. In accordance with this aspect of the disclosure, a competitive binding assay, such as Bio-Layer Interferometry (BLI) can be utilized to identify an antibody or binding portion thereof that competes for binding to a tau protein with the enumerated monoclonal antibody. Other competitive binding assays known in the art can also be utilized to identify a competitive binding antibody in accordance with this aspect of the disclosure.

(276) In another embodiment, the antibody or binding fragment thereof has a heavy chain variable region with a H-CDR1 having the amino acid sequence of SEQ ID NO: 206, or a modified amino acid sequence thereof containing 1, 2, 3, or more amino acid residue modifications as compared to SEQ ID NO: 206; a H-CDR2 having the amino acid sequence of SEQ ID NO: 207, or a modified amino acid sequence thereof containing 1, 2, 3, or more amino acid residue modifications as

compared to SEQ ID NO: 207; and a H-CDR3 comprising the amino acid sequence of SEQ ID NO: 208, or a modified amino acid sequence thereof, said modified amino acid sequence containing 1, 2, 3, or more amino acid modifications as compared to SEQ ID NO: 208.

(277) An exemplary single domain antibody having this heavy chain variable region is referred to herein as the T-sR2-2F12 antibody. This antibody comprises a V.sub.H chain amino acid sequence of SEQ ID NO: 209 as shown below.

(278) TABLE-US-00041 SEQ ID NO: 209

QVQLQESGGGLVQAGGSLRLSCAASGSIFRINGMGWHRQAPGKERELVA
TITRGGSSISYADSVKGRFTISRDNKNTLYLQMNNLKPEDTAVYYCAKY
RRPLFYSGSNYREGDFASWGQGTQVTVSS

(279) In another embodiment, the antibody or binding fragment thereof comprises a heavy chain variable region having an amino acid sequence that shares at least 60%, at least 65%, at least 70%, at least 75%, at least 80%, at least 85%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94% at least 95%, at least 96%, at least 97%, at least 98%, at least 99% sequence identity to SEQ ID NO: 209.

(280) In one embodiment, the antibody or binding fragment thereof of the present disclosure comprises a humanized variant of the heavy chain variable region of SEQ ID NO: 209, where the framework regions are humanized or replaced with human immunoglobulin framework sequences. As noted supra, suitable human or humanized framework sequences can be chosen based on their known structure, a consensus sequence, sequence homology to the framework sequences of donor antibody (e.g., the framework sequences of SEQ ID NO: 209), or a combination of these approaches. The humanized framework regions are designed to be similar in length and sequence to the parental framework sequences of SEQ ID NO: 209. In one embodiment, the humanized framework regions share 50%, 55%, 60%, 65%, 70%, 75%, 80%, 85%, 90%, 95% or more sequence identity to the framework regions of SEQ ID NO: 209. In another embodiment, the humanized framework regions are 60%, 65%, 70%, 75%, 80%, 85%, 90%, 95% or more similar in length to the framework regions of SEQ ID NO: 209. Humanized variants of the heavy chain variable region of SEQ ID NO: 209 share at least 50%, at least 55%, at least 60%, at least 65%, at least 70%, at least 75%, at least 80%, at least 85%, at least 90%, at least 95% sequence identity along the entire length of SEQ ID 209.

(281) Another aspect of the present disclosure relates to an antibody or binding portion thereof (e.g., a human antibody) that competes for binding to human tau protein with the anti-tau antibody comprising a heavy chain variable region with the amino acid sequence of SEQ ID NO: 209. In accordance with this aspect of the disclosure, a competitive binding assay, such as Bio-competes for binding to a tau protein with the enumerated monoclonal antibody. Other competitive binding assays known in the art can also be utilized to identify a competitive binding antibody in accordance with this aspect of the disclosure.

(282) In another embodiment, the antibody or binding fragment thereof has a heavy chain variable region with a H-CDR1 having the amino acid sequence of SEQ ID NO: 210, or a modified amino acid sequence thereof containing 1, 2, 3, or more amino acid residue modifications as compared to SEQ ID NO: 210; a H-CDR2 having the amino acid sequence of SEQ ID NO: 211, or a modified amino acid sequence thereof containing 1, 2, 3, or more amino acid residue modifications as compared to SEQ ID NO: 211; and a H-CDR3 comprising the amino acid sequence of SEQ ID NO: 212, or a modified amino acid sequence thereof, said modified amino acid sequence containing 1, 2, 3, or more amino acid modifications as compared to SEQ ID NO: 212.

(283) An exemplary single domain antibody having this heavy chain variable region is referred to herein as the T-sR2-2B3 antibody. This antibody comprises a V.sub.H chain amino acid sequence of SEQ ID NO: 213 as shown below.

(284) TABLE-US-00042 SEQ ID NO: 213

QVQLQASGGGLVQAGGSLRLSCAASGRFTFGSYTMGWFRQAPGKEREFVA

AI SRSGSTY YADSVKGRFTISRDN AKNTVYLQ MNSLKPEDTAVYYCNV
RGRPFILSKPFDSWGQGT LVT VSS

(285) In another embodiment, the antibody or binding fragment thereof comprises a heavy chain variable region having an amino acid sequence that shares at least 60%, at least 65%, at least 70%, at least 75%, at least 80%, at least 85%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94% at least 95%, at least 96%, at least 97%, at least 98%, at least 99% sequence identity to SEQ ID NO: 213.

(286) In one embodiment, the antibody or binding fragment thereof of the present disclosure comprises a humanized variant of the heavy chain variable region of SEQ ID NO: 213, where the framework regions are humanized or replaced with human immunoglobulin framework sequences. As noted supra, suitable human or humanized framework sequences can be chosen based on their known structure, a consensus sequence, sequence homology to the framework sequences of donor antibody (e.g., the framework sequences of SEQ ID NO: 213), or a combination of these approaches. The humanized framework regions are designed to be similar in length and sequence to the parental framework sequences of SEQ ID NO: 213. In one embodiment, the humanized framework regions share 50%, 55%, 60%, 65%, 70%, 75%, 80%, 85%, 90%, 95% or more sequence identity to the framework regions of SEQ ID NO: 213. In another embodiment, the humanized framework regions are 60%, 65%, 70%, 75%, 80%, 85%, 90%, 95% or more similar in length to the framework regions of SEQ ID NO: 213. Humanized variants of the heavy chain variable region of SEQ ID NO: 213 share at least 50%, at least 55%, at least 60%, at least 65%, at least 70%, at least 75%, at least 80%, at least 85%, at least 90%, at least 95% sequence identity along the entire length of SEQ ID 213.

(287) Another aspect of the present disclosure relates to an antibody or binding portion thereof (e.g., a human antibody) that competes for binding to human tau protein with the anti-tau antibody comprising a heavy chain variable region with the amino acid sequence of SEQ ID NO: 213. In accordance with this aspect of the disclosure, a competitive binding assay, such as Bio-Layer Interferometry (BLI) can be utilized to identify an antibody or binding portion thereof that competes for binding to a tau protein with the enumerated monoclonal antibody. Other competitive binding assays known in the art can also be utilized to identify a competitive binding antibody in accordance with this aspect of the disclosure.

(288) In another embodiment, the antibody or binding fragment thereof has a heavy chain variable region with a H-CDR1 having the amino acid sequence of SEQ ID NO: 214, or a modified amino acid sequence thereof containing 1, 2, 3, or more amino acid residue modifications as compared to SEQ ID NO: 214; a H-CDR2 having the amino acid sequence of SEQ ID NO: 215, or a modified amino acid sequence thereof containing 1, 2, 3, or more amino acid residue modifications as compared to SEQ ID NO: 215; and a H-CDR3 comprising the amino acid sequence of SEQ ID NO: 216, or a modified amino acid sequence thereof, said modified amino acid sequence containing 1, 2, 3, or more amino acid modifications as compared to SEQ ID NO: 216.

(289) An exemplary single domain antibody having this heavy chain variable region is referred to herein as the T-sR2-2C4 antibody. This antibody comprises a V.sub.H chain amino acid sequence of SEQ ID NO: 217 as shown below.

(290) TABLE-US-00043 SEQ ID NO: 217

QVQLQESGGGLVQAGNSLR LSCAASGR TFSNYAMGWFRQAPGKEREFVA
AITRNGGITYYAESVKGRFTISRDN AKNMVYLQ MNSLKPEDTAVYYCNI
KARRGSFFNPVNNYWGQGTQVT VSS

(291) In another embodiment, the antibody or binding fragment thereof comprises a heavy chain variable region having an amino acid sequence that shares at least 60%, at least 65%, at least 70%, at least 75%, at least 80%, at least 85%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94% at least 95%, at least 96%, at least 97%, at least 98%, at least 99% sequence identity to SEQ ID NO: 217.

(292) In one embodiment, the antibody or binding fragment thereof of the present disclosure comprises a humanized variant of the heavy chain variable region of SEQ ID NO: 217, where the framework regions are humanized or replaced with human immunoglobulin framework sequences. As noted supra, suitable human or humanized framework sequences can be chosen based on their known structure, a consensus sequence, sequence homology to the framework sequences of donor antibody (e.g., the framework sequences of SEQ ID NO: 217), or a combination of these approaches. The humanized framework regions are designed to be similar in length and sequence to the parental framework sequences of SEQ ID NO: 217. In one embodiment, the humanized framework regions share 50%, 55%, 60%, 65%, 70%, 75%, 80%, 85%, 90%, 95% or more sequence identity to the framework regions of SEQ ID NO: 217. In another embodiment, the humanized framework regions are 60%, 65%, 70%, 75%, 80%, 85%, 90%, 95% or more similar in length to the framework regions of SEQ ID NO: 217. Humanized variants of the heavy chain variable region of SEQ ID NO: 217 share at least 50%, at least 55%, at least 60%, at least 65%, at least 70%, at least 75%, at least 80%, at least 85%, at least 90%, at least 95% sequence identity along the entire length of SEQ ID 217.

(293) Another aspect of the present disclosure relates to an antibody or binding portion thereof (e.g., a human antibody) that competes for binding to human tau protein with the anti-tau antibody comprising a heavy chain variable region with the amino acid sequence of SEQ ID NO: 217. In accordance with this aspect of the disclosure, a competitive binding assay, such as Bio-Layer Interferometry (BLI) can be utilized to identify an antibody or binding portion thereof that competes for binding to a tau protein with the enumerated monoclonal antibody. Other competitive binding assays known in the art can also be utilized to identify a competitive binding antibody in accordance with this aspect of the disclosure.

(294) In another embodiment, the antibody or binding fragment thereof as disclosed herein comprises a heavy chain variable region having a H-CDR1 with an amino acid sequence selected from SEQ ID NOs: 153-164, 307, or a modified amino acid sequence thereof containing 1, 2, 3, 4, 5, 7, 8, or 9 amino acid residue modifications as compared to SEQ ID NOs: 153-164 and 307 that maintain or enhance binding specificity of the H-CDR1. In one embodiment, the amino acid sequence of the H-CDR1 contains no more than 1, 2, or 3 amino acid modifications as compared to SEQ ID NOs: 153-164 and 307, respectively. The HCVR further comprises a H-CDR2 with an amino acid sequence selected from SEQ ID NOs: 165-176, 308 or a modified amino acid sequence thereof containing 1, 2, 3, 4, 5, 6, 7, 8, 9, or 10, amino acid residue modifications as compared to SEQ ID NOs: 165-176 and 308 that maintain or enhance binding specificity of the H-CDR2. In one embodiment, the amino acid sequence of the H-CDR2 contains no more than 1, 2, or 3, amino acid modifications as compared to SEQ ID NOs: 165-176 and 308, respectively. The HCVR of the antibody or binding fragment thereof comprises a H-CDR3 with an amino acid sequence selected from SEQ ID NOs: 177-188, 309 or a modified amino acid sequence thereof containing 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, or 11 amino acid residue modifications as compared to SEQ ID NOs: 177-188 and 309 that maintain or enhance binding specificity of the H-CDR3. In one embodiment, the amino acid sequence of the H-CDR3 contains no more than 1, 2, or 3 amino acid modifications as compared to SEQ ID NOs: 177-188 and 309, respectively. The amino acid sequences of SEQ ID NOs: 153-188 and 307-309 are provided in Table 2 below.

(295) TABLE-US-00044 TABLE 2 Tau Single Domain Antibody Complementarity Determining Regions SEQ SEQ SEQ Antibody ID ID ID Description descr sequence NO: descr sequence NO: 01-T-bR3-1E2; CDR1 SGRIFSNNVM 153 CDR2 AISRSGGTTLYA 165 CDR3 GKVDEIRPTVSASYD 177 02-T-bR3- 1D10; 07-T- bR3-1C1; 09- T-bR3-1E4; 10- T-bR3-1D11; 14-T-bR3-1B1; 15-T-bR3-1C2; 17-T-bR3-1C5; 23-T-bR3-1C3; 25-T-bR3-1C6; 31-T-bR3-1C4; 33-T-bR3-1C9; 41-T-bR3- 1C10; 46-T- bR3-1B5; 49- T-bR3-1C12; 57-T-bR3-1D1; 65-T-bR3-1D5; 70-T-bR3-1B9; 73-T-bR3-1D6; 77-T-bR3-1A8; 81-T-bR3-1D7; 89-T-bR3-1D8; 93-T-bR3- 1A10; 94-T- bR3-1B12 06-T-bR3- CDR1 SSGPFSRYAM 154

CDR2 TISRSGSTTYG 166 CDR3 AVSRYYTAGASADTKTYD 178 1A11 08-T-bR3- CDR1 SGFNFGSFAI 155 CDR2 CISSTDDTSVYS 167 CDR3 ATVGQSCDLWDHPQVPVRYR 179 1E12; 16-T- GR bR3-1F8; 30-T- bR3-1B3; 39- T-bR3-1C7; 78-T-bR3- 1B10; 80-T- bR3-1H11; 85- T-bR3-1A9; 38T-bR3-1B4 22-T-bR3-1B2 CDR1 SGSISSINAI 156 CDR2 KITKGGTTIYT 168 CDR3 GADVNYGSPDYID 180 24-T-bR3- CDR1 SGRTFSTYTM 157 CDR2 AISWSSGTANYA 169 CDR3 NAWSPVGHD 181 1F10; 48-T- bR3-1G5 32-T-bR3-1F12 CDR1 SGRTFSNYAM 158 CDR2 AISWSGAYTFYG 170 CDR3 AAARGGRWYSTYD 182 40-T-bR3-1G3 CDR1 SGSTFSGYVI 159 CDR2 IISSSGSTNYA 171 CDR3 NARLWLNN 183 47-T-bR3-1C8 CDR1 SGFTFSSYWM 160 CDR2 TISPSGGTTYT 172 CDR3 AARRSGRYEYD 184 62-T-bR3-1B8; CDR1 SGFPFDDYPM 161 CDR2 SVSPNGGSTFYA 173 CDR3 AKVLDYYCSGYGICYASYD 185 71-T-bR3- 1D12 79-T-bR3-1E1 CDR1 SISIVSINTM 162 CDR2 GITSGGSTNYA 174 CDR3 NAGRYVPGAIVTN 186 86-T-bR3- CDR1 SMTTLGFKTM 163 CDR2 RISSGGQTNIA 175 CDR3 NARRYYSRLARYDYN 187 1B11 95-T-bR3-1E11 CDR1 GGGKSTAAVNGV 164 CDR2 AIRPGGRRDYL 176 CDR3 YAEGLLLPST 188 55-T-bR3-1D4 CDR1 SGRTFSGYS 307 CDR2 VGGISRSGGWT 308 CDR3 AAASRINDYAPALSRAYD 309

(296) In one embodiment, the antibody or binding fragment thereof has a heavy chain variable region with a H-CDR1 having the amino acid sequence of SEQ ID NO: 153, or a modified amino acid sequence thereof containing 1, 2, 3, or more amino acid residue modifications as compared to SEQ ID NO: 153; a H-CDR2 having the amino acid sequence of SEQ ID NO: 165, or a modified amino acid sequence thereof containing 1, 2, 3, or more amino acid residue modifications as compared to SEQ ID NO: 165; and a H-CDR3 comprising the amino acid sequence of SEQ ID NO: 177, or a modified amino acid sequence thereof, said modified amino acid sequence containing 1, 2, 3, or more amino acid modifications as compared to SEQ ID NO: 177.

(297) An exemplary single domain antibody having this heavy chain variable region is referred to herein as the 01-T-bR3-1E2 antibody. The 01-T-bR3-1E2 antibody comprises a V.sub.H chain amino acid sequence of SE ID NO: 189 as shown below.

(298) TABLE-US-00045 SEQ ID NO: 189

QVQLQESGGGLVQAGGSLRLSCAASGRIFSNVMGWFRQAPGKEREFVA
AISRSGGTTLYAESMKGRFTISRDNKNTLYLQMNSLKSEDTAMYYCGK
VDEIRPTVSASYDLWGQGTLVTVSS

(299) Other single domain antibodies described herein having the same V.sub.H chain amino acid sequence of SEQ ID NO: 189 include 02-T-bR3-1D10; 07-T-bR3-1C1; 09-T-bR3-1E4; 10-T-bR3-1D11; 14-T-bR3-1B1; 15-T-bR3-1C2; 17-T-bR3-1C5; 23-T-bR3-1C3; 25-T-bR3-1C6; 31-T-bR3-1C4; 33-T-bR3-1C9; 41-T-bR3-1C10; 46-T-bR3-1B5; 49-T-bR3-1C12; 57-T-bR3-1D1; 65-T-bR3-1D5; 70-T-bR3-1B9; 73-T-bR3-1D6; 77-T-bR3-1A8; 81-T-bR3-1D7; 89-T-bR3-1D8; 93-T-bR3-1A10; and 94-T-bR3-1B12.

(300) In another embodiment, the antibody or binding fragment thereof comprises a heavy chain variable region having an amino acid sequence that shares at least 60%, at least 65%, at least 70%, at least 75%, at least 80%, at least 85%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94% at least 95%, at least 96%, at least 97%, at least 98%, at least 99% sequence identity to SEQ ID NO: 189.

(301) In one embodiment, the antibody or binding fragment thereof of the present disclosure comprises a humanized variant of the heavy chain variable region of SEQ ID NO: 189, where the framework regions are humanized or replaced with human immunoglobulin framework sequences. As noted supra, suitable human or humanized framework sequences can be chosen based on their known structure, a consensus sequence, sequence homology to the framework sequences of donor antibody (e.g., the framework sequences of SEQ ID NO: 189), or a combination of these approaches. The humanized framework regions are designed to be similar in length and sequence to the parental framework sequences of SEQ ID NO: 189. In one embodiment, the humanized framework regions share 50%, 55%, 60%, 65%, 70%, 75%, 80%, 85%, 90%, 95% or more

sequence identity to the framework regions of SEQ ID NO: 189. In another embodiment, the humanized framework regions are 60%, 65%, 70%, 75%, 80%, 85%, 90%, 95% or more similar in length to the framework regions of SEQ ID NO: 189. Humanized variants of the heavy chain variable region of SEQ ID NO: 189 share at least 50%, at least 55%, at least 60%, at least 65%, at least 70%, at least 75%, at least 80%, at least 85%, at least 90%, at least 95% sequence identity along the entire length of SEQ ID 189.

(302) Another aspect of the present disclosure relates to an antibody or binding portion thereof (e.g., a human antibody) that competes for binding to human tau protein with the anti-tau antibody comprising a heavy chain variable region with the amino acid sequence of SEQ ID NO: 189. In accordance with this aspect of the disclosure, a competitive binding assay, such as Bio-Layer Interferometry (BLI) can be utilized to identify an antibody or binding portion thereof that competes for binding to a tau protein with the enumerated monoclonal antibody. Other competitive binding assays known in the art can also be utilized to identify a competitive binding antibody in accordance with this aspect of the disclosure.

(303) In one embodiment, the antibody or binding fragment thereof has a heavy chain variable region with a H-CDR1 having the amino acid sequence of SEQ ID NO: 154, or a modified amino acid sequence thereof containing 1, 2, 3, or more amino acid residue modifications as compared to SEQ ID NO: 154; a H-CDR2 having the amino acid sequence of SEQ ID NO: 166, or a modified amino acid sequence thereof containing 1, 2, 3, or more amino acid residue modifications as compared to SEQ ID NO: 166; and a H-CDR3 comprising the amino acid sequence of SEQ ID NO: 178, or a modified amino acid sequence thereof, said modified amino acid sequence containing 1, 2, 3, or more amino acid modifications as compared to SEQ ID NO: 178.

(304) An exemplary single domain antibody having this combination of heavy chain variable region CDRs is referred to herein as the 06-T-bR3-1A11 antibody. The 06-T-bR3-1A11 antibody comprises a V.sub.H chain amino acid sequence of SEQ ID NO: 190 as shown below.

(305) TABLE-US-00046 SEQ ID NO: 190

QVQLQESGGGLVQAGDSLRLSCAASSGPFSTRYAMGWFRQAPGKEREFVA
TISRSGSSTTYGDSVKGRFTISRDNKNTLYLEMNSLTPEDTAVYYCAV
SRYRTAGASADTKTYDYWGQGTLVTVSS

(306) In another embodiment, the antibody or binding fragment thereof comprises a heavy chain variable region having an amino acid sequence that shares at least 60%, at least 65%, at least 70%, at least 75%, at least 80%, at least 85%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94% at least 95%, at least 96%, at least 97%, at least 98%, at least 99% sequence identity to SEQ ID NO: 190.

(307) In one embodiment, the antibody or binding fragment thereof of the present disclosure comprises a humanized variant of the heavy chain variable region of SEQ ID NO: 190, where the framework regions are humanized or replaced with human immunoglobulin framework sequences. As noted supra, suitable human or humanized framework sequences can be chosen based on their known structure, a consensus sequence, sequence homology to the framework sequences of donor antibody (e.g., the framework sequences of SEQ ID NO: 190), or a combination of these approaches. The humanized framework regions are designed to be similar in length and sequence to the parental framework sequences of SEQ ID NO: 190. In one embodiment, the humanized framework regions share 50%, 55%, 60%, 65%, 70%, 75%, 80%, 85%, 90%, 95% or more sequence identity to the framework regions of SEQ ID NO: 190. In another embodiment, the humanized framework regions are 60%, 65%, 70%, 75%, 80%, 85%, 90%, 95% or more similar in length to the framework regions of SEQ ID NO: 190. Humanized variants of the heavy chain variable region of SEQ ID NO: 190 share at least 50%, at least 55%, at least 60%, at least 65%, at least 70%, at least 75%, at least 80%, at least 85%, at least 90%, at least 95% sequence identity along the entire length of SEQ ID NO: 190, respectively.

(308) Another aspect of the present disclosure relates to an antibody or binding portion thereof

(e.g., a human antibody) that competes for binding to human tau protein with the anti-tau antibody comprising a heavy chain variable region with the amino acid sequence of SEQ ID NO: 190. In accordance with this aspect of the disclosure, a competitive binding assay, such as Bio-Layer Interferometry (BLI) can be utilized to identify an antibody or binding portion thereof that competes for binding to a tau protein with the enumerated monoclonal antibody. Other competitive binding assays known in the art can also be utilized to identify a competitive binding antibody in accordance with this aspect of the disclosure.

(309) In one embodiment, the antibody or binding fragment thereof has a heavy chain variable region with a H-CDR1 having the amino acid sequence of SEQ ID NO: 155, or a modified amino acid sequence thereof containing 1, 2, 3, or more amino acid residue modifications as compared to SEQ ID NO: 155; a H-CDR2 having the amino acid sequence of SEQ ID NO: 167, or a modified amino acid sequence thereof containing 1, 2, 3, or more amino acid residue modifications as compared to SEQ ID NO: 167; and a H-CDR3 comprising the amino acid sequence of SEQ ID NO: 179, or a modified amino acid sequence thereof, said modified amino acid sequence containing 1, 2, 3, or more amino acid modifications as compared to SEQ ID NO: 179.

(310) An exemplary single domain antibody having this combination of heavy chain variable region CDRs is referred to herein as the 08-T-bR3-1E12 antibody. The 08-T-bR3-1E12 antibody comprises a V.sub.H chain amino acid sequence of SEQ ID NO: 191 as shown below.

(311) TABLE-US-00047 SEQ ID NO: 191

QVQLQASGGGLVQPGGSLKLSCQASGFNFGSFAIAWFRQAPGKGREGIS
CISSTDDTSVYSDAVKGRFAISRDNAAAYLQMNSLIPEDTANYYCAT
VGQSCDLWDHPQVPVRYRGRGTLTVSS

(312) Another exemplary single domain antibody having this combination of heavy chain variable region CDRs is referred to herein as the 16-T-bR3-1F8 antibody. The 16-T-bR3-1F8 antibody comprises a V.sub.H chain amino acid sequence of SEQ ID NO: 192 as shown below.

(313) TABLE-US-00048 SEQ ID NO: 192

QVQLQASGGGLVQPGGSLKLSCQASGFNFGSFAIAWFRQAPGKGREGIS
CISSTDDTSVYSDAVKGRFAISRDNAAAYLQMNSLIPEDTANYYCAT
VGQSCDLWDHPQVPVRYRGRGTQVTSS

(314) Other antibodies of the disclosure comprising the amino acid sequence of SEQ ID NO: 192 include 30-T-bR3-1B3; 39-T-bR3-1C7; 78-T-bR3-1B10; 80-T-bR3-1H11; 85-T-bR3-1A9; and 38T-bR3-1B4.

(315) In another embodiment, the antibody or binding fragment thereof comprises a heavy chain variable region having an amino acid sequence that shares at least 60%, at least 65%, at least 70%, at least 75%, at least 80%, at least 85%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94% at least 95%, at least 96%, at least 97%, at least 98%, at least 99% sequence identity to SEQ ID NO: 191 or SEQ ID NO: 192.

(316) In one embodiment, the antibody or binding fragment thereof of the present disclosure comprises a humanized variant of the heavy chain variable region of SEQ ID NO: 191 or SEQ ID NO: 192, where the framework regions are humanized or replaced with human immunoglobulin framework sequences. As noted supra, suitable human or humanized framework sequences can be chosen based on their known structure, a consensus sequence, sequence homology to the framework sequences of donor antibody (e.g., the framework sequences of SEQ ID NO: 191 or SEQ ID NO: 192), or a combination of these approaches. The humanized framework regions are designed to be similar in length and sequence to the parental framework sequences of SEQ ID NO: 191 or SEQ ID NO: 192. In one embodiment, the humanized framework regions share 50%, 55%, 60%, 65%, 70%, 75%, 80%, 85%, 90%, 95% or more sequence identity to the framework regions of SEQ ID NO: 191 or SEQ ID NO: 192. In another embodiment, the humanized framework regions are 60%, 65%, 70%, 75%, 80%, 85%, 90%, 95% or more similar in length to the framework regions of SEQ ID NO: 191 or SEQ ID NO: 192. Humanized variants of the heavy

chain variable region of SEQ ID NO: 191 or SEQ ID NO: 192 share at least 50%, at least 55%, at least 60%, at least 65%, at least 70%, at least 75%, at least 80%, at least 85%, at least 90%, at least 95% sequence identity along the entire length of SEQ ID NO: 191 or SEQ ID NO: 192, respectively.

(317) Another aspect of the present disclosure relates to an antibody or binding portion thereof (e.g., a human antibody) that competes for binding to human tau protein with the anti-tau antibody comprising a heavy chain variable region with the amino acid sequence of SEQ ID NO: 191 or SEQ ID NO: 192. In accordance with this aspect of the disclosure, a competitive binding assay, such as Bio-Layer Interferometry (BLI) can be utilized to identify an antibody or binding portion thereof that competes for binding to a tau protein with the enumerated monoclonal antibody. Other competitive binding assays known in the art can also be utilized to identify a competitive binding antibody in accordance with this aspect of the disclosure.

(318) In one embodiment, the antibody or binding fragment thereof has a heavy chain variable region with a H-CDR1 having the amino acid sequence of SEQ ID NO: 156, or a modified amino acid sequence thereof containing 1, 2, 3, or more amino acid residue modifications as compared to SEQ ID NO: 156; a H-CDR2 having the amino acid sequence of SEQ ID NO: 168, or a modified amino acid sequence thereof containing 1, 2, 3, or more amino acid residue modifications as compared to SEQ ID NO: 168; and a H-CDR3 comprising the amino acid sequence of SEQ ID NO: 180, or a modified amino acid sequence thereof, said modified amino acid sequence containing 1, 2, 3, or more amino acid modifications as compared to SEQ ID NO: 180.

(319) An exemplary single domain antibody having this heavy chain variable region is referred to herein as the 22-T-bR3-1B2 antibody. The 22-T-bR3-1B2 antibody comprises a V.sub.H chain amino acid sequence of SEQ ID NO: 193 as shown below.

(320) TABLE-US-00049 SEQ ID NO: 193

QVQLQESGGGFVQAGTSLRLSCAASGSISSINAI AWYRQGRGNQRELLA
KITKGGTTIYTNSVKGRFTISRDNKNTVYLQMDSLKPDDTAVYYCGAD
VNYGSPDYIDYWGQGTVTVSS

(321) In another embodiment, the antibody or binding fragment thereof comprises a heavy chain variable region having an amino acid sequence that shares at least 6000, at least 65%, at least 70%, at least 75%, at least 80%, at least 85%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94% at least 95%, at least 96%, at least 97%, at least 98%, at least 99% sequence identity to SEQ ID NO: 193.

(322) In one embodiment, the antibody or binding fragment thereof of the present disclosure comprises a humanized variant of the heavy chain variable region of SEQ ID NO: 193, where the framework regions are humanized or replaced with human immunoglobulin framework sequences. As noted supra, suitable human or humanized framework sequences can be chosen based on their known structure, a consensus sequence, sequence homology to the framework sequences of donor antibody (e.g., the framework sequences of SEQ ID NO: 193), or a combination of these approaches. The humanized framework regions are designed to be similar in length and sequence to the parental framework sequences of SEQ ID NO: 193. In one embodiment, the humanized framework regions share 50%, 55%, 60%, 65%, 70%, 75%, 80%, 85%, 90%, 95% or more sequence identity to the framework regions of SEQ ID NO: 193. In another embodiment, the humanized framework regions are 60%, 65%, 70%, 75%, 80%, 85%, 90%, 95% or more similar in length to the framework regions of SEQ ID NO: 193. Humanized variants of the heavy chain variable region of SEQ ID NO: 193 share at least 50%, at least 55%, at least 60%, at least 65%, at least 70%, at least 75%, at least 80%, at least 85%, at least 90%, at least 95% sequence identity along the entire length of SEQ ID 193.

(323) Another aspect of the present disclosure relates to an antibody or binding portion thereof (e.g., a human antibody) that competes for binding to human tau protein with the anti-tau antibody comprising a heavy chain variable region with the amino acid sequence of SEQ ID NO: 193. In

accordance with this aspect of the disclosure, a competitive binding assay, such as Bio-Layer Interferometry (BLI) can be utilized to identify an antibody or binding portion thereof that competes for binding to a tau protein with the enumerated monoclonal antibody. Other competitive binding assays known in the art can also be utilized to identify a competitive binding antibody in accordance with this aspect of the disclosure.

(324) In one embodiment, the antibody or binding fragment thereof has a heavy chain variable region with a H-CDR1 having the amino acid sequence of SEQ ID NO: 157, or a modified amino acid sequence thereof containing 1, 2, 3, or more amino acid residue modifications as compared to SEQ ID NO: 157; a H-CDR2 having the amino acid sequence of SEQ ID NO: 169, or a modified amino acid sequence thereof containing 1, 2, 3, or more amino acid residue modifications as compared to SEQ ID NO: 169; and a H-CDR3 comprising the amino acid sequence of SEQ ID NO: 181, or a modified amino acid sequence thereof, said modified amino acid sequence containing 1, 2, 3, or more amino acid modifications as compared to SEQ ID NO: 181.

(325) Exemplary single domain antibodies having this heavy chain variable region are referred to herein as the 24-T-bR3-1F10 and 48-T-bR3-1G5 antibodies. These antibodies comprise a V.sub.H chain amino acid sequence of SEQ ID NO: 194 as shown below.

(326) TABLE-US-00050 SEQ ID NO: 194

QVQLQESGGGLVQSGGSLRLSCAASGRTFSTYTMGWFRQAPGKEREFVA
AISWSSGTANYADSVKGRFTISRDSA KNTVYLQMNSLKPEDTAVYYCNA
WSPVGHDIYWGQGTQVTVSS

(327) In another embodiment, the antibody or binding fragment thereof comprises a heavy chain variable region having an amino acid sequence that shares at least 60%, at least 65%, at least 70%, at least 75%, at least 80%, at least 85%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94% at least 95%, at least 96%, at least 97%, at least 98%, at least 99% sequence identity to SEQ ID NO: 194

(328) In one embodiment, the antibody or binding fragment thereof of the present disclosure comprises a humanized variant of the heavy chain variable region of SEQ ID NO: 194, where the framework regions are humanized or replaced with human immunoglobulin framework sequences. As noted supra, suitable human or humanized framework sequences can be chosen based on their known structure, a consensus sequence, sequence homology to the framework sequences of donor antibody (e.g., the framework sequences of SEQ ID NO: 194), or a combination of these approaches. The humanized framework regions are designed to be similar in length and sequence to the parental framework sequences of SEQ ID NO: 194. In one embodiment, the humanized framework regions share 50%, 55%, 60%, 65%, 70%, 75%, 80%, 85%, 90%, 95% or more sequence identity to the framework regions of SEQ ID NO: 194. In another embodiment, the humanized framework regions are 60%, 65%, 70%, 75%, 80%, 85%, 90%, 95% or more similar in length to the framework regions of SEQ ID NO: 194. Humanized variants of the heavy chain variable region of SEQ ID NO: 194 share at least 50%, at least 55%, at least 60%, at least 65%, at least 70%, at least 75%, at least 80%, at least 85%, at least 90%, at least 95% sequence identity along the entire length of SEQ ID NO: 194, respectively.

(329) Another aspect of the present disclosure relates to an antibody or binding portion thereof (e.g., a human antibody) that competes for binding to human tau protein with the anti-tau antibody comprising a heavy chain variable region with the amino acid sequence of SEQ ID NO: 194. In accordance with this aspect of the disclosure, a competitive binding assay, such as Bio-Layer Interferometry (BLI) can be utilized to identify an antibody or binding portion thereof that competes for binding to a tau protein with the enumerated monoclonal antibody. Other competitive binding assays known in the art can also be utilized to identify a competitive binding antibody in accordance with this aspect of the disclosure.

(330) In one embodiment, the antibody or binding fragment thereof has a heavy chain variable region with a H-CDR1 having the amino acid sequence of SEQ ID NO: 158, or a modified amino

acid sequence thereof containing 1, 2, 3, or more amino acid residue modifications as compared to SEQ ID NO: 158; a H-CDR2 having the amino acid sequence of SEQ ID NO: 170, or a modified amino acid sequence thereof containing 1, 2, 3, or more amino acid residue modifications as compared to SEQ ID NO: 170; and a H-CDR3 comprising the amino acid sequence of SEQ ID NO: 182, or a modified amino acid sequence thereof, said modified amino acid sequence containing 1, 2, 3, or more amino acid modifications as compared to SEQ ID NO: 182.

(331) An exemplary single domain antibody having this heavy chain variable region is referred to herein as the 32-T-bR3-1F12 antibody. The 32-T-bR3-1F12 antibody comprises a V.sub.H chain amino acid sequence of SEQ ID NO: 195 as shown below.

(332) TABLE-US-00051 SEQ ID NO: 195

QVQLQQSGGGLVQAGGSLRLSCAASGRTFSNYAMGWFRQAPGKEREFVA
AISWSGAYTFYGNVSRGRFTISRDNPNNTVYLQMNSLKPEDTATYYCAA
ARGGRWYSTYDYWGQGTQVTVSS

(333) In another embodiment, the antibody or binding fragment thereof comprises a heavy chain variable region having an amino acid sequence that shares at least 60%, at least 65%, at least 70%, at least 75%, at least 80%, at least 85%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94% at least 95%, at least 96%, at least 97%, at least 98%, at least 99% sequence identity to SEQ ID NO: 195.

(334) In one embodiment, the antibody or binding fragment thereof of the present disclosure comprises a humanized variant of the heavy chain variable region of SEQ ID NO: 195, where the framework regions are humanized or replaced with human immunoglobulin framework sequences. As noted supra, suitable human or humanized framework sequences can be chosen based on their known structure, a consensus sequence, sequence homology to the framework sequences of donor antibody (e.g., the framework sequences of SEQ ID NO: 195), or a combination of these approaches. The humanized framework regions are designed to be similar in length and sequence to the parental framework sequences of SEQ ID NO: 195. In one embodiment, the humanized framework regions share 50%, 55%, 60%, 65%, 70%, 75%, 80%, 85%, 90%, 95% or more sequence identity to the framework regions of SEQ ID NO: 195. In another embodiment, the humanized framework regions are 60%, 65%, 70%, 75%, 80%, 85%, 90%, 95% or more similar in length to the framework regions of SEQ ID NO: 195. Humanized variants of the heavy chain variable region of SEQ ID NO: 195 share at least 50%, at least 55%, at least 60%, at least 65%, at least 70%, at least 75%, at least 80%, at least 85%, at least 90%, at least 95% sequence identity along the entire length of SEQ ID 195.

(335) Another aspect of the present disclosure relates to an antibody or binding portion thereof (e.g., a human antibody) that competes for binding to human tau protein with the anti-tau antibody comprising a heavy chain variable region with the amino acid sequence of SEQ ID NO: 195. In accordance with this aspect of the disclosure, a competitive binding assay, such as Bio-Layer Interferometry (BLI) can be utilized to identify an antibody or binding portion thereof that competes for binding to a tau protein with the enumerated monoclonal antibody. Other competitive binding assays known in the art can also be utilized to identify a competitive binding antibody in accordance with this aspect of the disclosure.

(336) In one embodiment, the antibody or binding fragment thereof has a heavy chain variable region with a H-CDR1 having the amino acid sequence of SEQ ID NO: 159, or a modified amino acid sequence thereof containing 1, 2, 3, or more amino acid residue modifications as compared to SEQ ID NO: 159; a H-CDR2 having the amino acid sequence of SEQ ID NO: 171, or a modified amino acid sequence thereof containing 1, 2, 3, or more amino acid residue modifications as compared to SEQ ID NO: 171; and a H-CDR3 comprising the amino acid sequence of SEQ ID NO: 183, or a modified amino acid sequence thereof, said modified amino acid sequence containing 1, 2, 3, or more amino acid modifications as compared to SEQ ID NO: 183.

(337) An exemplary single domain antibody having this heavy chain variable region is referred to

herein as the 40-T-bR3-1G3 antibody. The 40-T-bR3-1G3 antibody comprises a V.sub.H chain amino acid sequence of SEQ ID NO: 196 as shown below.

(338) TABLE-US-00052 SEQ ID NO: 196

QVQLQESGGGLVQAGGSLRLSCAASGSTFSGYVIGWYRQAPGKQREEVA
IISSSGSTNYADSVKGRFTISRDNANAKTTFYLQMNSLKPEDTAVYYCN
ARLWLNYYWGQGTLLVTVSS

(339) In another embodiment, the antibody or binding fragment thereof comprises a heavy chain variable region having an amino acid sequence that shares at least 60%, at least 65%, at least 70%, at least 75%, at least 80%, at least 85%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94% at least 95%, at least 96%, at least 97%, at least 98%, at least 99% sequence identity to SEQ ID NO: 196.

(340) In one embodiment, the antibody or binding fragment thereof of the present disclosure comprises a humanized variant of the heavy chain variable region of SEQ ID NO: 196, where the framework regions are humanized or replaced with human immunoglobulin framework sequences. As noted supra, suitable human or humanized framework sequences can be chosen based on their known structure, a consensus sequence, sequence homology to the framework sequences of donor antibody (e.g., the framework sequences of SEQ ID NO: 196), or a combination of these approaches. The humanized framework regions are designed to be similar in length and sequence to the parental framework sequences of SEQ ID NO: 196. In one embodiment, the humanized framework regions share 50%, 55%, 60%, 65%, 70%, 75%, 80%, 85%, 90%, 95% or more sequence identity to the framework regions of SEQ ID NO: 196. In another embodiment, the humanized framework regions are 60%, 65%, 70%, 75%, 80%, 85%, 90%, 95% or more similar in length to the framework regions of SEQ ID NO: 196. Humanized variants of the heavy chain variable region of SEQ ID NO: 196 share at least 50%, at least 55%, at least 60%, at least 65%, at least 70%, at least 75%, at least 80%, at least 85%, at least 90%, at least 95% sequence identity along the entire length of SEQ ID 196.

(341) Another aspect of the present disclosure relates to an antibody or binding portion thereof (e.g., a human antibody) that competes for binding to human tau protein with the anti-tau antibody comprising a heavy chain variable region with the amino acid sequence of SEQ ID NO: 196. In accordance with this aspect of the disclosure, a competitive binding assay, such as Bio-Layer Interferometry (BLI) can be utilized to identify an antibody or binding portion thereof that competes for binding to a tau protein with the enumerated monoclonal antibody. Other competitive binding assays known in the art can also be utilized to identify a competitive binding antibody in accordance with this aspect of the disclosure.

(342) In one embodiment, the antibody or binding fragment thereof has a heavy chain variable region with a H-CDR1 having the amino acid sequence of SEQ ID NO: 160, or a modified amino acid sequence thereof containing 1, 2, 3, or more amino acid residue modifications as compared to SEQ ID NO: 160; a H-CDR2 having the amino acid sequence of SEQ ID NO: 172, or a modified amino acid sequence thereof containing 1, 2, 3, or more amino acid residue modifications as compared to SEQ ID NO: 172; and a H-CDR3 comprising the amino acid sequence of SEQ ID NO: 184, or a modified amino acid sequence thereof, said modified amino acid sequence containing 1, 2, 3, or more amino acid modifications as compared to SEQ ID NO: 184.

(343) An exemplary single domain antibody having this heavy chain variable region is referred to herein as the 47-T-bR3-1C8 antibody. The 47-T-bR3-1C8 antibody comprises a V.sub.H chain amino acid sequence of SEQ ID NO: 197 as shown below.

(344) TABLE-US-00053 SEQ ID NO: 197

QVQLQESGGGLVQPGGSLRLSCAASGFTFSSYWMYWVRQAPGKGLEWVS
TISPSGGTTYTNSAKGRFTISRDNANKNTVYLVNNLKPEDTAVYYCAA
RRSGRYEYDYWGQGTQVTVSS

(345) In another embodiment, the antibody or binding fragment thereof comprises a heavy chain

variable region having an amino acid sequence that shares at least 60%, at least 65%, at least 70%, at least 75%, at least 80%, at least 85%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94% at least 95%, at least 96%, at least 97%, at least 98%, at least 99% sequence identity to SEQ ID NO: 197.

(346) In one embodiment, the antibody or binding fragment thereof of the present disclosure comprises a humanized variant of the heavy chain variable region of SEQ ID NO: 197, where the framework regions are humanized or replaced with human immunoglobulin framework sequences. As noted supra, suitable human or humanized framework sequences can be chosen based on their known structure, a consensus sequence, sequence homology to the framework sequences of donor antibody (e.g., the framework sequences of SEQ ID NO: 197), or a combination of these approaches. The humanized framework regions are designed to be similar in length and sequence to the parental framework sequences of SEQ ID NO: 197. In one embodiment, the humanized framework regions share 50%, 55%, 60%, 65%, 70%, 75%, 80%, 85%, 90%, 95% or more sequence identity to the framework regions of SEQ ID NO: 197. In another embodiment, the humanized framework regions are 60%, 65%, 70%, 75%, 80%, 85%, 90%, 95% or more similar in length to the framework regions of SEQ ID NO: 197. Humanized variants of the heavy chain variable region of SEQ ID NO: 197 share at least 50%, at least 55%, at least 60%, at least 65%, at least 70%, at least 75%, at least 80%, at least 85%, at least 90%, at least 95% sequence identity along the entire length of SEQ ID 197.

(347) Another aspect of the present disclosure relates to an antibody or binding portion thereof (e.g., a human antibody) that competes for binding to human tau protein with the anti-tau antibody comprising a heavy chain variable region with the amino acid sequence of SEQ ID NO: 197. In accordance with this aspect of the disclosure, a competitive binding assay, such as Bio-Layer Interferometry (BLI) can be utilized to identify an antibody or binding portion thereof that competes for binding to a tau protein with the enumerated monoclonal antibody. Other competitive binding assays known in the art can also be utilized to identify a competitive binding antibody in accordance with this aspect of the disclosure.

(348) In one embodiment, the antibody or binding fragment thereof has a heavy chain variable region with a H-CDR1 having the amino acid sequence of SEQ ID NO: 161, or a modified amino acid sequence thereof containing 1, 2, 3, or more amino acid residue modifications as compared to SEQ ID NO: 161; a H-CDR2 having the amino acid sequence of SEQ ID NO: 173, or a modified amino acid sequence thereof containing 1, 2, 3, or more amino acid residue modifications as compared to SEQ ID NO: 173; and a H-CDR3 comprising the amino acid sequence of SEQ ID NO: 185, or a modified amino acid sequence thereof, said modified amino acid sequence containing 1, 2, 3, or more amino acid modifications as compared to SEQ ID NO: 185.

(349) Exemplary single domain antibodies having this heavy chain variable region are referred to herein as the 62-T-bR3-1B8 and 71-T-bR3-1D12 antibodies. These antibodies comprise a V.sub.H chain amino acid sequence of SEQ ID NO: 198 as shown below.

(350) TABLE-US-00054 SEQ ID NO: 198

QVQLQESGGDLVQPGGSLRLSCVASGFPFDDYPMSWVRQAPGKGLEWVS
SVSPNGGSTFYADSLKGRFTISRDNAKNTLYLQINSLKSDDTAVYHCAK
VLDYYCSGYGCIYASYDLWGQGTQVTVSS

(351) In another embodiment, the antibody or binding fragment thereof comprises a heavy chain variable region having an amino acid sequence that shares at least 60%, at least 65%, at least 70%, at least 75%, at least 80%, at least 85%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94% at least 95%, at least 96%, at least 97%, at least 98%, at least 99% sequence identity to SEQ ID NO: 198.

(352) In one embodiment, the antibody or binding fragment thereof of the present disclosure comprises a humanized variant of the heavy chain variable region of SEQ ID NO: 198, where the framework regions are humanized or replaced with human immunoglobulin framework sequences.

As noted supra, suitable human or humanized framework sequences can be chosen based on their known structure, a consensus sequence, sequence homology to the framework sequences of donor antibody (e.g., the framework sequences of SEQ ID NO: 198), or a combination of these approaches. The humanized framework regions are designed to be similar in length and sequence to the parental framework sequences of SEQ ID NO: 198. In one embodiment, the humanized framework regions share 50%, 55%, 60%, 65%, 70%, 75%, 80%, 85%, 90%, 95% or more sequence identity to the framework regions of SEQ ID NO: 198. In another embodiment, the humanized framework regions are 60%, 65%, 70%, 75%, 80%, 85%, 90%, 95% or more similar in length to the framework regions of SEQ ID NO: 198. Humanized variants of the heavy chain variable region of SEQ ID NO: 198 share at least 50%, at least 55%, at least 60%, at least 65%, at least 70%, at least 75%, at least 80%, at least 85%, at least 90%, at least 95% sequence identity along the entire length of SEQ ID 198.

(353) Another aspect of the present disclosure relates to an antibody or binding portion thereof (e.g., a human antibody) that competes for binding to human tau protein with the anti-tau antibody comprising a heavy chain variable region with the amino acid sequence of SEQ ID NO: 198. In accordance with this aspect of the disclosure, a competitive binding assay, such as Bio-Layer Interferometry (BLI) can be utilized to identify an antibody or binding portion thereof that competes for binding to a tau protein with the enumerated monoclonal antibody. Other competitive binding assays known in the art can also be utilized to identify a competitive binding antibody in accordance with this aspect of the disclosure.

(354) In one embodiment, the antibody or binding fragment thereof has a heavy chain variable region with a H-CDR1 having the amino acid sequence of SEQ ID NO: 162, or a modified amino acid sequence thereof containing 1, 2, 3, or more amino acid residue modifications as compared to SEQ ID NO: 162; a H-CDR2 having the amino acid sequence of SEQ ID NO: 174, or a modified amino acid sequence thereof containing 1, 2, 3, or more amino acid residue modifications as compared to SEQ ID NO: 174; and a H-CDR3 comprising the amino acid sequence of SEQ ID NO: 186, or a modified amino acid sequence thereof, said modified amino acid sequence containing 1, 2, 3, or more amino acid modifications as compared to SEQ ID NO: 186.

(355) An exemplary single domain antibody having this heavy chain variable region is referred to herein as the 79-T-bR3-1E1 antibody. The 79-T-bR3-1E1 antibody comprises a V.sub.H chain amino acid sequence of SEQ ID NO: 199 as shown below.

(356) TABLE-US-00055 SEQ ID NO: 199

QVQLQESGGGLVQPGGSLRLSCAASISIVSINTMAWYRQAPGKQRELVA
GITSGGSTNYADSVKGRFTISRDNAKNTVSLQMNSLKPEDTAVYYCNAG
RYVPGAIVTNYWGQGTQVTVSS

(357) In another embodiment, the antibody or binding fragment thereof comprises a heavy chain variable region having an amino acid sequence that shares at least 60%, at least 65%, at least 70%, at least 75%, at least 80%, at least 85%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94% at least 95%, at least 96%, at least 97%, at least 98%, at least 99% sequence identity to SEQ ID NO: 199.

(358) In one embodiment, the antibody or binding fragment thereof of the present disclosure comprises a humanized variant of the heavy chain variable region of SEQ ID NO: 199, where the framework regions are humanized or replaced with human immunoglobulin framework sequences. As noted supra, suitable human or humanized framework sequences can be chosen based on their known structure, a consensus sequence, sequence homology to the framework sequences of donor antibody (e.g., the framework sequences of SEQ ID NO: 199), or a combination of these approaches. The humanized framework regions are designed to be similar in length and sequence to the parental framework sequences of SEQ ID NO: 199. In one embodiment, the humanized framework regions share 50%, 55%, 60%, 65%, 70%, 75%, 80%, 85%, 90%, 95% or more sequence identity to the framework regions of SEQ ID NO: 199. In another embodiment, the

humanized framework regions are 60%, 65%, 70%, 75%, 80%, 85%, 90%, 95% or more similar in length to the framework regions of SEQ ID NO: 199. Humanized variants of the heavy chain variable region of SEQ ID NO: 199 share at least 50%, at least 55%, at least 60%, at least 65%, at least 70%, at least 75%, at least 80%, at least 85%, at least 90%, at least 95% sequence identity along the entire length of SEQ ID 199.

(359) Another aspect of the present disclosure relates to an antibody or binding portion thereof (e.g., a human antibody) that competes for binding to human tau protein with the anti-tau antibody comprising a heavy chain variable region with the amino acid sequence of SEQ ID NO: 199. In accordance with this aspect of the disclosure, a competitive binding assay, such as Bio-Layer Interferometry (BLI) can be utilized to identify an antibody or binding portion thereof that competes for binding to a tau protein with the enumerated monoclonal antibody. Other competitive binding assays known in the art can also be utilized to identify a competitive binding antibody in accordance with this aspect of the disclosure.

(360) In one embodiment, the antibody or binding fragment thereof has a heavy chain variable region with a H-CDR1 having the amino acid sequence of SEQ ID NO: 163, or a modified amino acid sequence thereof containing 1, 2, 3, or more amino acid residue modifications as compared to SEQ ID NO: 163; a H-CDR2 having the amino acid sequence of SEQ ID NO: 175, or a modified amino acid sequence thereof containing 1, 2, 3, or more amino acid residue modifications as compared to SEQ ID NO: 175; and a H-CDR3 comprising the amino acid sequence of SEQ ID NO: 187, or a modified amino acid sequence thereof, said modified amino acid sequence containing 1, 2, 3, or more amino acid modifications as compared to SEQ ID NO: 187.

(361) An exemplary single domain antibody having this heavy chain variable region as described herein is the 86-T-bR3-1B11 antibody. This antibody comprises a V.sub.H chain amino acid sequence of SEQ ID NO: 200 as shown below.

(362) TABLE-US-00056 SEQ ID NO: 200

QVQLQEFGGGLVQAGGSLRLSCLASMTTLGFKTMGWYRQAPGKQRELVA
RISSGGQTNYADSVKGRFTISRDNKNTVYLMISLKPEDTAVYYCNAR
RYYSLARYDYNWYWGQGTQVTVSS

(363) In another embodiment, the antibody or binding fragment thereof comprises a heavy chain variable region having an amino acid sequence that shares at least 60%, at least 65%, at least 70%, at least 75%, at least 80%, at least 85%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94% at least 95%, at least 96%, at least 97%, at least 98%, at least 99% sequence identity to SEQ ID NO: 200.

(364) In one embodiment, the antibody or binding fragment thereof of the present disclosure comprises a humanized variant of the heavy chain variable region of SEQ ID NO: 200, where the framework regions are humanized or replaced with human immunoglobulin framework sequences. As noted supra, suitable human or humanized framework sequences can be chosen based on their known structure, a consensus sequence, sequence homology to the framework sequences of donor antibody (e.g., the framework sequences of SEQ ID NO: 200), or a combination of these approaches. The humanized framework regions are designed to be similar in length and sequence to the parental framework sequences of SEQ ID NO: 200. In one embodiment, the humanized framework regions share 50%, 55%, 60%, 65%, 70%, 75%, 80%, 85%, 90%, 95% or more sequence identity to the framework regions of SEQ ID NO: 200. In another embodiment, the humanized framework regions are 60%, 65%, 70%, 75%, 80%, 85%, 90%, 95% or more similar in length to the framework regions of SEQ ID NO: 200. Humanized variants of the heavy chain variable region of SEQ ID NO: 200 share at least 50%, at least 55%, at least 60%, at least 65%, at least 70%, at least 75%, at least 80%, at least 85%, at least 90%, at least 95% sequence identity along the entire length of SEQ ID NO: 200.

(365) Another aspect of the present disclosure relates to an antibody or binding portion thereof (e.g., a human antibody) that competes for binding to human tau protein with the anti-tau antibody

comprising a heavy chain variable region with the amino acid sequence of SEQ ID NO: 200. In accordance with this aspect of the disclosure, a competitive binding assay, such as Bio-Layer Interferometry (BLI) can be utilized to identify an antibody or binding portion thereof that competes for binding to a tau protein with the enumerated monoclonal antibody. Other competitive binding assays known in the art can also be utilized to identify a competitive binding antibody in accordance with this aspect of the disclosure.

(366) In one embodiment, the antibody or binding fragment thereof has a heavy chain variable region with a H-CDR1 having the amino acid sequence of SEQ ID NO: 164 or a modified amino acid sequence thereof containing 1, 2, 3, or more amino acid residue modifications as compared to SEQ ID NO: 164; a H-CDR2 having the amino acid sequence of SEQ ID NO: 176, or a modified amino acid sequence thereof containing 1, 2, 3, or more amino acid residue modifications as compared to SEQ ID NO: 176; and a H-CDR3 comprising the amino acid sequence of SEQ ID NO: 188, or a modified amino acid sequence thereof, said modified amino acid sequence containing 1, 2, 3, or more amino acid modifications as compared to SEQ ID NO: 188.

(367) An exemplary single domain antibody having this heavy chain variable region is referred to herein as the 95-T-bR3-1E11 antibody. This antibody comprises a V.sub.H chain amino acid sequence of SEQ ID NO: 201 as shown below.

(368) TABLE-US-00057 SEQ ID NO: 201

QVQLQAFGGAAQAQAGDSLRLSCVVGKSTAAVNGVGWYRQAPGRQREFV
AAIRPGGRRDYLDVKGFRFSVFRNKTTVYLRMNDLRIEDTAVYYCYAEG
LLLPSTYWGGQGLTVTVSS

(369) In another embodiment, the antibody or binding fragment thereof comprises a heavy chain variable region having an amino acid sequence that shares at least 60%, at least 65%, at least 70%, at least 75%, at least 80%, at least 85%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94% at least 95%, at least 96%, at least 97%, at least 98%, at least 99% sequence identity to SEQ ID NO: 201.

(370) In one embodiment, the antibody or binding fragment thereof of the present disclosure comprises a humanized variant of the heavy chain variable region of SEQ ID NO: 201, where the framework regions are humanized or replaced with human immunoglobulin framework sequences. As noted supra, suitable human or humanized framework sequences can be chosen based on their known structure, a consensus sequence, sequence homology to the framework sequences of donor antibody (e.g., the framework sequences of SEQ ID NO: 201), or a combination of these approaches. The humanized framework regions are designed to be similar in length and sequence to the parental framework sequences of SEQ ID NO: 201. In one embodiment, the humanized framework regions share 50%, 55%, 60%, 65%, 70%, 75%, 80%, 85%, 90%, 95% or more sequence identity to the framework regions of SEQ ID NO: 201. In another embodiment, the humanized framework regions are 60%, 65%, 70%, 75%, 80%, 85%, 90%, 95% or more similar in length to the framework regions of SEQ ID NO: 201. Humanized variants of the heavy chain variable region of SEQ ID NO: 201 share at least 50%, at least 55%, at least 60%, at least 65%, at least 70%, at least 75%, at least 80%, at least 85%, at least 90%, at least 95% sequence identity along the entire length of SEQ ID 201.

(371) In one embodiment, the antibody or binding fragment thereof has a heavy chain variable region with a H-CDR1 having the amino acid sequence of SEQ ID NO: 307 or a modified amino acid sequence thereof containing 1, 2, 3, or more amino acid residue modifications as compared to SEQ ID NO: 307; a H-CDR2 having the amino acid sequence of SEQ ID NO: 308, or a modified amino acid sequence thereof containing 1, 2, 3, or more amino acid residue modifications as compared to SEQ ID NO: 308; and a H-CDR3 comprising the amino acid sequence of SEQ ID NO: 309, or a modified amino acid sequence thereof, said modified amino acid sequence containing 1, 2, 3, or more amino acid modifications as compared to SEQ ID NO: 309.

(372) An exemplary single domain antibody having this heavy chain variable region is referred to

herein as the 55-T-bR3-1D4 antibody. This antibody comprises a V.sub.H chain amino acid sequence of SEQ ID NO: 310 as shown below.

(373) TABLE-US-00058 SEQ ID NO: 310

QVQLQEFGGGLVQAGGSLRLSCAASGRTFSGYSMGWFRQAPGKEREFVG
GISRSGGWTTYADSVKGRFTISRDNANKNTVYLMNSLKPEDTAVYYCAA
AVSRINDYAPALSRAYDYWGQGTQVTVSS

(374) In another embodiment, the antibody or binding fragment thereof comprises a heavy chain variable region having an amino acid sequence that shares at least 60%, at least 65%, at least 70%, at least 75%, at least 80%, at least 85%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94% at least 95%, at least 96%, at least 97%, at least 98%, at least 99% sequence identity to SEQ ID NO: 310.

(375) In one embodiment, the antibody or binding fragment thereof of the present disclosure comprises a humanized variant of the heavy chain variable region of SEQ ID NO: 310, where the framework regions are humanized or replaced with human immunoglobulin framework sequences. As noted supra, suitable human or humanized framework sequences can be chosen based on their known structure, a consensus sequence, sequence homology to the framework sequences of donor antibody (e.g., the framework sequences of SEQ ID NO: 310), or a combination of these approaches. The humanized framework regions are designed to be similar in length and sequence to the parental framework sequences of SEQ ID NO: 310. In one embodiment, the humanized framework regions share 50%, 55%, 60%, 65%, 70%, 75%, 80%, 85%, 90%, 95% or more sequence identity to the framework regions of SEQ ID NO: 310. In another embodiment, the humanized framework regions are 60%, 65%, 70%, 75%, 80%, 85%, 90%, 95% or more similar in length to the framework regions of SEQ ID NO: 310. Humanized variants of the heavy chain variable region of SEQ ID NO: 310 share at least 50%, at least 55%, at least 60%, at least 65%, at least 70%, at least 75%, at least 80%, at least 85%, at least 90%, at least 95% sequence identity along the entire length of SEQ ID 310.

(376) Another aspect of the present disclosure relates to an antibody or binding portion thereof (e.g., a human antibody) that competes for binding to human tau protein with the anti-tau antibody comprising a heavy chain variable region with the amino acid sequence of SEQ ID NO: 310. In accordance with this aspect of the disclosure, a competitive binding assay, such as Bio-Layer Interferometry (BLI) can be utilized to identify an antibody or binding portion thereof that competes for binding to a tau protein with the enumerated monoclonal antibody. Other competitive binding assays known in the art can also be utilized to identify a competitive binding antibody in accordance with this aspect of the disclosure.

(377) Antibody "specificity" refers to selective recognition of the antibody or binding portion thereof as described herein for a particular epitope of the tau protein. The term "epitope" includes any protein determinant capable of specific binding to an immunoglobulin or T-cell receptor or otherwise interacting with a molecule. Epitopic determinants generally consist of chemically active surface groupings of molecules such as amino acids or carbohydrate or sugar side chains and generally have specific three dimensional structural characteristics, as well as specific charge characteristics. The epitope of the antibodies described herein may be "linear" or "conformational". In a linear epitope, all of the points of interaction between the protein and the antibody occur linearly along the primary amino acid sequence of the protein. In a conformational epitope, the points of interaction occur across amino acid residues on the protein that are separated from one another, i.e., noncontiguous amino acids juxtaposed by tertiary folding of a protein. Epitopes formed from contiguous amino acids are typically retained on exposure to denaturing solvents, whereas epitopes formed by tertiary folding are typically lost on treatment with denaturing solvents. An epitope typically includes at least 3, and more usually, at least 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20 or more amino acids in a unique spatial conformation. Antibodies that recognize the same epitope can be verified in a simple immunoassay showing the ability of one

antibody to block the binding of another antibody to a target antigen.

(378) In one embodiment, the epitope recognized by the isolated antibody of the present invention is a non-linear or conformational epitope, i.e. a tertiary or quaternary structure that is shared by pathological proteins. For example, the non-linear or conformational epitope recognized by the antibodies described herein is a conformational epitope that is common or shared with one or more, or all, amyloidogenic proteins, e.g., tau, synuclein, A β , prion, etc. Accordingly, in one embodiment, the antibodies described herein have antigenic specificity for a shared conformational epitope common to all amyloidogenic proteins known in the art.

(379) Another aspect of the present disclosure is directed to an antibody mimetic that binds tau protein. An “antibody mimetic” as referred to herein encompasses any organic compound, e.g., a peptide or polypeptide, that can specifically bind an antigen like an antibody, and is about 3-20 kDa. In one embodiment, the antibody mimetic comprises a scaffold which binds its antigen via amino acids in exposed loops similar to the CDR loops of an antibody. These antibody mimetics include, without limitation, adnectins, lipocalins, Kunitz domain-based binders, avimers, knottins, fynomers, atrimers, and cytotoxic T-lymphocyte associated protein-4 (CTLA4)-based binders (reviewed in Weidle et al., “The Emerging Role of New Protein Scaffold-based Agents for the Treatment of Cancer,” *Cancer Genomics & Proteomics* 10:155-168 (2013), which is hereby incorporated by reference in its entirety). In accordance with this aspect of the present disclosure, the loop binding regions of the antibody mimetic are adapted to comprise one or more of the heavy chain CDRs of the antibodies disclosed herein. For example, an antibody mimetic of the present disclosure may comprise a first loop region having an amino acid sequence of any one of SEQ ID NOs: 1-38, 202, 206, 210, 214, or a modified amino acid sequence of any one of SEQ ID NOs: 1-38, 202, 206, 210, and 214 said modified sequence containing 1, 2, 3, 4, 5, 6, 7, 8, 9, or 10 amino acid residue modifications as compared to any one of SEQ ID NOs: 1-38, 202, 206, 210, and 214. The antibody mimetic may comprise another loop region having an amino acid sequence of any one of SEQ ID NOs: 39-76, 203, 207, 211, 215 or a modified amino acid sequence of any one of SEQ ID NOs: 39-76, 203, 207, 211, and 215 said modified sequences containing 1, 2, 3, 4, 5, 6, or 7, amino acid residue modifications as compared to any one of SEQ ID NOs: 39-76, 203, 207, 211, and 215. The antibody mimetic may comprise another loop region having an amino acid sequence any one of SEQ ID NOs: 77-114, 204, 208, 212, 216, or a modified amino acid sequence of any one of SEQ ID NOs: 77-114, 204, 208, 212, and 216 said modified sequence containing 1, 2, 3, 4, 5, 6, or 7 amino acid residue modifications as compared to any one of SEQ ID NOs: 77-114, 204, 208, 212, and 216.

(380) In another embodiment, an antibody mimetic of the present disclosure may comprise a first loop region having an amino acid sequence of any one of SEQ ID NOs: 153-164 and 307 or a modified amino acid sequence of any one of SEQ ID NOs: 153-164 and 307, said modified sequence containing 1, 2, 3, 4, 5, 6, 7, 8, or 9 amino acid residue modifications as compared to any one of SEQ ID NOs: 153-164 and 307. The antibody mimetic may comprise another loop region having an amino acid sequence of any one of SEQ ID NOs: 165-176 and 308, or a modified amino acid sequence of any one of SEQ ID NOs: 165-176 and 308, said modified sequences containing 1, 2, 3, 4, 5, 6, 7, 8, 9, or 10, amino acid residue modifications as compared to any one of SEQ ID NOs: 165-176 and 308. The antibody mimetic may comprise another loop region having an amino acid sequence any one of SEQ ID NOs: 177-188 and 309, or a modified amino acid sequence of any one of SEQ ID NOs: 177-188 and 309, said modified sequence containing 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, or 11 amino acid residue modifications as compared to any one of SEQ ID NOs: 177-188 and 309.

(381) In one embodiment, the antibody mimetic comprises one or more modified fibronectin type III (FN3) domains (e.g., an adnectin or centyrin molecule), where each modified FN3 domain has one or more loop regions that comprise one or more CDR sequences or modified CDR sequences as disclosed herein.

(382) The FN3 domain is an evolutionarily conserved protein domain that is about 100 amino acids in length and possesses a beta sandwich structure. The beta sandwich structure of human FN3 comprises seven beta-strands, referred to as strands A, B, C, D, E, F, G, with six connecting loops, referred to as loops AB, BC, CD, DE, EF, and FG that exhibit structural homology to immunoglobulin binding domains. Three of the six loops, i.e., loops DE, BC, and FG, correspond topologically to the complementarity determining regions of an antibody, i.e., CDR1, CDR2, and CDR3. The remaining three loops are surface exposed in a manner similar to antibody CDR3. In accordance with the present disclosure, one or more of the loop regions of each FN3 domain of the binding molecule are modified to comprise one or more CDR sequences disclosed herein.

(383) The modified FN3 domain can be a FN3 domain derived from any of the wide variety of animal, yeast, plant, and bacterial extracellular proteins containing these domains. In one embodiment, the FN3 domain is derived from a mammalian FN3 domain. Exemplary FN3 domains include, for example and without limitation, any one of the 15 different FN3 domains present in human tenascin C, or the 15 different FN3 domains present in human fibronectin (FN) (e.g., the 10^{sup}.th fibronectin type III domain). Exemplary FN3 domains also include non-natural synthetic FN3 domains, such as those described in U.S. Pat. Publ. No. 2010/0216708 to Jacobs et al., which is hereby incorporated by reference in its entirety. Individual FN3 domains are referred to by domain number and protein name, e.g., the 3^{sup}.rd FN3 domain of tenascin (TN3), or the 10^{sup}.th FN3 domain of fibronectin (FN10).

(384) Another aspect of the present disclosure is directed to isolated polynucleotides encoding the antibody or binding fragment thereof or antibody mimetic as described herein. The nucleic acid molecules described herein include isolated polynucleotides, portions of expression vectors or portions of linear DNA sequences, including linear DNA sequences used for in vitro transcription/translation, and vectors compatible with prokaryotic, eukaryotic or filamentous phage expression, secretion, and/or display of the antibodies or binding fragments thereof described herein.

(385) In one embodiment, an isolated polynucleotide encodes any one or more of a H-CDR1 of any one of SEQ ID NOs: 1-38, 202, 206, 210, 214, and variants thereof, a H-CDR2 of any one of SEQ ID NOs: 39-76, 203, 207, 211, 215, and variants thereof, and a H-CDR3 of any one of SEQ ID NOs: 77-114, 204, 208, 212, 216, and variants thereof. In another embodiment, an isolated polynucleotide as described herein encodes any one or more of a H-CDR1 of any one of SEQ ID NOs: 153-164, 307, and variants thereof, a H-CDR2 of any one of SEQ ID NOs: 165-176, 308, and variants thereof, and a H-CDR3 of any one of SEQ ID NOs: 177-188, 309, and variants thereof.

(386) In another embodiment, an isolated polynucleotide as described herein encodes a heavy chain variable region of a tau antibody having the amino acid sequence of any one of SEQ ID NOs: 115-152, 205, 209, 213, 217, and variants thereof. The nucleotide sequences of these isolated polynucleotides are enumerated in Table 3 herein and include SEQ ID NOs: 218-260. In another embodiment, the isolated polynucleotide encodes a heavy chain variable region having the amino acid sequence of any one of SEQ ID NOs: 189-201, 310 and variants thereof. The nucleotide sequences of these isolated polynucleotides are enumerated in Table 4 herein and include SEQ ID NOs: 261-306. Nucleic acid molecules having nucleotide sequences that differ from SEQ ID NOs: 218-306, which as a result of the degeneracy of the genetic code, also encode the tau antibody described herein are also encompassed by the present disclosure. Such nucleic acid molecules may share 80%, 85%, 90%, or 95% sequence identity to any one of the sequences of SEQ ID NOs: 218-306.

(387) The polynucleotides of the invention may be produced by chemical synthesis such as solid phase polynucleotide synthesis on an automated polynucleotide synthesizer and assembled into complete single or double stranded molecules. Alternatively, the polynucleotides of the invention may be produced by other techniques such as PCR followed by routine cloning. Techniques for producing or obtaining polynucleotides of a known sequence are well known in the art.

(388) The polynucleotides of the invention may comprise at least one non-coding sequence, such as a promoter or enhancer sequence, intron, polyadenylation signal, a cis sequence facilitating RepA binding, and the like. The polynucleotide sequences may also comprise additional sequences encoding additional amino acids that provide, for example, a marker or a tag sequence such as a histidine tag or an HA tag to facilitate purification or detection of the protein, a signal sequence, a fusion protein partner such as RepA, Fc or bacteriophage coat protein such as pIX or pIII.

(389) Another embodiment of the disclosure is directed to a vector comprising at least one polynucleotide encoding the antibody or binding fragment thereof or antibody mimetic as described herein. Such vectors include, without limitation, plasmid vectors, viral vectors, including without limitation, vaccinia vector, lentiviral vector, adenoviral vector, adeno-associated viral vector, vectors for baculovirus expression, transposon based vectors or any other vector suitable for introduction of the polynucleotides described herein into a given organism or genetic background by any means to facilitate expression of the encoded antibody polypeptide. The polynucleotide sequences encoding the heavy chain variable domains as described herein are combined with sequences of a promoter, a translation initiation segment (e.g., a ribosomal binding sequence and start codon), a 3' untranslated region, polyadenylation signal, a termination codon, and transcription termination to form one or more expression vector constructs.

(390) In one embodiment, the vector is an adenoviral-associated viral (AAV) vector. A number of therapeutic AAV vectors suitable for delivery of the polynucleotides encoding tau antibodies described herein to the central nervous system are known in the art. See e.g., Deverman et al., "Gene Therapy for Neurological Disorders: Progress and Prospects," *Nature Rev.* 17:641-659 (2018), which is hereby incorporated by reference in its entirety. Suitable AAV vectors include serotypes AAV1, AAV2, AAV3, AAV4, AAV5, AAV6, AAV7, AAV8, AAV9, AAV10, or AAV 11 in their native form or engineered for enhanced tropism. AAV vectors known to have tropism for the CNS that are particularly suited for therapeutic expression of the tau antibodies described herein include, AAV1, AAV2, AAV4, AAV5, AAV8 and AAV9 in their native form or engineered for enhanced tropism. In one embodiment, the AAV vector is an AAV2 vector. In another embodiment, the AAV vector is an AAV5 vector as described by Vitale et al., "Anti-tau Conformational scFv MC1 Antibody Efficiently Reduces Pathological Tau Species in Adult JNPL3 Mice," *Acta Neuropathol. Commun.* 6:82 (2018), optionally containing the GFAP or CAG promoter and the Eoodchuck hepatitis virus (WPRE) post-translational regulatory element. In another embodiment, the AAV vector is an AAV9 vector as described by Haiyan et al., "Targeting Root Cause by Systemic scAAV9-hIDS Gene Delivery: Functional Correction and Reversal of Severe MPSII in Mice," *Mol. Ther. Methods Clin. Dev.* 10:327-340 (2018), which is hereby incorporated by reference in its entirety. In another embodiment, the AAV vector is an AAVrh10 vector as described by Liu et al., "Vectored Intracerebral Immunizations with the Anti-Tau Monoclonal Antibody PHF1 Markedly Reduces Tau Pathology in Mutant Transgenic Mice," *J. Neurosci.* 36(49): 12425-35 (2016), which is hereby incorporated by reference in its entirety.

(391) In another embodiment the AAV vector is a hybrid vector comprising the genome of one serotype, e.g., AAV2, and the capsid protein of another serotype, e.g., AAV1 or AAV3-9 to control tropism. See e.g., Broekman et al., "Adeno-associated Virus Vectors Serotyped with AAV8 Capsid are More Efficient than AAV-1 or -2 Serotypes for Widespread Gene Delivery to the Neonatal Mouse Brain," *Neuroscience* 138:501-510 (2006), which is hereby incorporated by reference in its entirety. In one embodiment, the AAV vector is an AAV2/8 hybrid vector as described by Ising et al., "AAV-mediated Expression of Anti-Tau ScFv Decreases Tau Accumulation in a Mouse Model of Tauopathy," *J. Exp. Med.* 214(5):1227 (2017), which is hereby incorporated by reference in its entirety. In another embodiment the AAV vector is an AAV2/9 hybrid vector as described by Simon et al., "A Rapid Gene Delivery-Based Mouse Model for Early-Stage Alzheimer Disease-Type Tauopathy," *J. Neuropath. Exp. Neurol.* 72(11): 1062-71 (2013), which is hereby incorporated by reference in its entirety.

(392) In another embodiment, the AAV vector is one that has been engineered or selected for its enhanced CNS transduction after intraparenchymal administration, e.g., AAV-DJ (Grimm et al., *J. Virol.* 82:5887-5911 (2008), which is hereby incorporated by reference in its entirety); increased transduction of neural stem and progenitor cells, e.g., SCH9 and AAV4.18 (Murlidharan et al., *J. Virol.* 89: 3976-3987 (2015) and Ojala et al., *Mol. Ther.* 26:304-319 (2018), which are hereby incorporated by reference in their entirety); enhanced retrograde transduction, e.g., rAAV2-retro (Muller et al., *Nat. Biotechnol.* 21:1040-1046 (2003), which is hereby incorporated by reference in its entirety); selective transduction into brain endothelial cells, e.g., AAV-BRI (Korbelin et al., *EMBO Mol. Med.* 8: 609-625 (2016), which is hereby incorporated by reference in its entirety); or enhanced transduction of the adult CNS after IV administration, e.g., AAV-PHP.B and AAVPHP.eB (Deverman et al., *Nat. Biotechnol.* 34: 204-209 (2016) and Chan et al., *Nat. Neurosci.* 20: 1172-1179 (2017), which are hereby incorporated by reference in their entirety).

(393) In accordance with this embodiment, the expression vector construct encoding the anti-tau antibody or binding portion thereof can include the nucleic acid encoding the heavy chain variable region polypeptide, a fragment thereof, a variant thereof, or combinations thereof. In one embodiment, the heavy chain variable region polynucleotide encodes only a variable heavy chain (VH) region. In another embodiment, the heavy chain variable region polynucleotide is engineered to further comprise a region encoding at least one constant heavy chain (CH) region. The at least one constant heavy chain region can include a constant heavy chain region 1 (CH1), a constant heavy chain region 2 (CH2), and a constant heavy chain region 3 (CH3), and/or a hinge region.

(394) The promoter sequence of the expression vector construct is suitable for driving expression of the antibody or binding fragment thereof. The promoter can be inducible or constitutive. Suitable promoter sequences include, without limitation, the elongation factor 1-alpha promoter (EF1a) promoter, a phosphoglycerate kinase-1 promoter (PGK) promoter, a cytomegalovirus immediate early gene promoter (CMV), artificial CMV-chicken β -actin promoter with β -globin splice acceptor (CAG), chicken β -actin (CBA) promoter, a chimeric liver-specific promoter (LSP) a cytomegalovirus enhancer/chicken beta-actin promoter (CAG), a tetracycline responsive promoter (TRE), a transthyretin promoter (TTR), a simian virus 40 promoter (SV40) and a CK6 promoter. Other promoters suitable for driving gene expression in mammalian cells that are known in the art are also suitable for incorporation into the expression constructs disclosed herein. The expression construct can further encode enhance-promoter elements that control or restrict expression of the encoded antibody. Enhancer-promoter elements that restrict expression to several neuronal and glial cell types in the CNS are known in the art and suitable for inclusion in the vector expression constructs described herein. See e.g., Lee et al., "GFAP Promoter Elements Required for Region-Specific and Astrocyte-Specific Expression," *Glia* 56: 481-493 (2008); Dimidschstein et al., "A Viral Strategy for Targeting and Manipulating Interneurons Across Vertebrate Species," *Nat. Neuroscience* 19:1743-49 (2016); and de Leeuw et al., "rAAV-compatible MiniPromoters for Restricted Expression in the Brain and Eye," *Mol. Brain* 9:52 (2016), which are hereby incorporated by reference in their entirety.

(395) The expression construct can further encode a linker sequence. The linker sequence can encode an amino acid sequence that spatially separates and/or links the one or more components of the expression construct.

(396) Another embodiment of the invention is a host cell comprising the vectors described herein. The antibodies and binding fragments thereof described herein can be optionally produced by a cell line, a mixed cell line, an immortalized cell or clonal population of immortalized cells, as well known in the art (see e.g., Ausubel et al., ed., *Current Protocols in Molecular Biology*, John Wiley & Sons, Inc., NY, N.Y. (1987-2001); Sambrook et al., *Molecular Cloning: A Laboratory Manual*, 2.sup.nd Edition, Cold Spring Harbor, N.Y. (1989); Harlow and Lane, *Antibodies, a Laboratory Manual*, Cold Spring Harbor, N.Y. (1989); Colligan et al., eds., *Current Protocols in Immunology*, John Wiley & Sons, Inc., NY (1994-2001); Colligan et al., *Current Protocols in Protein Science*,

John Wiley & Sons, NY, N.Y., (1997-2001), which are hereby incorporated by reference in their entirety).

(397) The host cell chosen for expression may be of mammalian origin or may be selected from COS-1, COS-7, HEK293, BHK21, CHO, BSC-1, He G2, SP2/0, HeLa, myeloma, lymphoma, yeast, insect, or plant cells, or any derivative, immortalized or transformed cell thereof.

Alternatively, the host cell may be selected from a species or organism incapable of glycosylating polypeptides, e.g., a prokaryotic cell or organism, such as BL21, BL21(DE3), BL21-GOLD(DE3), XL1-Blue, JM109, HMS174, HMS174(DE3), and any of the natural or engineered *E. coli* spp, *Klebsiella* spp., or *Pseudomonas* spp strains.

(398) The antibodies described herein can be prepared by any of a variety of techniques using the isolated polynucleotides, vectors, and host cells described supra. In general, antibodies can be produced by cell culture techniques, including the generation of monoclonal antibodies via conventional techniques, or via transfection of antibody genes, heavy chains and/or light chains into suitable bacterial or mammalian cell hosts, in order to allow for the production of antibodies, wherein the antibodies may be recombinant. Standard molecular biology techniques are used to prepare the recombinant expression vector, transfect the host cells, select for transformants, culture the host cells and recover the antibody from the culture medium. Transfecting the host cell can be carried out using a variety of techniques commonly used for the introduction of exogenous DNA into a prokaryotic or eukaryotic host cell, e.g., by electroporation, calcium-phosphate precipitation, DEAE-dextran transfection and the like. Although it is possible to express the antibodies described herein in either prokaryotic or eukaryotic host cells, expression of antibodies in eukaryotic cells is sometimes preferable, and sometimes preferable in mammalian host cells, because such eukaryotic cells (and in particular mammalian cells) are more likely than prokaryotic cells to assemble and secrete a properly folded and immunologically active antibody.

(399) As noted above, exemplary mammalian host cells for expressing the recombinant antibodies of the invention include Chinese Hamster Ovary (CHO cells) (including dhfr-CHO cells, described in Urlaub and Chasin, *Proc. Natl. Acad. Sci. USA*, 77: 4216-4220 (1980), which is hereby incorporated by reference in its entirety). Other suitable mammalian host cells include, without limitation, NS0 myeloma cells, COS cells, and SP2 cells. When recombinant expression vectors encoding antibody genes are introduced into mammalian host cells, the antibodies are produced by culturing the host cells for a period of time sufficient to allow for expression of the antibody in the host cells or, more preferably, secretion of the antibody into the culture medium in which the host cells are grown.

(400) Host cells can also be used to produce functional antibodies and fragments thereof. It is understood that variations on the above procedure are within the scope of the present disclosure. For example, it may be desirable to transfect a host cell with DNA encoding functional fragments the heavy chain of an antibody described herein. Recombinant DNA technology may also be used to remove some or all of the DNA encoding portions of the heavy chain that are not necessary for binding to the antigens of interest. The molecules expressed from such truncated DNA molecules are also encompassed by the antibodies described herein.

(401) The antibodies and antibody binding fragments are recovered and purified from recombinant cell cultures by known methods including, but not limited to, protein A purification, ammonium sulfate or ethanol precipitation, acid extraction, anion or cation exchange chromatography, phosphocellulose chromatography, hydrophobic interaction chromatography, affinity chromatography, hydroxylapatite chromatography and lectin chromatography. High performance liquid chromatography ("HPLC") can also be used for purification.

(402) In another embodiment, the antibody or binding fragment thereof, or the polynucleotide encoding the antibody or binding fragment thereof is a component of a pharmaceutical composition. In one embodiment, the pharmaceutical composition comprises a monoclonal antibody composition. In another embodiment, the pharmaceutical composition comprises two or

more different single-domain antibodies, e.g., a polyclonal antibody composition. In another embodiment, the pharmaceutical composition comprises polynucleotides encoding the monoclonal antibody composition. In another embodiment, the pharmaceutical composition comprises polynucleotides encoding two or more different single-domain antibodies as described herein. In another embodiment, the pharmaceutical composition comprises one or more antibodies or polynucleotides encoding the same as described herein and one or more prophylactic or therapeutic agents other than the antibodies described herein that are useful for preventing or treating a condition mediated by a toxic tau protein.

(403) The therapeutically effective amount of antibody present in the pharmaceutical composition or formulation is determined by taking into account the desired dose volumes and mode(s) of administration. Exemplary antibody concentrations in the pharmaceutical compositions of the present disclosure include from about 0.1 mg/mL to about 50 mg/mL, from about 0.5 mg/mL to about 25 mg/mL, and from about 2 mg/mL to about 10 mg/mL.

(404) An aqueous formulation is prepared comprising the antibody in a pH-buffered solution. The buffer has a pH in the range from about 4.5 to about 10, from about 5 to about 9, or from about 6 to 8. Examples of buffers include phosphate buffers (e.g., phosphate buffered saline), acetate (e.g. sodium acetate), succinate (such as sodium succinate), gluconate, histidine, citrate and other organic acid buffers.

(405) A polyol, which acts as a tonicifier and may stabilize the antibody, may be included in the formulation. In one embodiment, the tonicifying polyol is a salt such as sodium chloride. In another embodiment, the polyol is a non-reducing sugar, such as sucrose or trehalose. The polyol is added to the formulation in an amount which may vary with respect to the desired isotonicity of the formulation. Preferably the aqueous formulation is isotonic, in which case suitable concentrations of the polyol in the formulation are in the range from about 1% to about 15% w/v, or in the range from about 2% to about 10% w/v, for example. However, hypertonic or hypotonic formulations may also be suitable. The amount of polyol added may also alter with respect to the molecular weight of the polyol. For example, a lower amount of a monosaccharide (e.g. mannitol) may be added, compared to a disaccharide (such as trehalose).

(406) A surfactant may also be added to the pharmaceutical composition containing the antibody. Exemplary surfactants include nonionic surfactants such as polysorbates (e.g. polysorbates 20, 80 etc), poloxamers (e.g. poloxamer 188), Pluronic F68, and PEG (polyethylene glycol). The amount of surfactant added is such that it reduces aggregation of the formulated antibody and/or minimizes the formation of particulates in the formulation and/or reduces adsorption. For example, the surfactant may be present in the formulation in an amount from about 0.001% to about 0.5%, from about 0.005% to about 0.2%, or from about 0.01% to about 0.1%.

(407) In one embodiment, the pharmaceutical composition contains the above-identified agents (i.e. antibody, buffer, polyol and surfactant) and is essentially free of one or more preservatives, such as benzyl alcohol, phenol, m-cresol, chlorobutanol and benzethonium Cl. In another embodiment, a preservative may be included in the pharmaceutical composition, particularly where the formulation is a multidose formulation. Suitable preservatives include, without limitation phenol, m-cresol, p-cresol, o-cresol, chlorocresol, benzyl alcohol, phenylmercuric nitrite, phenoxyethanol, formaldehyde, chlorobutanol, magnesium chloride (e.g., hexahydrate), alkylparaben (methyl, ethyl, propyl, butyl and the like), benzalkonium chloride, benzethonium chloride, sodium dehydroacetate and thimerosal, or mixtures thereof in an aqueous diluent. The concentration of preservative may be in the range from about 0.01% to about 5%, from about 0.5% to about 2% and any range or value therein. Non-limiting examples include, no preservative, 0.1-2% m-cresol, 0.1-3% benzyl alcohol, 0.001-0.5% thimerosal, 0.001-2.0% phenol, 0.0005-1.0% alkylparaben(s), and the like. One or more other pharmaceutically acceptable carriers, excipients or stabilizers such as those described in *Remington's Pharmaceutical Sciences* 16th edition, Osol, A. Ed. (1980), which is hereby incorporated by reference in its entirety, may be included in the composition provided that

they do not adversely affect the desired characteristics of the formulation. Acceptable carriers, excipients or stabilizers are nontoxic to recipients at the dosages and concentrations employed and include; additional buffering agents; co-solvents; antioxidants including ascorbic acid and methionine; chelating agents such as EDTA; metal complexes (e.g. Zn-protein complexes); biodegradable polymers such as polyesters; and/or salt-forming counterions such as sodium. (408) The pharmaceutical compositions to be used for in vivo administration must be sterile. This is readily accomplished by filtration through sterile filtration membranes, prior to, or following, preparation of the composition.

(409) The pharmaceutical compositions comprising antibodies or binding fragments thereof, or polynucleotides encoding the antibodies or binding fragments thereof, are for use in, but not limited to, preventing, treating, managing, or ameliorating one or more symptoms of a condition involving a pathological tau protein, e.g., Alzheimer's disease or other tauopathy.

(410) In another aspect of the present disclosure the anti-tau antibodies described herein, binding fragments thereof, or a pharmaceutical composition containing the same, are employed in a method of inhibiting onset of one or more symptoms of a condition involving pathological tau protein in a subject. This method involves administering to the subject the anti-tau antibodies described, polynucleotides encoding the anti-tau antibodies described herein, or a pharmaceutical composition containing the same to the subject in an amount effective to inhibit the onset of one or more symptoms of the condition involving pathological tau protein in the subject.

(411) In another aspect of the present disclosure the tau antibodies described herein, polynucleotides encoding the tau antibodies described herein, or a pharmaceutical composition containing the same, are employed in a method of treating a subject having a condition involving pathological tau protein. This method involves selecting a subject having one or more symptoms of the condition involving a pathological tau protein and administering to the subject the tau antibody, a polynucleotide encoding a tau antibody, or a pharmaceutical composition containing the same in an amount effective to treat the condition involving the pathological tau protein in the subject.

(412) In accordance with these embodiments, the condition involving a pathological tau protein may include Alzheimer's disease, Pick's disease, progressive supranuclear palsy, corticobasal degeneration, argyrophilic grain disease, primary age-related tauopathy, globular glial tauopathy, frontotemporal dementia, parkinsonism linked to chromosome 17, chronic traumatic encephalopathy, Huntington's disease, Amyotrophic Lateral Sclerosis (ALS), Niemann-Pick-C, Guam-ALS-Parkinson's dementia, post-encephalitic Parkinson's disease, aluminum toxicity, or prion disease.

(413) In accordance with these embodiments, the "subject" is typically a human. However, other non-human mammals amenable to treatment in accordance with the methods described herein include, without limitation, primates, dogs, cats, rodents (e.g., mouse, rat, guinea pig), horses, deer, cervids, cattle and cows, sheep, and pigs.

(414) In prophylactic applications, the pharmaceutical compositions of the present invention are administered to a subject that is susceptible to, or otherwise at risk of developing a particular condition mediated by a pathological form of tau protein, e.g., Pick's disease, progressive supranuclear palsy, corticobasal degeneration, argyrophilic grain disease, primary age-related tauopathy, globular glial tauopathy, frontotemporal dementia, parkinsonism linked to chromosome 17, chronic traumatic encephalopathy, Huntington's disease, Amyotrophic Lateral Sclerosis (ALS), Niemann-Pick-C, Guam-ALS-Parkinson's dementia, post-encephalitic Parkinson's disease, aluminum toxicity, and prion disease. In accordance with this embodiment, the composition is administered in an amount sufficient to eliminate or reduce the risk of the condition or to delay, inhibit, or prevent the onset of the condition. Prophylactic application also includes the administration of an antibody composition to prevent or delay the recurrence or relapse of a condition mediated by the tau protein or peptide in its pathological form. The present methods and compositions are especially suitable for prophylactic treatment of individuals who have a known

genetic risk of developing a condition involving pathological tau protein. For example, several mutations in tau protein are known to be associated with the frontotemporal dementia and parkinsonism linked to chromosome 17 (FTDP-17) (see Goedert and Jakes, "Mutations Causing Neurodegenerative Tauopathies," *Biochimica et Biophysica Acta* 1739(2-3): 240-50 (2005), which is hereby incorporated by reference in its entirety). Genetic markers of other tau related diseases, such as Alzheimer's disease are also known in the art. For example, mutations in the APP gene, particularly mutations at position 717 and positions 670 and 671 referred to as the Hardy and Swedish mutations respectively. Other markers of risk are mutations in the presenilin genes, PS1 and PS2, and ApoE4, family history of AD, hypercholesterolemia or atherosclerosis.

(415) In therapeutic applications, pharmaceutical compositions are administered to a subject suspected of, or already suffering from a condition associated with or caused by a pathological form of tau protein in an amount sufficient to cure, or at least partially arrest or alleviate, one or more symptoms of the condition and its complications. An amount adequate to accomplish this is defined as a therapeutically- or pharmaceutically-effective dose. In both prophylactic and therapeutic regimes, agents are usually administered in several dosages until a sufficient response has been achieved. An effective dose of the composition of the present invention, for the treatment of the above described conditions will vary depending upon many different factors, including means of administration, target site, physiological state of the patient, whether the patient is human or an animal, other medications administered, and whether treatment is prophylactic or therapeutic.

(416) In accordance with the prophylactic and therapeutic methods described herein, compositions comprising the antibody or binding fragments thereof are administered in a dosage ranging from about 0.0001 to 100 mg/kg, and more usually 0.01 to 10 mg/kg of the recipient's body weight. For example, the antibody or binding fragment thereof is administered in a dosage of 1, 2, 3, 4, 5, 6, 7, 8, 9 or 10 mg/kg, or higher, for example 15, 20, 25, 30, 35, 40, 45, 50, 55, 60, 65, 70, 75, 80, 85, 90, 95, or 100 mg/kg. An exemplary treatment regime entails administration once per every two weeks or once a month or once every 3 to 6 months. Intervals between single dosages can be weekly, monthly or yearly. Intervals can also be irregular as indicated by measuring blood levels of antibody in the patient. Alternatively, antibody can be administered as a sustained release formulation, in which case less frequent administration is required. Dosage and frequency vary depending on the half-life of the antibody in the patient. In general, human antibodies show the longest half life, followed by humanized antibodies, chimeric antibodies, and nonhuman antibodies. The dosage and frequency of administration can vary depending on whether the treatment is prophylactic or therapeutic. In prophylactic applications, a relatively low dosage is administered at relatively infrequent intervals over a long period of time. Some patients continue to receive treatment for the rest of their lives. In therapeutic applications, a relatively high dosage at relatively short intervals is sometimes required until progression of the disease is reduced or terminated, and preferably until the patient shows partial or complete amelioration of symptoms of disease. Thereafter, the patient can be administered a prophylactic regime.

(417) The mode of administration of the antibody, binding fragment thereof, or pharmaceutical composition described herein may be any suitable route that delivers the compositions to the host, such as parenteral administration, e.g., intradermal, intramuscular, intraperitoneal, intravenous or subcutaneous, pulmonary; transmucosal (e.g., oral, intranasal); using a formulation in a tablet, capsule, solution, powder, gel, particle; and contained in a syringe, an implanted device, osmotic pump, cartridge, micropump; or other means appreciated by the skilled artisan, as well known in the art. Site specific administration may be achieved by, for example, intraarticular, intrabronchial, intraabdominal, intracapsular, intracartilaginous, intracavitary, intracelial, intracerebellar, intracerebroventricular, intracolic, intracervical, intragastric, intrahepatic, intracardial, intraosteal, intrapericardiac, intraperitoneal, intrapleural, intraprostatic, intrapulmonary, intrarenal, intraretinal, intraspinal, intrasynovial, intrathoracic, intravascular, intravesical, intralesional, sublingual, intranasal, or transdermal delivery.

(418) Administration can be systemic or local. In one embodiment, it may be desirable to administer the antibodies of the invention locally to the area in need of treatment; this may be achieved by, for example, and not by way of limitation, local infusion, by injection, or by means of an implant, said implant being of a porous or non-porous material, including membranes and matrices, such as sialastic membranes, polymers, fibrous matrices (e.g., Tissuel®), or collagen matrices.

(419) In another embodiment, compositions containing the antibody or binding fragment thereof are delivered in a controlled release or sustained release system. In one embodiment, a pump is used to achieve controlled or sustained release. In another embodiment, polymeric materials can be used to achieve controlled or sustained release of the antibody compositions described herein. Examples of polymers used in sustained release formulations include, but are not limited to, poly(2-hydroxy ethyl methacrylate), poly(methyl methacrylate), poly(acrylic acid), poly(ethylene-co-vinyl acetate), poly(methacrylic acid), polyglycolides (PLG), polyanhydrides, poly(N-vinyl pyrrolidone), poly(vinyl alcohol), polyacrylamide, poly(ethylene glycol), polylactides (PLA), poly(lactide-co-glycolides) (PLGA), and polyorthoesters. The polymer used in a sustained release formulation is preferably inert, free of leachable impurities, stable on storage, sterile, and biodegradable. Also, the use of liposomes, microcapsules or microspheres, inclusion complexes, or other types of carriers known in the art are also contemplated.

(420) In yet another embodiment, a controlled or sustained release system can be placed in proximity of the prophylactic or therapeutic target, thus requiring only a fraction of the systemic dose. Controlled and/or release systems for delivery of antibodies known in the art are suitable for use and delivery of compositions containing the antibodies and binding fragments thereof as described herein, see e.g., Song et al, "Antibody Mediated Lung Targeting of Long-Circulating Emulsions," *PDA Journal of Pharmaceutical Science & Technology* 50:372-397 (1995); Cleek et al, "Biodegradable Polymeric Carriers for a bFGF Antibody for Cardiovascular Application," *Pro. Int'l. Symp. Control. Rel. Bioact. Mater.* 24:853-854 (1997); and Lam et al., "Microencapsulation of Recombinant Humanized Monoclonal Antibody for Local Delivery," *Proc. Int'l. Symp. Control Rel. Bioact. Mater.* 24:759-760 (1997), each of which is incorporated herein by reference in their entireties.

(421) In embodiments where the pharmaceutical composition comprises polynucleotides encoding the antibody or binding fragment thereof as described herein, the nucleic acid can be administered in vivo to promote expression of its encoded antibody, by constructing it as part of an appropriate nucleic acid expression vector and administering it so that it becomes intracellular, e.g., the by use of a retroviral vector (see e.g., U.S. Pat. No. 4,980,286 to Morgan et al., which is hereby incorporated by reference in its entirety). In one embodiment, polynucleotides encoding the tau antibody as described herein are incorporated into an AAV vector as described supra (e.g., AAV2, AAV4, AAV5, AAV7, AABV8, AAV9, AAVrh10, AAV2/8, AAV2/9, etc.) and delivered via intraparenchymal administration, including convection enhanced delivery (CED), intrathecal administration, intracerebroventricular administration, subpial administration, intramuscular administration, or intravenous administration. Other forms of nucleic acid delivery can also be employed, e.g., direct injection, use of microparticle bombardment (see e.g., a gene gun; Biolistic, Dupont), coating with lipids or cell-surface receptors or transfecting agents, or by administering it in linkage to a homeobox-like peptide which is known to enter the nucleus (see e.g., Joliot et al, *Proc. Natl. Acad. Sci. USA* 88: 1864-1868 (1991), which is hereby incorporated by reference in its entirety). Alternatively, a nucleic acid can be introduced intracellularly and incorporated within host cell DNA for expression by homologous recombination.

(422) If the methods described herein involve intranasal administration of the antibody composition, the composition can be formulated in an aerosol form, spray, mist or in the form of drops. In particular, prophylactic or therapeutic agents for use according to the present invention can be conveniently delivered in the form of an aerosol spray presentation from pressurized packs

or a nebulizer, with the use of a suitable propellant (e.g., dichlorodifluoromethane, trichlorofluoromethane, dichlorotetrafluoroethane, carbon dioxide or other suitable gas). In the case of a pressurized aerosol the dosage unit may be determined by providing a valve to deliver a metered amount. Capsules and cartridges (composed of, e.g., gelatin) for use in an inhaler or insufflator may be formulated containing a powder mix of the compound and a suitable powder base such as lactose or starch.

(423) If the methods described herein involve oral administration of the antibody compositions described herein, the compositions can be formulated orally in the form of tablets, capsules, cachets, gelcaps, solutions, suspensions, and the like. Tablets or capsules can be prepared by conventional means with pharmaceutically acceptable excipients such as binding agents (e.g., pregelatinised maize starch, polyvinylpyrrolidone, or hydroxypropyl methylcellulose); fillers (e.g., lactose, microcrystalline cellulose, or calcium hydrogen phosphate); lubricants (e.g., magnesium stearate, talc, or silica); disintegrants (e.g., potato starch or sodium starch glycolate); or wetting agents (e.g., sodium lauryl sulphate). The tablets may be coated by methods well-known in the art. Liquid preparations for oral administration may take the form of, but not limited to, solutions, syrups or suspensions, or they may be presented as a dry product for constitution with water or other suitable vehicle before use. Such liquid preparations may be prepared by conventional means with pharmaceutically acceptable additives such as suspending agents (e.g., sorbitol syrup, cellulose derivatives, or hydrogenated edible fats); emulsifying agents (e.g., lecithin or acacia); non-aqueous vehicles (e.g., almond oil, oily esters, ethyl alcohol, or fractionated vegetable oils); and preservatives (e.g., methyl or propyl-p-hydroxybenzoates or sorbic acid). The preparations may also contain buffer salts, flavoring, coloring, and sweetening agents as appropriate.

(424) In another embodiment, a pharmaceutical composition comprising a recombinant nucleic acid sequence encoding an antibody or binding portion thereof as described herein, is administered to a subject to facilitate in vivo expression and formation of the antibody for the treatment or prevention of conditions mediated by toxic oligomeric proteins or peptides in a subject. Expression vector constructs suitable for use in this embodiment of the disclosure are described supra.

(425) The polynucleotide compositions can result in the generation of the antibody in the subject within at least about 1 hour, 2 hours, 3 hours, 4 hours, 5 hours, 6 hours, 7 hours, 8 hours, 9 hours, 10 hours, 11 hours, 12 hours, 13 hours, 14 hours, 15 hours, 20 hours, 25 hours, 30 hours, 35 hours, 40 hours, 45 hours, 50 hours, or 60 hours of administration of the composition to the subject. The composition can result in generation of the synthetic antibody in the subject within at least about 1 day, 2 days, 3 days, 4 days, 5 days, 6 days, 7 days, 8 days, 9 days, or 10 days of administration of the composition to the subject. The composition can result in generation of the antibody in the subject within about 1 hour to about 6 days, about 1 hour to about 5 days, about 1 hour to about 4 days, about 1 hour to about 3 days, about 1 hour to about 2 days, about 1 hour to about 1 day, about 1 hour to about 72 hours, about 1 hour to about 60 hours, about 1 hour to about 48 hours, about 1 hour to about 36 hours, about 1 hour to about 24 hours, about 1 hour to about 12 hours, or about 1 hour to about 6 hours of administration of the composition to the subject.

(426) The composition, when administered to the subject in need thereof, can result in the persistent generation of the antibody in the subject. The composition can result in the generation of the antibody in the subject for at least about 1 day, 2 days, 3 days, 4 days, 5 days, 6 days, 7 days, 8 days, 9 days, 10 days, 11 days, 12 days, 13 days, 14 days, 15 days, 16 days, 17 days, 18 days, 19 days, 20 days, 21 days, 22 days, 23 days, 24 days, 25 days, 26 days, 27 days, 28 days, 29 days, 30 days, 31 days, 32 days, 33 days, 34 days, 35 days, 36 days, 37 days, 38 days, 39 days, 40 days, 41 days, 42 days, 43 days, 44 days, 45 days, 46 days, 47 days, 48 days, 49 days, 50 days, 51 days, 52 days, 53 days, 54 days, 55 days, 56 days, 57 days, 58 days, 59 days, or 60 days.

(427) Formulations for injection may be presented in unit dosage form (e.g., in ampoules or in multi-dose containers) with an added preservative. The compositions may take such forms as suspensions, solutions or emulsions in oily or aqueous vehicles, and may contain formulatory

agents such as suspending, stabilizing and/or dispersing agents. Alternatively, the active ingredient may be in powder form for constitution with a suitable vehicle (e.g., sterile pyrogen-free water) before use. The methods of the invention may additionally comprise of administration of compositions formulated as depot preparations. Such long acting formulations may be administered by implantation (e.g., subcutaneously or intramuscularly) or by intramuscular injection. Thus, for example, the compositions may be formulated with suitable polymeric or hydrophobic materials (e.g., as an emulsion in an acceptable oil) or ion exchange resins, or as sparingly soluble derivatives (e.g., as a sparingly soluble salt).

(428) The methods of the invention encompass administration of compositions formulated as neutral or salt forms. Pharmaceutically acceptable salts include those formed with anions such as those derived from hydrochloric, phosphoric, acetic, oxalic, tartaric acids, etc., and those formed with cations such as those derived from sodium, potassium, ammonium, calcium, ferric hydroxides, isopropylamine, triethylamine, 2-ethylamino ethanol, histidine, procaine, etc.

(429) Generally, the ingredients of compositions are supplied either separately or mixed together in unit dosage form, for example, as a dry lyophilized powder or water free concentrate in a hermetically sealed container such as an ampoule or sachette indicating the quantity of active agent. Where the mode of administration is infusion, composition can be dispensed with an infusion bottle containing sterile pharmaceutical grade water or saline. Where the mode of administration is by injection, an ampoule of sterile water for injection or saline can be provided so that the ingredients may be mixed prior to administration.

(430) The antibodies, binding fragments thereof, or pharmaceutical compositions containing the same can be packaged in hermetically sealed containers such as an ampoule or sachette indicating the quantity of the antibody or binding fragment thereof. In one embodiment, one or more of the antibodies, or pharmaceutical compositions of the invention is supplied as a dry sterilized lyophilized powder or water free concentrate in a hermetically sealed container and can be reconstituted (e.g., with water or saline) to the appropriate concentration for administration to a subject. In one embodiment, one or more of the antibodies or pharmaceutical compositions of the invention is supplied as a dry sterile lyophilized powder in a hermetically sealed container at a unit dosage of at least 5 mg, for example at least 10 mg, at least 15 mg, at least 25 mg, at least 35 mg, at least 45 mg, at least 50 mg, at least 75 mg, or at least 100 mg. The lyophilized antibodies or pharmaceutical compositions of the invention should be stored at between 2° C. and 8° C. in its original container and the antibodies, or pharmaceutical compositions of the invention should be administered within 1 week, for example within 5 days, within 72 hours, within 48 hours, within 24 hours, within 12 hours, within 6 hours, within 5 hours, within 3 hours, or within 1 hour after being reconstituted. In an alternative embodiment, one or more of the antibodies or pharmaceutical compositions of the invention are supplied in liquid form in a hermetically sealed container indicating the quantity and concentration of the antibody. In a further embodiment, the liquid form of the administered composition is supplied in a hermetically sealed container at least 0.25 mg/ml, for example at least 0.5 mg/ml, at least 1 mg/ml, at least 2.5 mg/ml, at least 5 mg/ml, at least 8 mg/ml, at least 10 mg/ml, at least 15 mg/ml, at least 25 mg/ml, at least 50 mg/ml, at least 75 mg/ml or at least 100 mg/ml. The liquid form should be stored at between 2° C. and 8° C. in its original container.

(431) The antibodies and binding fragments described herein can be incorporated into a pharmaceutical composition suitable for parenteral administration. In one aspect, antibodies will be prepared as an injectable solution containing 0.1-250 mg/ml antibody. The injectable solution can be composed of either a liquid or lyophilized dosage form in a flint or amber vial, ampule or pre-filled syringe. The buffer can be L-histidine (1-50 mM), optimally 5-10 mM, at pH 5.0 to 7.0 (optimally pH 6.0). Other suitable buffers include but are not limited to, sodium succinate, sodium citrate, sodium phosphate or potassium phosphate. Sodium chloride can be used to modify the tonicity of the solution at a concentration of 0-300 mM (optimally 150 mM for a liquid dosage

form).

(432) Cryoprotectants can be included for a lyophilized dosage form, principally 0-10% sucrose (optimally 0.5-1.0%). Other suitable cryoprotectants include trehalose and lactose. Bulking agents can be included for a lyophilized dosage form, principally 1-10% mannitol (optimally 2-4%). Stabilizers can be used in both liquid and lyophilized dosage forms, principally 1-50 mM L-Methionine (optimally 5-10 mM). Additional surfactants include but are not limited to polysorbate 20 and BRIJ surfactants. The pharmaceutical composition comprising the antibodies described herein prepared as an injectable solution for parenteral administration, can further comprise an agent useful as an adjuvant, such as those used to increase the absorption, or dispersion of the antibody. A particularly useful adjuvant is hyaluronidase, such as Hylanex® (recombinant human hyaluronidase). Addition of hyaluronidase in the injectable solution improves human bioavailability following parenteral administration, particularly subcutaneous administration. It also allows for greater injection site volumes (i.e. greater than 1 ml) with less pain and discomfort, and minimum incidence of injection site reactions (see WO 04/078140 to Bookbinder et al., and U.S. Patent Appl. Publication No. US2006104968 to Bookbinder et al., which are hereby incorporated herein by reference in their entirety).

(433) The antibodies and binding fragments described herein can also be employed in a number of diagnostic, prognostic and research applications.

(434) Another aspect of the present disclosure is directed to a method of diagnosing Alzheimer's disease or a tauopathy in a subject. This method involves detecting, in the subject, the presence of tau protein or peptide using a diagnostic reagent, wherein the diagnostic reagent comprises an antibody or binding fragment described herein. The diagnosis of Alzheimer's disease or a tauopathy in the subject is based on the detection of tau protein or peptide in the subject.

(435) In one embodiment, the method of diagnosing Alzheimer's disease and/or a tauopathy in a subject involves the detection of tau protein or peptide in the subject. In one embodiment, the method of diagnosing Alzheimer's disease and/or a tauopathy in the subject involves the detection of accumulated tau protein or peptide. In another embodiment, the method of diagnosing Alzheimer's disease and/or a tauopathy in the subject involves the detection of pathological tau protein in the subject.

(436) Detecting the presence of tau protein or peptide in a subject using the antibodies or antibody fragments thereof as described herein can be achieved by obtaining a biological sample from the subject (e.g., blood, urine, cerebral spinal fluid, ocular lacrimal secretion, saliva, feces, nasal brushings and tissue or organ biopsy), contacting the biological sample with the diagnostic antibody reagent, and detecting binding of the diagnostic antibody reagent to tau protein or peptide if present in the sample from the subject. Assays for carrying out the detection of tau protein or peptide in a biological sample using a diagnostic antibody are well known in the art and include, without limitation, ELISA, immunohistochemistry, SIMOA (single molecule array), and Western blot.

(437) In accordance with this and other embodiments described herein, the tau antibody or binding fragments described herein are coupled to a detectable label. The label can be any detectable moiety known and used in the art. Suitable labels include, without limitation, radioisotopes or radionuclides (e.g., ³H, ¹⁴C, ¹⁸F, ³⁵S, ⁹⁰Y, ⁹⁹Tc, ¹¹¹In, ¹²⁴I, ¹¹³I, ¹⁷⁷Lu, ¹⁶⁶Ho, ⁸⁹Zr, or ¹⁵³Sm); fluorescent labels (e.g., FITC, rhodamine, lanthanide phosphors), enzymatic labels (e.g., horseradish peroxidase, luciferase, alkaline phosphatase); chemiluminescent markers; biotinyl groups; predetermined polypeptide epitopes recognized by a secondary reporter (e.g., leucine zipper pair sequences, binding sites for secondary antibodies, metal binding domains, epitope tags); and magnetic agents, such as gadolinium chelates.

(438) Detecting the presence of tau in a subject using the diagnostic antibody reagent of the present invention can also be achieved using in vivo imaging techniques. In vivo imaging involves

administering to the subject the antibody or binding fragments thereof described herein, and detecting the binding of the antibody or binding fragment thereof to the tau in vivo.

(439) Diagnostic antibodies or similar reagents can be administered by intravenous injection into the body of the patient, by injection into the cerebrospinal fluid, by intracranial injection directly into the brain, or by drilling a hole through the skull. The dosage of antibody should be within the same ranges as for treatment methods. In accordance with this embodiment, the antibody or binding fragment is coupled to an imaging agent to facilitate in vivo imaging. The imaging agent can be any agent known to one of skill in the art to be useful for imaging, preferably being a medical imaging agent. Examples of medical imaging agents include, but are not limited to, single photon emission computed tomography (SPECT) agents, positron emission tomography (PET) agents, magnetic resonance imaging (MRI) agents, nuclear magnetic resonance imaging (NMR) agents, x-ray agents, optical agents (e.g., fluorophores, bioluminescent probes, near infrared dyes, quantum dots), ultrasound agents and neutron capture therapy agents, computer assisted tomography agents, two photon fluorescence microscopy imaging agents, and multi-photon microscopy imaging agents. Exemplary detectable markers include radioisotopes (e.g., ^{18}F , ^{11}C , ^{13}N , ^{64}Cu , ^{124}I , ^{76}Br , ^{82}Rb , ^{68}Ga , $^{99\text{m}}\text{Tc}$, ^{111}In , ^{125}I , ^{201}Tl , ^{89}Zr , or ^{15}O , which are suitable for PET and/or SPECT use) and ultra-small superparamagnetic particles of iron oxide (USPIO) which are suitable for MRI.

(440) Diagnosis of a condition involving pathological tau is performed by comparing the amount, size, and/or intensity of detected tau in a sample from the subject or in the subject, to corresponding baseline values. An appropriate baseline value can be the average level of tau found in a population of undiseased individuals. Alternatively, an appropriate baseline value may be the level of tau in the same subject determined at an earlier time.

(441) The diagnostic methods described herein can also be used to monitor a subject's response to therapy. In this embodiment, detection of tau in the subject is determined prior to or concurrent with the commencement of treatment. The level of tau in the subject at this timepoint is used as a baseline value. At various times during the course of treatment the detection of tau is repeated, and the measured values thereafter compared with the baseline values. A decrease in values relative to baseline signals a positive response to treatment. No change or an increase in values relative to baseline signals an inadequate response to treatment. The treatment plan of an individual can be modified based on the results of monitoring the levels of tau in the subject.

(442) A related aspect of the disclosure is directed to a method of identifying a subject's risk for developing a condition mediated by or associated with a pathological form of tau. This method involves detecting, in the subject, the presence of accumulated tau protein or peptide using a diagnostic reagent comprising the antibody or binding fragment thereof described herein, and identifying the subject's risk of developing a condition mediated by or associated with accumulated tau based on the results of the detecting step.

(443) Methods of detecting the presence of tau in the subject or in a sample from the subject include the in vitro and in vivo methods described supra. In one embodiment, the subject is not exhibiting any definitive signs or symptoms of a condition involving a pathological tau protein, and employment of this method serves as an early diagnostic. In another embodiment, the subject is not exhibiting any signs or symptoms of a condition involving a pathological tau protein, but has a genetic predisposition to a condition and employment of this method serves to predict the likelihood that the individual will develop the condition involving a pathological tau protein in the future. In either embodiment, appropriate therapeutic and/or prophylactic intervention can be employed, e.g., administration of a therapeutic composition containing an antibody or polynucleotides encoding an antibody in an amount effective to slow or prevent the onset or progression of the condition.

(444) Another aspect of the present disclosure is directed to a diagnostic kit that comprises the antibody or binding fragment thereof as described herein and a detectable label.

(445) A suitable detectable label is any moiety attached to an antibody or an analyte to render the reaction between the antibody and the analyte detectable. A label can produce a signal that is detectable by visual or instrumental means. Various labels include signal-producing substances, such as chromogens, fluorescent compounds, chemiluminescent compounds, radioactive compounds, and the like. Representative examples of detectable labels include moieties that produce light, e.g., acridinium compounds, and moieties that produce fluorescence, e.g., fluorescein. In this regard, the moiety itself may not be detectable, but becomes detectable upon reaction with yet another moiety.

(446) Other suitable detectable labels include radioactive labels (e.g., H, I, S, C, F, and P), enzymatic labels (e.g., horseradish peroxidase, alkaline peroxidase, glucose 6-phosphate dehydrogenase, and the like), chemiluminescent labels (e.g., acridinium esters, thioesters, or sulfonamides; luminol, isoluminol, phenanthridinium esters, and the like), fluorescent labels (such as fluorescein (e.g., 5-fluorescein, 6-carboxyfluorescein, 3'-carboxyfluorescein, 5(6)-carboxyfluorescein, 6-hexachloro-fluorescein, 6-tetrachlorofluorescein, fluorescein isothiocyanate, and the like)), rhodamine, phycobiliproteins, R-phycoerythrin, quantum dots (e.g., zinc sulfide-capped cadmium selenide), a thermometric label, or an immuno-polymerase chain reaction label.

(447) TABLE-US-00059 TABLE 3 Polynucleotide Sequences of Tan Antibodies Enriched by Solid Phase Panning

NA	AA	SEQ	SEQ	ID	ID	Antibody	ID	Nucleotide
Sequence	NO:	NO:	03-T-sR2-1D10					

ATGAAATACCTATTGCCTACGGCGGCCGCTGGATTGTTATTACTCGC 218 115
GGCCCAGCCGGCCATGGCGCAGGTGCAGCTGCAGGAGTCTGGGGG
AGGATTGGTGCAGGCTGGGGGGCTCTCTGAGACTCTCCTGTACAGCC
TCTGGACGCACCTTCCGTGCCTATGCCATGGGGTGGTTCCGCCAGGC
TCCAGGGAAGGAGCGTGAGTTGGTAGCAGCTATTAGCCGCACTGGT
GGTGTCAACCTATGCAGACTCCGTGAAGGGCCGATTCACCATCTC
CAGAGACAACGCCAAGAATACGGTGTATCTGCAAATGAACAACCTG
AAGACTGAGGACACGGCCGTCTATTATTGTAATGCATACTTCCGTTG
GGGTACTCGCTACTGGGGGCCAGGGGACCCAGGTCACCGTCTCCTCA
ACTAGTGGCCCCGGGAGGCCAACACCATCACCAACCATCATGGCGCAG
AACAAAACTCATCTCAGAAGAGGATCTGTCTTAG 04-T-sR2-2A11

ATGAAATACCTATTGCCTACGGCGGCCGCTGGATTGTTATTACTCGC 219 116
GGCCCAGCCGGCCATGGCGCAGGTGCAGCTGCAGGAGTCTGGGGG
AGGCTTGGTGCAGCCTGGGGGGTCTCTGAGACTCTCCTGTGCAGCC
TCTGGAAGGATCTTCAGTATTTGGACCATGGGCTGGTACCGCCAGG
CTCCAGGGAAGCAGCGCGAGTTGGTCGCGGCTATTACTAGTGTTGG
TAACACAGACTATGCAGACTCCGTGAAGGGCCGATTCACCATCTCCA
GAGAGACTGCCAAGAAGACGGTGTATCTGCAAATGAACAGCCTGAA
ACCTGAGGACACGGCCGTCTATTACTGTAATGTAGAATCTCGGCGTG
GTATAGGCTTCCTACGTAAACGTATAGCTACTGGGGGCCAGGGGAC
CCAGGTCACCGTCTCCTCAACTAGTGGCCCCGGGAGGCCAACACCATC
ACCACCATCATGGCGCAGAACAAAACTCATCTCAGAAGAGGATCT GTCTTAG 11-T-

sR2-1E4 ATGAAATACCTATTGCCTACGGCGGCCGCTGGATTGTTATTACTCGC 220 117
GGCCCAGCCGGCCATGGCGCAGGTGCAGCTGCAGGAGTCTGGGGG
AGGCTTGGTGCAGCCTGGGGGGTCTCTGACACTCTCCTGTGCAGCCT
CTGGAAGCATCTTCAGAATCAATAACATGGGCTGGTTCCGCCAGGCT
CCAGGGAAGCAGCGCGAGTTGGTCGCAACTATTACTCGTGGTGGTA
ACACAACTATGCAGACTCCGTGAAGGGCCGATTCACCATCTCCAGA
GACAACGCCAAGAAAACGCTGTATCTGCAAATGAACAGCCTGAAAC
CTGAGGACACGGCCGTCTATTACTGTAATGCAAATTATCTTATTCGAT
CATACTGGGGGCCAGGGGACCCAGGTCACCGTCTCCTCAACTAGTGG

CCCGGGAGCCGACCAAGGATCTGTCTTAG 12-T-sR2-2B4
ATGAAATACCTATTGCCTACGGCGGCCGCTGGATTGTTATTACTCGC 221 118
GGCCCAGCCGGCCATGGCGCAGGTGCAGTTGCAGGAGTCTGGGGG
AGGCTTGGTGCAGCCTGGGGGGTCTCTGACACTCTCCTGTGCAGCCT
CTGGAAGCATCTTCAGAATCAATAACATGGGCTGGTTCCGCCAGGCT
CCAGGGAAGCAGCGCGAGTTGGTCGCAACTATTACTCGTGGTGGTA
ACACAAACTATGCAGACTCCGTGAAGGGCCGATTCACCATCTCCAGA
GACAACGCCAAGAACACGGTGTATCTGCAAATGGACAGCCTGAAAC
CTGAGGACACGGCCGTCTATTACTGTAATGTAAAAAAGCATTTCGGC
ATTCGATATGACTACTGGGGCCAGGGGACCCAGGTCACCGTCTCCTC
AACTAGTGGCCCCGGGAGGCCAACACCATCACCAACCATCATGGCGCA
GAACAAAAACTCATCTCAGAAGAGGATCTGTCTTAG 13-T-sR2-2F8
ATGAAATACCTATTGCCTACGGCGGCCGCTGGATTGTTATTACTCGC 222 119
GGCCCAGCCGGCCATGGCGCAGGTGCAGCTGCAGGAGTCTGGGGG
AGGCTTGGTGCAGGCTGGGGGGTCTCTGAGACTCTCCTGCGCAGCC
TCTGGACGCACCTTCAGCAACTATGCCTTGGCCTGGTTCCGCCAGGC
TCCAGGGTTGGAGCGTGAGTTTGTGTCAAGTATTAGCTGGAGTGGT
GGTGTACTATACTATGCAGACTCCGTGAAGGGCCGATTCACCATGTC
CAGAGACAACGGCAAGAACACGGTGTACCTGCAAATGAACAGCCTG
AAACCTGAGGACACGGCCGTTTATTACTGTGCAGCGAACGCCGGGT
TGAGTTTATTAAGGAATTGGAGGACTAATGAGTATGCCTACTGGGG
CCAGGGGACCCAGGTCACCGTCTCCTCAACTAGTGGCCCCGGGAGGC
CAACACCATCACCAACCATCATGGCGCAGAACAAAAACTCATCTCAGA
AGAGGATCTGTCTTAG 18-T-sR2-1A8
ATGAAATACCTATTGCCTACGGCGGCCGCTGGATTGTTATTACTCGC 223 120
GGCCCAGCCGGCCATGGCGCAGGTGCAGTTGCAGGAGTCTGGGGG
AGGCTTGGTGCAGGCTGGGGGGTCTCTGAGACTCTCCTGTGCAGCC
TCTGGAAGCATCTTCCGTATCAATGCCATGGCCTGGTACCGCCAGGC
TCCAGGGAAGCAGCGCGAGTTGGTCGCTAGAATTAATACTGGTGGT
AACACAAACTATGCAGGCTCCGTGAAGGGCCGATTCACCATCTCCAG
AGACAACGGCAAGAACACGGTGTATCTGCAAATGAACAGCCTGAAA
CCTGAGGACACGGGCGTCTATTACTGTAATGTACAGAGATTCATCAC
TACCTATTGGGGCCAGGGGACCCAGGTCACCGTCTCCTCAACTAGTG
GCCCCGGGAGGCCAACACCATCACCAACCATCATGGCGCAGAACAAAA
ACTCATCTCAGAAGAGGATCTGTCTTAG 20-T-sR2-2B7
ATGAAATACCTATTGCCTACGGCGGCCGCTGGATTGTTATTACTCGC 224 121
GGCCCAGCCGGCCATGGCGCAGGTGCAGTTGCAGGAGTCTGGGGG
AGGATTGGTGCAGGCTGGGGGGTCTCTGAGACTCTCCTGTGCAGCC
TCTGGACGCACCTTCAGTACCTATAGGATGGGCTGGTTCCGCCAGGC
TCCAGGGAAGGAGCGTGAGTTTGTAGCAGCTATTAGGTGGAGTACA
AGTTATGCAGATTCCGTGAAGGGCCGATTCATCATCTCCAGAGACAA
CGCCAAGAACACGGTGTATCTGCAAATGAACAGCCTGAAACCTGAG
GACACGGCCGTTTATTACTGTGCAACAGGGAGGGCCTGGAGTACAC
TGGCCACGACATATGTTTACTGGGGCCAGGGGACCCAGGTCACCGT
CTCCTCAACTAGTGGCCCCGGGAGGCCAACACCATCACCAACCATCATG
GCGCAGAACAAAAACTCATCTCAGAAGAGGATCTGTCTTAG 21-T-sR2-2G7
ATGAAATACCTATTGCCTACGGCGGCCGCTGGATTGTTATTACTCGC 225 122
GGCCCAGCCGGCCATGGCGCAGGTGCAGCTGCAGGAGTCTGGGGG
AGGATTGGTGCAGGCCGGGGGGTCTCTGAGACTCTCCTGTGCAGCC

TCTGGACGACACGACCTTACGATATGAGGCGCTTGGTTCCGCGGCGGCTCCAGGGG
TCCAGGGAAGGAGCGTGAGTTTGTAGCAGGTATTAGCTGGAGTGG
AACATCGTATGCAGACTCCGTGAAGGGCCGATTACCATCTCCAGAG
ACAACGCCAAGAACACGCTGTATCTGCAAATGAACAGTCTGAAATCT
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(448) TABLE-US-00060 TABLE 4 Polynucleotide Sequences of Tan Antibodies
Enriched by Solution Phase Panning NA AA SEQ SEQ ID ID Antibody ID Nucleotide
Sequence NO: NO: 01-T-bR3-1E2

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AGAGGATCTGTCTTAG

EXAMPLES

(449) Examples are provided herein to illustrate the present invention. These examples are not meant to constrain the present invention to any particular application or theory of operation.

Example 1—Production of Llama Single Domain Antibodies Against Human Tau

(450) Immunizations: A llama was immunized with Tau441 according to the following schedule. In general, immunizations occurred once per month, with serum and/or PBMCs taken 1 week after boosts.

(451) TABLE-US-00061 Tau441 Immunization Schedule Pre-bleed Day 0 Immunization 1 200 ug

CFA, SQ Day 0 Immunization 2 100 ug IF A, SQ Day 21 Bleed 1 Day 28 Immunization 3 100 ug IF A 100 ug Adjuplex, Day 49 Bleed 2 SQ Day 56 Immunization 4 100 ug IF A 100 ug Adjuplex, Day 77 Bleed 3 SQ Day 84 Immunization 5 100 ug IF A 100 ug Adjuplex, Day 105 Bleed 4 SQ Day 112~1 × 10^{sup.8} PBMCs Immunization 6 200 ug Adjuplex split IM/SQ Day 142 Bleed 5 Day 149 Immunization 7 200 ug Adjuplex split IM/SQ Day 188 Bleed 6 Day 196~1 × 10^{sup.8} PBMCs Complete Freund's Adjuvant (CFA), Incomplete Freund's Adjuvant (IFA), Adjuplex Adjuvant (Sigma), Subcutaneous (SQ), Intramuscular (IM).

(452) Serum titer levels were determined by indirect ELISA using the following protocol: 1) Antigen was coated in 96 well Corning 9018 high-binding polystyrene assay plates at 2 µg/mL in PBS, 4 degrees ON. 2) Wash plate 4 times with PBS containing 0.05% Tween20 (PBST). 3) Block with PBST/5% non-fat dried milk (PBST/milk) for 2 hr at room temperature. 4) Wash 4 times with PBST. 5) Dilutions of sera were made in PBST/milk and incubated in plate for 1 hr at room temperature. 6) Wash 4 times with PBST. 7) Incubate with Rabbit anti-Llama IgG(H+L) (Life Technologies), 1:10,000 dilution in PBST/milk, 1 hr room temperature. 8) Wash 4 times with PBST. 9) Bound antibody was detected with Gt anti-Rb IgG(H+L)-HRP (Jackson), 1:10,000 in PBST/milk, 45 min room temperature.

(453) Library Construction: Approximately 100 mL of llama blood was collected in EDTA-coated tubes (Becton Dickenson) and peripheral blood mononuclear cells (PBMCs) were isolated using Hisotpaque-1077 (Sigma). Approximately 1×10^{sup.8} PBMCs were isolated from the specific bleeds listed above and RNA prepared using RNeasy (Qiagen). cDNA was synthesized from a total of 50 µg of RNA (25 ug from each bleed) using random hexanucleotide primers and SuperScript reverse transcriptase (Life Technologies). Single domain antibodies (VHH) were cloned in a two step PCR process using the cDNA as template. The first PCR amplified the variable domains of all immunoglobulin heavy chains, both single chain (VHH) and conventional antibodies (VH) using CaL1/CaL2 primers:

(454) TABLE-US-00062 (SEQ ID NO: 311) 5'-GTCCTGGCTGCTCTTCTACAAGG-3' (SEQ ID NO: 312) 5'-GGTACGTGCTGTTGAACTGTTCC-3'.

(455) The second nested PCR specifically amplified llama single domain antibodies (ProSci proprietary primers). PCR products were gel purified and the DNA fragments encoding the VHH domains were cloned into phage display vector pADL-23c (Antibody Design Labs) and electroporated into *E. coli* TG1 cells, yielding a library of approximately 1×10^{sup.9} in size.

Example 2: Selection of Single Domain Antibodies Against Tau

(456) Phage displaying VHH antibodies were rescued from individual libraries with helper phage M13K07 and subjected to biopanning in two different ways: solid-phase and solution-phase panning. For solid-phase panning, Tau was adsorbed on 2 um diameter polystyrene beads (Polysciences, Inc), blocked with PBST containing 5% non-fat dried milk, and incubated with 2×10^{sup.11} phage in 1 mL PBST/milk for 1 hr at room temperature. For solution-phase panning, proteins were biotinylated using Lightning-Link Rapid Biotin (Innova Biosciences) and lug of biotinylated Tau incubated with 2×10^{sup.11} phage in 1 mL PBST/milk for 1 hr at room temperature. Polystyrene beads were captured by centrifugation and washed extensively with PBST. Biotinylated proteins bound to phage were captured with streptavidin-coated magnetic Dynabeads (Life Technologies) and extensively washed with PBST. In both cases, bound phage were eluted with both high pH (100 mM triethylamine, pH~10) and low pH (100 mM glycine, pH~2.5) for 5 min and neutralized with 1M Tris pH 7.5. *E. coli* strains TG1 and SS320 were then infected with eluted phage and used for a subsequent round of panning (TG1) or to express antibodies (SS320).

Example 3—Expression of Single Domain Antibodies for Screening

(457) Individual SS320 clones were grown in 96 well plate format in 100 uL of 2YT/1% glucose/100 µg/mL ampicillin overnight at 37° C. 10 uL of the overnight cultures were inoculated into 1 mL each of 2YT/0.1% glucose/100 µg/mL ampicillin in deep 96 well blocks and grown at

37° C., 200 rpm for 3 hr until cells were in log phase. Single domain antibody expression was then induced with IPTG (1 mM final) and the temperature reduced to 30° C. overnight. The next day, bacterial cultures were centrifuged at 3200 g for 10 min and supernatant removed. The remaining bacterial pellets were frozen at -70° C. Bacterial pellets were then thawed, resuspended in 300 uL of PBS, and incubated for 30 min at room temperature. Cellular debris was removed by centrifugation at 3200 g for 10 min and the antibody-containing supernatants transferred to a fresh 96 well plate for storage at -70° C. until used.

(458) ELISA screening of the supernatants was performed using the following protocol: 1) coat antigen at 2 µg/mL in PBS, 4° C. ON (use 96 well Corning 9018 high-binding polystyrene assay plates). 2) wash 4 times with PBST 3) 2 hr block with PBST/5% milk, room temp 4) wash 4 times PBST 5) add dilutions of sdAb supes (1:1) in PBST/5% milk, 1 hr room temp 6) wash 4 times PBST 7) add ProSci's anti-c-myc-tag antibody (Cat. No. PM-7669), 1:1,000 in PBST/5% milk, 1 hr room temp 8) wash 4 times PBST 9) detect with goat anti-mouse IgG-HRP (Jackson Cat. No. 115-035-164), 1:5,000 in PBST/5% milk, incubate 30 to 45 min room temperature 10) wash 4 times PBST 11) develop 10 min to 1 hr depending on signal

Example 4—Single Domain Antibody Purification

(459) Single Domain Antibodies were purified from 50 mL cultures as follows. Inoculated 3 mL of 2YT/1% glucose/100 µg/mL Amp with specific clone in SS320 cells and grew overnight 37 degrees, 200 rpm. Next day, 500 uL of overnight culture was added to 50 mL of 2YT/0.1% glucose/100 µg/mL ampicillin and grown at 37 degrees, 200 rpm. After OD.sub.600 reached 0.7, induced the culture with IPTG (1 mM final) and grown overnight at 30 degrees.

(460) The next day, bacteria were collected by centrifugation for 15 min, 3500 g at room temperature. The cell pellet was resuspended in 2.5 mL of ice-cold TES (20 mM Tris, 0.5 mM EDTA, 17% sucrose), and incubated for 1 hr on ice. 5 mL of TES/4 (TES diluted 1 to 4 in water) was added, and incubation on ice continued for 45 min mixing occasionally. The suspension was centrifuged at 10,000 g for 30 min, 4 degrees and the supernatant was collected. The His-tagged single domain antibodies were purified by using IMAC (Immobilized Metal Affinity Chromatography) according to the manufacturer's instructions (Qiagen).

Example 5—Single Domain Antibody Characterization

(461) Single domain antibody (sdAb) titer following administration of tau441 as described in Example 1 was measured by ELISA. ELISA plates were coated with 2 µg/ml of the longest tau isoform (441 aa), and reacted with various bleeds obtained after several immunizations with tau441. The hydrogen peroxidase (HRP)-linked secondary antibody recognizes single-domain llama antibodies. Some auto-sdAbs against tau were detected in the pre-bleed. FIG. 1A shows that sdAb titer in the tau protein immunized llama peaks at bleed 2 and remains stable through bleed 6.

(462) To test the reactivity of the tau sdAb to human tauopathy brains, paired helical filament (PHF) enriched tau fractions from two different Progressive Supranuclear Palsy (PSP) human tauopathy cases were spotted onto nitrocellulose membranes and incubated with plasma from each of the seven time points, pre-treatment through bleed 6, at a 1:1000 dilution overnight. Following this, blots were washed and incubated with HRP-conjugated llama secondary antibody at 1:2000 and developed together to ensure that time in the developing fluid and exposure would be identical. For both cases, signal increased reaching a maximum at bleed 4, followed by a small decrease in signal at bleed 5 and a larger decrease at bleed 6 (FIG. 1). sdAb clones were obtained from B-cells from bleeds 4 and 5 using the services of ProSci Inc. (Poway, CA).

(463) Phage display library panning against biotinylated solution phase (FIG. 2A) and solid phase (FIG. 2B) recTau441 was used to enrich for clones (Abs>0.1) with 85% (3.sup.rd round: one plate) and 24% (2.sup.nd round—2 plates) positivity, respectively.

(464) The sdAb clones have a diverse binding profile as shown in FIGS. 3A-3D. Some preferably bind to paired helical filament (PHF) enriched fraction from Alzheimer's brain, whereas others bind better to recTau441 or bind well to both PHF and recTau441. Note that these are culture

supernatants and clone expression may vary, which will affect absorbance values. Supernatants can be expected to contain 0.1-1 µg/ml. X and Y axes are absorbance values for recTau441 and PHF, respectively.

(465) The anti-tau sdAbs recognize tau pathology in human tauopathy brains. A monoclonal sdAb, 2B8, stains numerous tangles and pretangles in a human tauopathy brain as shown in the image of FIG. 4A. Its epitope has yet to be characterized. Several other anti-tau sdAbs recognize pathological tau on tissue sections with varying reactivity towards tangles, pretangles, somal, axonal and dendritic tau, reflecting their diversity. For comparison, PHF1 staining of a phospho-tau epitope of adjacent section of the same brain reveals mostly mature tangles, dystrophic neurites and possibly glial cells (FIG. 4B).

(466) To examine whether the tau sdAb can clear tau from neurons, primary neurons were prepared from day 0 Tg JNPL3 tauopathy pups, incubated with 1 µg/ml of the 2B8 anti-tau sdAb, and compared to untreated neurons from the same animal. Immunoblotting was carried out for neuronal marker NeuN and total tau levels. NeuN levels were unchanged between control and treated cells, indicating that the sdAb is not toxic to neurons (FIG. 5A). Neurons treated with 2B8 for 3 days had significantly lower tau levels compared to untreated controls ($p=0.03$) as shown in the graph of FIG. 5B.

(467) The ability of monoclonal (m) and polyclonal (p) tau sdAbs to prevent toxicity of AD brain derived pathological tau protein was investigated. To examine extracellular effects, antibodies (1 µg/ml) were administered with enriched human paired helical filaments (PHF; 1 µg/ml) derived from AD brain to primary tauopathy JNPL3 mouse cultures (PHF+Ab) for 7 days. Under these conditions, m-sdAb (sdAb1=2B8) and p-sdAbs work equally well extracellularly in preventing PHF neurotoxicity (FIG. 6A) and in clearing tau pathology (FIG. 6B). To examine intracellular effects, antibodies are administered 24 h after PHF (PHF.fwdarw.Ab), which by then has been taken up into the neurons. Under these conditions (PHF.fwdarw.Ab), both the m-sdAb (sdAb1) and p-sdAbs were equally effective in preventing PHF neurotoxicity (FIG. 6A), and in clearing tau pathology (FIG. 6B), examined 7 days later, with comparable efficacy to the extracellular condition.

(468) The anti-tau sdAbs also prevent toxicity of and clear human Alzheimer's brain derived tau protein in tauopathy human neuron-like cultures. Differentiated SH-SY5Y human neuroblastoma cells were treated with PHF-enriched tau for 24 h, washed to remove extracellular PHF, treated with sdAb for 72 h, and then processed for western blots. PHF showed a dose-dependent neurotoxicity that was partially alleviated by the sdAbs as shown in FIG. 7A. Likewise, PHF treatment led to a dose-dependant increase in total- (FIG. 7B) and phospho-tau (FIG. 7C) that was strongly attenuated by the sdAbs. #, ##, ###, ####: $p<0.05$, 0.01, 0.001, 0.0001, compared to untreated control. **, ***, ****: $p<0.01$, 0.001, 0.0001, compared to PHF alone. sdAb1: 2B8; p-sdAbs: polyclonal sdAb.

(469) To test distribution of sdAb upon systemic administration, mice received an intravenous injection of 250 µg of sdAb labeled with a near infrared dye as previously described for anti-tau scFc (Krishnaswamy et al., "Antibody-derived In vivo Imaging of tau Pathology," *J Neurosci* 10:34(50): 16835-50 (2014), which is hereby incorporated by reference in its entirety). In vivo imaging system (IVIS) signal was obtained from the anesthetized mice several times within the first hour. Brain signal (blue circle) was strong (yellow) in JNPL3 mouse G47 (FIG. 8A; 17 min), and G48 (FIG. 8B; 10 min) with strong signal (yellow) seen also presumably in the kidneys (FIG. 8A) reflecting probe clearance and in the spinal cord (FIGS. 8A and 8B), which in the JNPL3 model has extensive tau pathology. A quantitative analysis of IVIS brain signal over time (FIG. 8C), shows strong (G48 and G47) and moderate (rest of animals) specific signal from the tauopathy mice, presumably reflecting their degree of tau pathology, and minimal signal in wt mice (WT1 and WT2). One mouse (F45) received both sdAb 2B8 (sdAb1) and polyclonal sdAbs (p-sdAbs) resulting in a similar brain signal. The signal peaks within a few minutes and is stable within the

first hour. The sdAb signal is cleared faster (within the first day) than with scFv or mAb that takes several days to dissipate (Krishnaswamy et al., "Antibody-derived In vivo Imaging of tau Pathology," *JNeurosci* 10:34(50): 16835-50 (2014), which is hereby incorporated by reference in its entirety).

(470) Distribution of the tau sdAb in the brain after intravenous administration was also determined. In a subset of IVIS imaged mice (near-infrared dye labeled sdAb 2B8 (SdAb1) (250 µg) injected intravenously (i.v)), brains were removed 1 h after injection, sectioned and stained with a nuclear stain (Hoechst) (FIG. 9, far left images), and an antibody against early endosomes (EEA1; FIG. 9, top panel, second image from left) or late endosomes/lysosomes (Rab7; FIG. 9, bottom panel, second image from left). Merged images and their magnification (as per white boxes; FIG. 9, top and bottom panel, last two images on the right) revealed that sdAb1 enters the brain following intravenous injection and is taken up into neurons into the endosomal-lysosomal system.

(471) Binding of anti-tau sdAb to pathological tau in the brain after intravenous injection in tauopathy mice was also examined. In a subset of IVIS imaged mice (near-infrared dye labeled sdAb 2B8 (sdAb1) (250 µg) injected intravenously (i.v)), brains were removed 1 h after injection, sectioned and stained with a nuclear stain (Hoechst; FIG. 10, far left images), and an antibody against hyperphosphorylated tau epitope (PHF1; FIG. 10, top panel, second from left) or a conformational tau epitope (MC1; FIG. 10, bottom panel, second from left). Merged images and their magnification (as per white boxes; FIG. 10, top and bottom panel, last two images on the right) revealed that sdAb1 enters the brain following intravenous injection and is taken up into neurons, where it binds to pathological tau protein (in the endosomal-lysosomal system as per FIG. 9).

(472) Although preferred embodiments have been depicted and described in detail herein, it will be apparent to those skilled in the relevant art that various modifications, additions, substitutions, and the like can be made without departing from the spirit of the invention and these are therefore considered to be within the scope of the invention as defined in the claims which follow.

Claims

1. An antibody or fragment thereof, said antibody or binding fragment thereof comprising a heavy chain variable region selected from the group consisting of: (i) a heavy chain variable region comprising the CDR-H1 of SEQ ID NO: 1, the CDR-H2 of SEQ ID NO: 39, and the CDR-H3 of SEQ ID NO: 77; (ii) a heavy chain variable region comprising the CDR-H1 of SEQ ID NO: 2, the CDR-H2 of SEQ ID NO: 40, and the CDR-H3 of SEQ ID NO: 78; (iii) a heavy chain variable region comprising the CDR-H1 of SEQ ID NO: 3, the CDR-H2 of SEQ ID NO: 41, and the CDR-H3 of SEQ ID NO: 79; (iv) a heavy chain variable region comprising the CDR-H1 of SEQ ID NO: 4, the CDR-H2 of SEQ ID NO: 42, and the CDR-H3 of SEQ ID NO: 80; (v) a heavy chain variable region comprising the CDR-H1 of SEQ ID NO: 5, the CDR-H2 of SEQ ID NO: 43, and the CDR-H3 of SEQ ID NO: 81; (vi) a heavy chain variable region comprising the CDR-H1 of SEQ ID NO: 6, the CDR-H2 of SEQ ID NO: 44, and the CDR-H3 of SEQ ID NO: 82; (vii) a heavy chain variable region comprising the CDR-H1 of SEQ ID NO: 7, the CDR-H2 of SEQ ID NO: 45, and the CDR-H3 of SEQ ID NO: 83; (viii) a heavy chain variable region comprising the CDR-H1 of SEQ ID NO: 8, the CDR-H2 of SEQ ID NO: 46, and the CDR-H3 of SEQ ID NO: 84; (ix) a heavy chain variable region comprising the CDR-H1 of SEQ ID NO: 9, the CDR-H2 of SEQ ID NO: 47, and the CDR-H3 of SEQ ID NO: 85; (x) a heavy chain variable region comprising the CDR-H1 of SEQ ID NO: 10, the CDR-H2 of SEQ ID NO: 48, and the CDR-H3 of SEQ ID NO: 86; (xi) a heavy chain variable region comprising the CDR-H1 of SEQ ID NO: 11, the CDR-H2 of SEQ ID NO: 49, and the CDR-H3 of SEQ ID NO: 87; (xii) a heavy chain variable region comprising the CDR-H1 of SEQ ID NO: 12, the CDR-H2 of SEQ ID NO: 50, and the CDR-H3 of SEQ ID NO: 88; (xiii) a heavy chain variable region comprising the CDR-H1 of SEQ ID NO: 13, the CDR-H2

of SEQ ID NO: 51, and the CDR-H3 of SEQ ID NO: 89; (xiv) a heavy chain variable region comprising the CDR-H1 of SEQ ID NO: 14, the CDR-H2 of SEQ ID NO: 52, and the CDR-H3 of SEQ ID NO: 90; (xv) a heavy chain variable region comprising the CDR-H1 of SEQ ID NO: 15, the CDR-H2 of SEQ ID NO: 53, and the CDR-H3 of SEQ ID NO: 91; (xvi) a heavy chain variable region comprising the CDR-H1 of SEQ ID NO: 16, the CDR-H2 of SEQ ID NO: 54, and the CDR-H3 of SEQ ID NO: 92; (xvii) a heavy chain variable region comprising the CDR-H1 of SEQ ID NO: 17, the CDR-H2 of SEQ ID NO: 55, and the CDR-H3 of SEQ ID NO: 93; (xviii) a heavy chain variable region comprising the CDR-H1 of SEQ ID NO: 18 the CDR-H2 of SEQ ID NO: 56, and the CDR-H3 of SEQ ID NO: 94; (xix) a heavy chain variable region comprising the CDR-H1 of SEQ ID NO: 19, the CDR-H2 of SEQ ID NO: 57, and the CDR-H3 of SEQ ID NO: 95; (xx) a heavy chain variable region comprising the CDR-H1 of SEQ ID NO: 20, the CDR-H2 of SEQ ID NO: 58, and the CDR-H3 of SEQ ID NO: 96; (xxi) a heavy chain variable region comprising the CDR-H1 of SEQ ID NO: 21, the CDR-H2 of SEQ ID NO: 59, and the CDR-H3 of SEQ ID NO: 97; (xxii) a heavy chain variable region comprising the CDR-H1 of SEQ ID NO: 22, the CDR-H2 of SEQ ID NO: 60, and the CDR-H3 of SEQ ID NO: 98; (xxiii) a heavy chain variable region comprising the CDR-H1 of SEQ ID NO: 23, the CDR-H2 of SEQ ID NO: 61, and the CDR-H3 of SEQ ID NO: 99; (xxiv) a heavy chain variable region comprising the CDR-H1 of SEQ ID NO: 24, the CDR-H2 of SEQ ID NO: 62, and the CDR-H3 of SEQ ID NO: 100; (xxv) a heavy chain variable region comprising the CDR-H1 of SEQ ID NO: 25, the CDR-H2 of SEQ ID NO: 63, and the CDR-H3 of SEQ ID NO: 101; (xxvi) a heavy chain variable region comprising the CDR-H1 of SEQ ID NO: 26, the CDR-H2 of SEQ ID NO: 64, and the CDR-H3 of SEQ ID NO: 102; (xxvii) a heavy chain variable region comprising the CDR-H1 of SEQ ID NO: 27, the CDR-H2 of SEQ ID NO: 65, and the CDR-H3 of SEQ ID NO: 103; (xxviii) a heavy chain variable region comprising the CDR-H1 of SEQ ID NO: 28, the CDR-H2 of SEQ ID NO: 66, and the CDR-H3 of SEQ ID NO: 104; (xxix) a heavy chain variable region comprising the CDR-H1 of SEQ ID NO: 29, the CDR-H2 of SEQ ID NO: 67, and the CDR-H3 of SEQ ID NO: 105; (xxx) a heavy chain variable region comprising the CDR-H1 of SEQ ID NO: 30, the CDR-H2 of SEQ ID NO: 68, and the CDR-H3 of SEQ ID NO: 106; (xxxi) a heavy chain variable region comprising the CDR-H1 of SEQ ID NO: 31, the CDR-H2 of SEQ ID NO: 69, and the CDR-H3 of SEQ ID NO: 107; (xxxii) a heavy chain variable region comprising the CDR-H1 of SEQ ID NO: 32, the CDR-H2 of SEQ ID NO: 70, and the CDR-H3 of SEQ ID NO: 108; (xxxiii) a heavy chain variable region comprising the CDR-H1 of SEQ ID NO: 33, the CDR-H2 of SEQ ID NO: 71, and the CDR-H3 of SEQ ID NO: 109; (xxxiv) a heavy chain variable region comprising the CDR-H1 of SEQ ID NO: 34 the CDR-H2 of SEQ ID NO: 72, and the CDR-H3 of SEQ ID NO: 110; (xxxv) a heavy chain variable region comprising the CDR-H1 of SEQ ID NO: 35, the CDR-H2 of SEQ ID NO: 73, and the CDR-H3 of SEQ ID NO: 111; (xxxvi) a heavy chain variable region comprising the CDR-H1 of SEQ ID NO: 36, the CDR-H2 of SEQ ID NO: 74, and the CDR-H3 of SEQ ID NO: 112; (xxxvii) a heavy chain variable region comprising the CDR-H1 of SEQ ID NO: 37, the CDR-H2 of SEQ ID NO: 75, and the CDR-H3 of SEQ ID NO: 113; (xxxviii) a heavy chain variable region comprising the CDR-H1 of SEQ ID NO: 38, the CDR-H2 of SEQ ID NO: 76, and the CDR-H3 of SEQ ID NO: 114; (xxxix) a heavy chain variable region comprising the CDR-H1 of SEQ ID NO: 202, the CDR-H2 of SEQ ID NO: 203, and the CDR-H3 of SEQ ID NO: 204; (xl) a heavy chain variable region comprising the CDR-H1 of SEQ ID NO: 206, the CDR-H2 of SEQ ID NO: 207, and the CDR-H3 of SEQ ID NO: 208; (xli) a heavy chain variable region comprising the CDR-H1 of SEQ ID NO: 210, the CDR-H2 of SEQ ID NO: 211, and the CDR-H3 of SEQ ID NO: 212; and (xlii) a heavy chain variable region comprising the CDR-H1 of SEQ ID NO: 214, the CDR-H2 of SEQ ID NO: 215, and the CDR-H3 of SEQ ID NO: 216.

2. The antibody or binding fragment thereof of claim 1, wherein said antibody or binding fragment thereof binds to human tau protein.

3. The antibody or binding fragment thereof of claim 1, wherein said antibody or binding fragment

thereof is a camelid antibody or binding fragment thereof.

4. The antibody or binding fragment thereof of claim 1, wherein said antibody or binding fragment thereof is a monoclonal antibody or binding fragment thereof.

5. The antibody or binding fragment thereof of claim 1, wherein said heavy chain variable region further comprises human or a humanized immunoglobulin heavy chain framework regions.

6. The antibody or binding fragment thereof of claim 5, wherein the heavy chain variable region comprises an amino acid sequence that is at least 80% identical to an amino acid sequence selected from the group consisting of SEQ ID NOs: 115-152, 205, 209, 213, and 217/216.

7. The antibody or binding fragment thereof of claim 5, wherein the heavy chain variable region comprises an amino acid sequence selected from the group consisting of SEQ ID NOs: 115-152, 205, 209, 213, and 217.

8. The antibody or binding fragment thereof of claim 1, wherein said antibody or binding fragment thereof is selected from the group consisting of: a single domain antibody (Fv), a minibody, and a bispecific antibody.

9. A polyclonal antibody composition comprising two or more antibodies or fragments thereof of claim 1.

10. A pharmaceutical composition comprising: the antibody or binding fragment thereof of claim 6 and a pharmaceutical carrier.

11. A method of treating a subject to inhibit onset of one or more symptoms of a condition involving pathological tau protein, said method comprising: administering to the subject the pharmaceutical composition of claim 10, wherein said composition is administered in an amount effective to inhibit onset of one or more symptoms of the condition involving pathological tau protein.

12. A method of treating a subject for a condition involving a pathological tau protein, said method comprising: administering to the subject the pharmaceutical composition of claim 10, wherein said composition is administered in an amount effective to treat the condition involving pathological tau protein.

13. A method of diagnosing Alzheimer's disease and/or a tauopathy in a subject, said method comprising: administering the antibody or binding fragment thereof according to claim 1 to the subject; detecting, in the subject, binding of the antibody or binding fragment thereof to accumulated tau protein; and diagnosing the subject for Alzheimer's disease and/or tauopathy based on said detecting.

14. A method of monitoring the progression of Alzheimer's disease or a tauopathy in a subject, said method comprising: administering the antibody or binding fragment thereof according to claim 1 to the subject; detecting, in the subject, binding of the antibody or binding fragment thereof to accumulated tau protein; repeating said detecting periodically; and monitoring the progression of Alzheimer's disease or tauopathy in the subject based on said repeated detecting.

15. A diagnostic kit comprising: the antibody or binding fragment thereof of claim 1 and a detectable label.

16. An antibody or fragment thereof, said antibody or binding fragment thereof comprising a heavy chain variable region selected from the group consisting of: (i) a heavy chain variable region comprising the CDR-H1 of SEQ ID NO: 153, the CDR-H2 of SEQ ID NO: 165, and the CDR-H3 of SEQ ID NO: 177; (ii) a heavy chain variable region comprising the CDR-H1 of SEQ ID NO: 154, the CDR-H2 of SEQ ID NO: 166, and the CDR-H3 of SEQ ID NO: 178; (iii) a heavy chain variable region comprising the CDR-H1 of SEQ ID NO: 155, the CDR-H2 of SEQ ID NO: 167, and the CDR-H3 of SEQ ID NO: 179; (iv) a heavy chain variable region comprising the CDR-H1 of SEQ ID NO: 156, the CDR-H2 of SEQ ID NO: 168, and the CDR-H3 of SEQ ID NO: 180; (v) a heavy chain variable region comprising the CDR-H1 of SEQ ID NO: 157, the CDR-H2 of SEQ ID NO: 169, and the CDR-H3 of SEQ ID NO: 181; (vi) a heavy chain variable region comprising the CDR-H1 of SEQ ID NO: 158, the CDR-H2 of SEQ ID NO: 170, and the CDR-H3 of SEQ ID NO:

182; (vii) a heavy chain variable region comprising the CDR-H1 of SEQ ID NO: 159, the CDR-H2 of SEQ ID NO: 171, and the CDR-H3 of SEQ ID NO: 183; (viii) a heavy chain variable region comprising the CDR-H1 of SEQ ID NO: 160, the CDR-H2 of SEQ ID NO: 172, and the CDR-H3 of SEQ ID NO: 184; (ix) a heavy chain variable region comprising the CDR-H1 of SEQ ID NO: 161, the CDR-H2 of SEQ ID NO: 173, and the CDR-H3 of SEQ ID NO: 185; (x) a heavy chain variable region comprising the CDR-H1 of SEQ ID NO: 162, the CDR-H2 of SEQ ID NO: 174, and the CDR-H3 of SEQ ID NO: 186; (xi) a heavy chain variable region comprising the CDR-H1 of SEQ ID NO: 163, the CDR-H2 of SEQ ID NO: 175, and the CDR-H3 of SEQ ID NO: 187; (xii) a heavy chain variable region comprising the CDR-H1 of SEQ ID NO: 164, the CDR-H2 of SEQ ID NO: 176, and the CDR-H3 of SEQ ID NO: 188; and (xiii) a heavy chain variable region comprising the CDR-H1 of SEQ ID NO: 307, the CDR-H2 of SEQ ID NO: 308, and the CDR-H3 of SEQ ID NO: 309.

17. The antibody or binding fragment thereof of claim 16, wherein said antibody or binding fragment thereof binds to human tau protein.

18. The antibody or binding fragment thereof of claim 16, wherein said antibody or binding fragment thereof is a camelid antibody or binding fragment thereof.

19. The antibody or binding fragment thereof of claim 16, wherein said antibody or binding fragment thereof is a monoclonal antibody or binding fragment thereof.

20. The antibody or binding fragment thereof of claim 16, wherein said heavy chain variable region further comprises human or a humanized immunoglobulin heavy chain framework regions.

21. The antibody or binding fragment thereof of claim 20, wherein the heavy chain variable region comprises an amino acid sequence that is at least 80% identical to an amino acid sequence selected from the group consisting of SEQ ID NOs: 189-201 and 310.

22. The antibody or binding fragment thereof of claim 20, wherein the heavy chain variable region comprises an amino acid sequence selected from the group consisting of SEQ ID NOs: 189-201 and 310.

23. The antibody or binding fragment thereof of claim 16, wherein said antibody or binding fragment thereof is selected from the group consisting of: a single domain antibody (Fv), a minibody, and a bispecific antibody.

24. A polyclonal antibody composition comprising two or more antibodies or fragments thereof of claim 16.

25. A pharmaceutical composition comprising: the antibody or binding fragment thereof of claim 16 and a pharmaceutical carrier.

26. A method of treating a subject to inhibit onset of one or more symptoms of a condition involving pathological tau protein, said method comprising: administering to the subject the pharmaceutical composition of claim 25, wherein said composition is administered in an amount effective to inhibit onset of one or more symptoms of the condition involving pathological tau protein in the subject.

27. A method of treating a subject for a condition involving a pathological tau protein, said method comprising: administering to the subject the pharmaceutical composition of claim 25, wherein said composition is administered in an amount effective to treat the condition involving pathological tau protein.

28. A method of diagnosing Alzheimer's disease and/or a tauopathy in a subject, said method comprising: administering the antibody or binding fragment thereof according to claim 16 to the subject; detecting, in the subject, binding of the antibody or binding fragment thereof to accumulated tau protein; and diagnosing the subject for Alzheimer's disease and/or tauopathy based on said detecting.

29. A method of monitoring the progression of Alzheimer's disease or a tauopathy in a subject, said method comprising: administering the antibody or binding fragment thereof according to claim 16 to the subject; detecting, in the subject, binding of the antibody or binding fragment thereof to

accumulated tau protein; repeating said detecting periodically; and monitoring the progression of Alzheimer's disease or tauopathy in the subject based on said repeated detecting.

30. A diagnostic kit comprising: the antibody or binding fragment thereof of claim 16 and a detectable label.

31. An isolated polynucleotide encoding an antibody or binding fragment thereof comprising a heavy chain variable region selected from the group consisting of: (i) a heavy chain variable region comprising the CDR-H1 of SEQ ID NO: 1, the CDR-H2 of SEQ ID NO: 39, and the CDR-H3 of SEQ ID NO: 77; (ii) a heavy chain variable region comprising the CDR-H1 of SEQ ID NO: 2, the CDR-H2 of SEQ ID NO: 40, and the CDR-H3 of SEQ ID NO: 78; (iii) a heavy chain variable region comprising the CDR-H1 of SEQ ID NO: 3, the CDR-H2 of SEQ ID NO: 41, and the CDR-H3 of SEQ ID NO: 79; (iv) a heavy chain variable region comprising the CDR-H1 of SEQ ID NO: 4, the CDR-H2 of SEQ ID NO: 42, and the CDR-H3 of SEQ ID NO: 80; (v) a heavy chain variable region comprising the CDR-H1 of SEQ ID NO: 5, the CDR-H2 of SEQ ID NO: 43, and the CDR-H3 of SEQ ID NO: 81; (vi) a heavy chain variable region comprising the CDR-H1 of SEQ ID NO: 6, the CDR-H2 of SEQ ID NO: 44, and the CDR-H3 of SEQ ID NO: 82; (vii) a heavy chain variable region comprising the CDR-H1 of SEQ ID NO: 7, the CDR-H2 of SEQ ID NO: 45, and the CDR-H3 of SEQ ID NO: 83; (viii) a heavy chain variable region comprising the CDR-H1 of SEQ ID NO: 8, the CDR-H2 of SEQ ID NO: 46, and the CDR-H3 of SEQ ID NO: 84; (ix) a heavy chain variable region comprising the CDR-H1 of SEQ ID NO: 9, the CDR-H2 of SEQ ID NO: 47, and the CDR-H3 of SEQ ID NO: 85; (x) a heavy chain variable region comprising the CDR-H1 of SEQ ID NO: 10, the CDR-H2 of SEQ ID NO: 48, and the CDR-H3 of SEQ ID NO: 86; (xi) a heavy chain variable region comprising the CDR-H1 of SEQ ID NO: 11, the CDR-H2 of SEQ ID NO: 49, and the CDR-H3 of SEQ ID NO: 87; (xii) a heavy chain variable region comprising the CDR-H1 of SEQ ID NO: 12, the CDR-H2 of SEQ ID NO: 50, and the CDR-H3 of SEQ ID NO: 88; (xiii) a heavy chain variable region comprising the CDR-H1 of SEQ ID NO: 13, the CDR-H2 of SEQ ID NO: 51, and the CDR-H3 of SEQ ID NO: 89; (xiv) a heavy chain variable region comprising the CDR-H1 of SEQ ID NO: 14, the CDR-H2 of SEQ ID NO: 52, and the CDR-H3 of SEQ ID NO: 90; (xv) a heavy chain variable region comprising the CDR-H1 of SEQ ID NO: 15, the CDR-H2 of SEQ ID NO: 53, and the CDR-H3 of SEQ ID NO: 91; (xvi) a heavy chain variable region comprising the CDR-H1 of SEQ ID NO: 16, the CDR-H2 of SEQ ID NO: 54, and the CDR-H3 of SEQ ID NO: 92; (xvii) a heavy chain variable region comprising the CDR-H1 of SEQ ID NO: 17, the CDR-H2 of SEQ ID NO: 55, and the CDR-H3 of SEQ ID NO: 93; (xviii) a heavy chain variable region comprising the CDR-H1 of SEQ ID NO: 18, the CDR-H2 of SEQ ID NO: 56, and the CDR-H3 of SEQ ID NO: 94; (xix) a heavy chain variable region comprising the CDR-H1 of SEQ ID NO: 19, the CDR-H2 of SEQ ID NO: 57, and the CDR-H3 of SEQ ID NO: 95; (xx) a heavy chain variable region comprising the CDR-H1 of SEQ ID NO: 20, the CDR-H2 of SEQ ID NO: 58, and the CDR-H3 of SEQ ID NO: 96; (xxi) a heavy chain variable region comprising the CDR-H1 of SEQ ID NO: 21, the CDR-H2 of SEQ ID NO: 59, and the CDR-H3 of SEQ ID NO: 97; (xxii) a heavy chain variable region comprising the CDR-H1 of SEQ ID NO: 22, the CDR-H2 of SEQ ID NO: 60, and the CDR-H3 of SEQ ID NO: 98; (xxiii) a heavy chain variable region comprising the CDR-H1 of SEQ ID NO: 23, the CDR-H2 of SEQ ID NO: 61, and the CDR-H3 of SEQ ID NO: 99; (xxiv) a heavy chain variable region comprising the CDR-H1 of SEQ ID NO: 24, the CDR-H2 of SEQ ID NO: 62, and the CDR-H3 of SEQ ID NO: 100; (xxv) a heavy chain variable region comprising the CDR-H1 of SEQ ID NO: 25, the CDR-H2 of SEQ ID NO: 63, and the CDR-H3 of SEQ ID NO: 101; (xxvi) a heavy chain variable region comprising the CDR-H1 of SEQ ID NO: 26, the CDR-H2 of SEQ ID NO: 64, and the CDR-H3 of SEQ ID NO: 102; (xxvii) a heavy chain variable region comprising the CDR-H1 of SEQ ID NO: 27, the CDR-H2 of SEQ ID NO: 65, and the CDR-H3 of SEQ ID NO: 103; (xxviii) a heavy chain variable region comprising the CDR-H1 of SEQ ID NO: 28, the CDR-H2 of SEQ ID NO: 66, and the CDR-H3 of SEQ ID NO: 104; (xxix) a heavy chain variable region comprising the CDR-H1 of SEQ ID NO: 29, the

CDR-H2 of SEQ ID NO: 67, and the CDR-H3 of SEQ ID NO: 105; (xxx) a heavy chain variable region comprising the CDR-H1 of SEQ ID NO: 30, the CDR-H2 of SEQ ID NO: 68, and the CDR-H3 of SEQ ID NO: 106; (xxxi) a heavy chain variable region comprising the CDR-H1 of SEQ ID NO: 31, the CDR-H2 of SEQ ID NO: 69, and the CDR-H3 of SEQ ID NO: 107; (xxxii) a heavy chain variable region comprising the CDR-H1 of SEQ ID NO: 32, the CDR-H2 of SEQ ID NO: 70, and the CDR-H3 of SEQ ID NO: 108; (xxxiii) a heavy chain variable region comprising the CDR-H1 of SEQ ID NO: 33, the CDR-H2 of SEQ ID NO: 71, and the CDR-H3 of SEQ ID NO: 109; (xxxiv) a heavy chain variable region comprising the CDR-H1 of SEQ ID NO: 34 the CDR-H2 of SEQ ID NO: 72, and the CDR-H3 of SEQ ID NO: 110; (xxxv) a heavy chain variable region comprising the CDR-H1 of SEQ ID NO: 35, the CDR-H2 of SEQ ID NO: 73, and the CDR-H3 of SEQ ID NO: 111; (xxxvi) a heavy chain variable region comprising the CDR-H1 of SEQ ID NO: 36, the CDR-H2 of SEQ ID NO: 74, and the CDR-H3 of SEQ ID NO: 112; (xxxvii) a heavy chain variable region comprising the CDR-H1 of SEQ ID NO: 37, the CDR-H2 of SEQ ID NO: 75, and the CDR-H3 of SEQ ID NO: 113; (xxxviii) a heavy chain variable region comprising the CDR-H1 of SEQ ID NO: 38, the CDR-H2 of SEQ ID NO: 76, and the CDR-H3 of SEQ ID NO: 114; (xxxix) a heavy chain variable region comprising the CDR-H1 of SEQ ID NO: 202, the CDR-H2 of SEQ ID NO: 203, and the CDR-H3 of SEQ ID NO: 204; (xl) a heavy chain variable region comprising the CDR-H1 of SEQ ID NO: 206, the CDR-H2 of SEQ ID NO: 207, and the CDR-H3 of SEQ ID NO: 208; (xli) a heavy chain variable region comprising the CDR-H1 of SEQ ID NO: 210, the CDR-H2 of SEQ ID NO: 211, and the CDR-H3 of SEQ ID NO: 212; (xlii) a heavy chain variable region comprising the CDR-H1 of SEQ ID NO: 214, the CDR-H2 of SEQ ID NO: 215, and the CDR-H3 of SEQ ID NO: 216; (xliii) a heavy chain variable region comprising the CDR-H1 of SEQ ID NO: 153, the CDR-H2 of SEQ ID NO: 165, and the CDR-H3 of SEQ ID NO: 177; (xliv) a heavy chain variable region comprising the CDR-H1 of SEQ ID NO: 154, the CDR-H2 of SEQ ID NO: 166, and the CDR-H3 of SEQ ID NO: 178; (xlv) a heavy chain variable region comprising the CDR-H1 of SEQ ID NO: 155, the CDR-H2 of SEQ ID NO: 167, and the CDR-H3 of SEQ ID NO: 179; (xlvi) a heavy chain variable region comprising the CDR-H1 of SEQ ID NO: 156, the CDR-H2 of SEQ ID NO: 168, and the CDR-H3 of SEQ ID NO: 180; (xlvii) a heavy chain variable region comprising the CDR-H1 of SEQ ID NO: 157, the CDR-H2 of SEQ ID NO: 169, and the CDR-H3 of SEQ ID NO: 181; (xlviii) a heavy chain variable region comprising the CDR-H1 of SEQ ID NO: 158, the CDR-H2 of SEQ ID NO: 170, and the CDR-H3 of SEQ ID NO: 182; (xlix) a heavy chain variable region comprising the CDR-H1 of SEQ ID NO: 159, the CDR-H2 of SEQ ID NO: 171, and the CDR-H3 of SEQ ID NO: 183; (l) a heavy chain variable region comprising the CDR-H1 of SEQ ID NO: 160, the CDR-H2 of SEQ ID NO: 172, and the CDR-H3 of SEQ ID NO: 184; (li) a heavy chain variable region comprising the CDR-H1 of SEQ ID NO: 161, the CDR-H2 of SEQ ID NO: 173, and the CDR-H3 of SEQ ID NO: 185; (lii) a heavy chain variable region comprising the CDR-H1 of SEQ ID NO: 162, the CDR-H2 of SEQ ID NO: 174, and the CDR-H3 of SEQ ID NO: 186; (liii) a heavy chain variable region comprising the CDR-H1 of SEQ ID NO: 163, the CDR-H2 of SEQ ID NO: 175, and the CDR-H3 of SEQ ID NO: 187; (liv) a heavy chain variable region comprising the CDR-H1 of SEQ ID NO: 164, the CDR-H2 of SEQ ID NO: 176, and the CDR-H3 of SEQ ID NO: 188; and (lv) a heavy chain variable region comprising the CDR-H1 of SEQ ID NO: 307, the CDR-H2 of SEQ ID NO: 308, and the CDR-H3 of SEQ ID NO: 309.

32. A vector comprising the isolated polynucleotide of claim 31.

33. The vector of claim 32, wherein said vector is an adeno-associated viral vector.

34. A host cell comprising the vector of claim 33.
